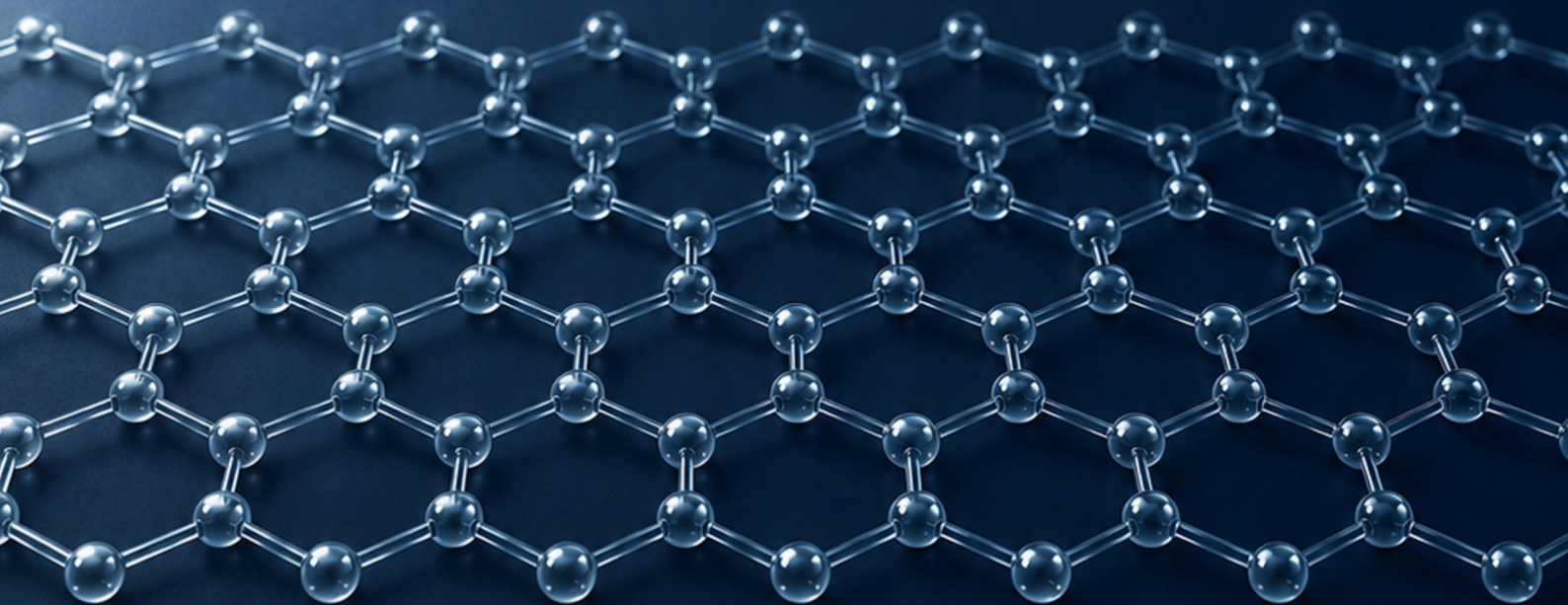


Antonio Cammarata, Elliot Perviz

Elements of Atomistic Simulations



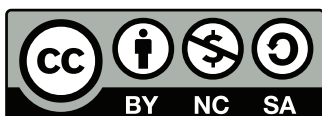
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Preface

These lecture notes accompany the course “Elements of atomistic simulations” held at the Department of Control Engineering, Faculty of Electrical Engineering, Czech Technical University in Prague, and are intended to provide a coherent written support to the topics discussed during the lectures. The course develops along a progressive path, starting from classical descriptions of matter and advancing toward quantum-mechanical approaches, with the common goal of understanding, modelling, and predicting the structural, dynamical, and electronic properties of condensed-matter systems from an atomistic perspective.

We begin by introducing classical molecular dynamics, where atoms are treated as point particles evolving under Newton’s equations of motion. This framework allows the simulation of large systems over long time scales and provides access to thermodynamic, structural, and dynamical properties through statistical averaging. Within this context, we discuss interatomic potentials, equilibration, computation of observables, and the interpretation of quantities such as radial and pair distribution functions, pressure, and temperature. We then extend the discussion to periodic systems, reciprocal space, Brillouin zones, supercells, and $\mathbf{k}$ -point sampling, which are essential concepts for modelling crystalline materials and defects. The course then transitions to a quantum-mechanical description of matter, motivated by the limitations of classical approaches when electronic degrees of freedom play a crucial role. We introduce the fundamental concepts of quantum mechanics, the role of the Schrödinger equation, basis sets, and the self-consistent field procedure as a unifying framework underlying several electronic-structure methods. Emphasis is placed on how total energies, forces, and other physical quantities are obtained in practice, and on the general structure of first-principles calculations, independently of the specific level of approximation adopted.

Building on this foundation, we address key applications such as geometry optimisation, reaction and transformation paths, and phase stability through formation energies. Particular attention is devoted to lattice dynamics and phonons, including their physical interpretation, computational determination, and connection to measurable properties such as vibrational spectra, thermal stability, and heat transport. Both finite-displacement and linear-response approaches are discussed, together with practical aspects of convergence and numerical accuracy. Finally, the course introduces electron–phonon coupling as a central concept linking electronic structure and lattice dynamics, highlighting its importance for transport phenomena, superconductivity, and temperature-dependent material properties.

These notes aim to provide a structured reference that students can consult during and after the course, helping to consolidate definitions, equations, and conceptual links between different topics. However, they are not meant to be a self-contained textbook, nor can they substitute for the oral lectures. Many explanations, physical interpretations, examples, and methodological insights are developed interactively during the lectures and through discussion, and cannot be fully captured in written form. The notes should therefore be regarded as a complementary tool, to be read alongside attentive participation

in the lectures and active engagement with exercises and computational examples. Several books exist covering all the topics here discussed; as an example, we suggest the reading of “Understanding Molecular Simulation: From Algorithms to Applications” by D. Frenkel and B. Smit,¹ “Molecular Quantum Mechanics” by P. W. Atkins and R. S. Friedman,² “Quantum Mechanics, Volume 1” by C. Cohen-Tannoudji, B. Diu and F. Laloe,³ and “Electrons and Phonons: The Theory of Transport Phenomena in Solids” by J. M. Ziman.⁴

Ultimately, the objective of the course is not only to present specific methods, but also to develop a critical understanding of their assumptions, capabilities, and limitations, opening the students the path towards the creation of new materials for target applications.

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Chapter 1

Introduction

The practice of atomistic simulations relies fundamentally on a computational environment that enables efficient numerical experimentation, data management, and reproducible scientific workflows. In this course, the Linux operating system is adopted as the reference platform because it provides a stable, transparent, and highly configurable environment that underpins the majority of modern scientific computing infrastructures. A working knowledge of Linux is therefore not merely a technical convenience but an essential component of the methodological toolkit required for molecular dynamics and quantum mechanical simulations. The aim of this introductory chapter is to establish the conceptual and practical foundations needed to interact productively with Linux-based systems and High Performance Computing facilities, which are the natural habitat of atomistic simulation codes. Some concepts, like molecular dynamics or basis set, mentioned in this chapter, will be discussed in detail later on.

1.1 The Linux computational environment

Linux systems⁵ are typically accessed through a command-line interface, most commonly via a Unix shell such as `bash`, which acts as an interpreter between the user and the operating system kernel. The shell provides mechanisms for launching programs, redirecting input and output streams, manipulating files and directories, and composing complex workflows through scripting. In the context of atomistic simulations, the shell is routinely used to prepare input files, automate parameter sweeps, parse output data, and chain together preprocessing and postprocessing steps. Understanding the semantics of environment variables, file permissions, and process control is particularly important when simulations are executed on shared systems or remote clusters.

A central feature of the Linux environment is the compilation of scientific software from source code, which is commonly written in languages such as C, C++, and Fortran for reasons of performance and numerical control.⁶ Compilation can be performed in a static or dynamic manner, corresponding respectively to the inclusion of all required collection of pre-compiled pieces of code (i.e. *libraries*) within the executable or to their resolution at runtime through shared objects. Static compilation can enhance portability and reproducibility at the cost of larger executables, whereas dynamic linking reduces memory usage. For atomistic simulation codes, the choice between static and dynamic linking often depends on the policies of the target computing facility and on the availability of optimized numerical libraries.

Linear algebra libraries play a crucial role, as they provide efficient implementations of

matrix operations that dominate the computational cost of many algorithms. The Linear Algebra PACKage^{7,8} (LAPACK) and the Basic Linear Algebra Subprograms⁹ (BLAS) define standardised interfaces for dense linear algebra routines, enabling simulation codes to leverage highly optimized vendor-specific implementations. On modern systems, these libraries are frequently provided by packages such as OpenBLAS,¹⁰ Intel MKL,¹¹ or AMD AOCL,¹² each of which exploits low-level architectural features such as vector instructions and cache hierarchies. A conceptual understanding of how simulation codes link against these libraries is essential for diagnosing performance bottlenecks and ensuring numerical reliability.

1.2 High Performance Computing facilities

While small-scale atomistic simulations can be executed on personal workstations, realistic systems and long time scales require the use of High Performance Computing infrastructures. Such facilities consist of large numbers of compute nodes interconnected by high-speed networks and managed by specialized software that allocates resources among users. Access to these systems is typically obtained through secure remote connections using the Secure Shell protocol (ssh), which provides encrypted communication and authentication mechanisms. Through ssh, users can log into remote systems (*computational clusters*), execute commands, and interact with batch schedulers as if they were working on a local terminal. Access to a remote cluster is typically established via an ssh command of the form `ssh username@cluster.address.cz`.

Efficient data transfer between local machines and remote clusters is another critical aspect of practical simulation work. Tools such as `rsync` enable incremental file synchronisation, minimizing data transfer by copying only modified portions of files or directory trees. This functionality is particularly valuable when dealing with large trajectory files or checkpoint data generated during the simulations. The use of such tools supports reproducibility by ensuring that input parameters, scripts, and results remain consistent across computing environments and local data.

The execution of jobs on shared HPC systems is governed by workload managers, among which SLURM (Simple Linux Utility for Resource Management) is one of the most widely deployed.¹³ A scheduler such as SLURM mediates access to compute resources by queuing user jobs, enforcing allocation policies, and launching applications on assigned nodes. Users specify their computational requirements, such as the number of processors, memory, and wall-clock time, through *job submission scripts*, which are essentially text files containing commands specifying the kind and amount of resource to ask for; an example of such script is the following:

```
[user@login_node.cz ~]$ cat runscript.job
```

```
#!/bin/bash
```

```
#SBATCH --job-name myjob
#SBATCH --output job.out
#SBATCH --error job.err
#SBATCH --mail-user=user@fel.cvut.cz
#SBATCH --mail-type=END,FAIL,TIME_LIMIT
#SBATCH --account OPEN-01-02
#SBATCH --nodes=1
```

```
#SBATCH --tasks-per-node=128
#SBATCH --time 48:00:00
#SBATCH --partition qcpu

cd $SLURM_SUBMIT_DIR

mycommand input.in

exit 0
```

Understanding the interaction between simulation codes and the scheduler is essential to achieve efficient resource utilisation and avoid premature job termination.

1.3 Parallelism in atomistic simulations

The computational demands of atomistic simulations necessitate the use of parallel computing paradigms, which distribute work across multiple processing units. Message Passing Interface (MPI)¹⁴ is a standardised framework that enables parallel programs to communicate by explicitly exchanging messages between processes. MPI is particularly well suited for distributed-memory architectures, where each process has its own address space and communication occurs over a network. Atomistic simulation codes^{15–19} employ MPI to decompose the simulated system across processes, allowing scaling to thousands of cores. In contrast, OpenMP²⁰ provides a shared-memory parallel programming model based on compiler directives that annotate regions of code to be executed concurrently by multiple threads. OpenMP is often used within a single compute node to exploit multicore processors, complementing MPI in hybrid parallelization strategies. Such hybrid approaches are increasingly common, as they balance the low-latency communication of shared memory with the scalability of distributed-memory systems. A conceptual understanding of these models allows users to select appropriate parallelization parameters and to interpret performance metrics produced by simulation codes.

1.4 Precision, accuracy and errors in atomistic simulations

The reliability of atomistic simulations depends critically on the precision with which the underlying equations (e.g. Newton or Schrödinger equations) are solved, as well as on the interpretation of the resulting numerical data. In practice, every simulation involves a set of methodological and numerical choices that determine the attainable precision, including integration time steps in molecular dynamics, basis sets in quantum mechanical calculations, and numerical grids or meshes used to evaluate integrals. These choices control the extent to which the computed quantities approximate the ideal solution of the theoretical model being employed. For example, in molecular dynamics, the integration of Newton’s equations of motion is performed using finite time steps, which introduce a systematic discretization error into the trajectories. As the time step is reduced, the numerical integration converges toward the exact continuous-time solution, and conserved quantities such as the total energy approach asymptotic values characteristic of the chosen force field or potential energy surface. Similarly, in quantum mechanical calculations based on density functional theory or wavefunction methods, the completeness of the basis set determines how accurately the electronic wavefunction can be represented. Increasing the

size and flexibility of the basis set leads to a systematic convergence of the total energy, a behaviour formalized in the variational principle for ground-state calculations.²¹

Numerical integration meshes, such as real-space grids or reciprocal-space sampling points, play an analogous role in controlling precision. For example, the density integration grid in density functional theory or the $\mathbf{k}$ -point sampling of the Brillouin zone in periodic systems introduces discretization errors that vanish only in the limit of infinite resolution. In all these cases, the notion of *numerical precision* refers to the closeness of the computed result to the exact solution of the chosen theoretical model as the numerical parameters are systematically refined. The asymptotic value obtained in this limit is therefore model-dependent and does not necessarily coincide with the true physical value observed in nature.

It is essential to distinguish numerical precision from *statistical error*, particularly in simulations that involve sampling over configurations, such as molecular dynamics or Monte Carlo methods. Statistical error arises from the finite length of a data set, and reflects the uncertainty associated with estimating ensemble averages from a limited number of samples.¹ Unlike numerical discretization errors, statistical errors decrease with increased sampling size rather than with changes in numerical parameters such as time step or basis set. As a result, a simulation may be numerically precise, in the sense that discretization errors are negligible, yet still exhibit large statistical uncertainties due to insufficient sampling. The separation of numerical precision and statistical error is crucial for the correct interpretation of simulation results. Numerical convergence tests, such as reducing the time step or increasing the basis set, are designed to assess whether systematic errors are under control. Statistical analysis techniques, including block averaging and error estimation from autocorrelation times, are required to quantify sampling uncertainty. Only when both sources of error are adequately characterised can simulation data be meaningfully compared across different computational setups.

A further layer of complexity arises when comparing simulation results with experimental measurements. Experimental values are subject to their own uncertainties, which may originate from instrumental resolution, sample preparation, or data analysis procedures. Moreover, experiments often probe macroscopic observables under conditions that may differ subtly from those assumed in simulations, such as finite temperature effects, impurities, or external fields. Consequently, agreement between simulation and experiment must be evaluated within the combined error bars of both approaches rather than through direct equality of numerical values.²² When comparing physical quantities obtained from simulations and experiments, it is therefore essential to assess how different sources of error propagate into the final results. A systematic numerical error in a simulation setup can shift all computed values in the same direction, while statistical errors introduce fluctuations around a mean value. Experimental uncertainties may exhibit similar distinctions between systematic and random components. A meaningful comparison requires that these contributions be identified and, where possible, quantified on a common footing.

From a methodological standpoint, the goal of atomistic simulations is often not to reproduce experimental values exactly, but to capture trends, relative differences, and mechanistic insights that are robust with respect to reasonable variations in numerical setup. By analyzing how computed quantities change as numerical precision is increased, one can assess the internal consistency of the simulation and establish confidence intervals for predicted observables. This approach enables comparison among different simulations and between simulations and experiments, grounded in an explicit understanding of numerical precision, statistical uncertainty, and physical modelling assumptions.

1.5 Atomic structure representation and interpretation

Atomistic simulations are defined at their most fundamental level by a specification of the atomic structure and of the interactions governing the behaviour of the constituent particles. Typical input parameters therefore include the positions of all atoms in the system and a mathematical description of their interactions, which may take the form of an empirical force field in classical molecular dynamics or a Hamiltonian operator in quantum mechanical approaches. These elements fully determine the interactions between the atoms, thus the system properties and its response to external stimuli.

Atomic positions are always defined with respect to a finite simulation domain, commonly referred to as the *simulation box* or *unit cell*. The unit cell is specified by a set of *lattice vectors* that define the geometry and size of the domain in which the atoms are embedded. Within this framework, atomic coordinates may be expressed either in Cartesian form, corresponding to absolute positions in the space, or in *reduced* coordinates defined relative to the lattice vectors; in this case, the space is said to be *real*. The choice of the coordinate representation is a matter of convenience, but it has important practical implications for a suitable treatment of the system geometry (finite or infinite). An alternative internal-coordinate description of atomic structures is the *Z-matrix* representation, in which the position of each atom is defined relative to previously specified atoms through a set of bond lengths, bond angles, and dihedral angles rather than absolute Cartesian coordinates.²³ In this representation, the first atom is placed at the origin, the second along a reference axis at a given distance, and subsequent atoms are positioned by specifying their distance to a reference atom, the angle formed with a second reference atom, and the dihedral angle with respect to a third, thereby constructing the molecular geometry sequentially. The Z-matrix is particularly well suited for describing isolated molecules and small molecular fragments, as it provides a compact and chemically intuitive parametrization of molecular structure that aligns naturally with common notions of bonding and molecular geometry.²¹ It is typically used in quantum chemistry calculations for geometry optimisation, constrained structural scans, and the systematic exploration of potential energy surfaces, where direct control over internal degrees of freedom is advantageous.²² However, the Z-matrix representation is less convenient for extended or periodic systems, for large flexible molecules, or for molecular dynamics simulations, where frequent changes in connectivity and the need for uniform treatment of all atoms favor Cartesian or fractional coordinate descriptions.¹

Boundary conditions determine how the simulation box interacts with its surroundings and thus strongly influence the physical interpretation of the model. *Periodic boundary conditions* replicate the unit cell infinitely in one, two, or three spatial directions, providing an effective representation of bulk materials or extended phases. For example, the application of periodic boundary conditions in all three spatial directions is commonly used to model bulk crystalline solids such as silicon, sodium chloride, or aluminium, where the unit cell is repeated indefinitely to mimic an infinite crystal lattice. Periodic boundary conditions applied in two spatial directions and combined with a non-periodic direction are typically employed to simulate surfaces or thin films, such as a metal slab exposed to vacuum or an oxide surface interacting with adsorbates. When periodicity is imposed in only one spatial direction, the resulting model is suitable for quasi-one-dimensional systems such as polymer chains, nanowires, or carbon nanotubes, where translational symmetry exists along a single axis while the system remains finite in the perpendicular directions. Non-periodic boundary conditions, by contrast, describe isolated or finite systems such as *clusters* or *molecules* in vacuum. The terms molecule and cluster are both used to

describe finite aggregates of atoms, but they carry distinct conceptual meanings that reflect differences in bonding and typical theoretical treatment. A molecule is traditionally defined as a chemically bound entity composed of a finite number of atoms held together by well-defined covalent, ionic, or metallic bonds, with a discrete electronic structure that gives rise to characteristic spectroscopic signatures and a specific stoichiometry. Molecules are commonly studied as isolated systems and are often described using quantum chemical methods that emphasize localized bonding patterns and internal degrees of freedom.²² A cluster, instead, refers to an aggregate of atoms or molecules whose size lies between that of a single molecule and that of a macroscopic condensed phase. Clusters may consist of a few to several thousands of atoms and do not necessarily exhibit the fixed connectivity or unique bonding patterns associated with molecules. Their stability often arises from a balance of many weak or partially delocalized interactions, and their structural and electronic properties can evolve smoothly with size toward bulk-like behaviour.¹

The distinction between molecules and clusters is, therefore, not absolute but reflects differences in physical behaviour and modelling assumptions. In atomistic simulations, clusters are frequently used to model nanoparticles, catalyst precursors, or fragments of extended materials under non-periodic boundary conditions, whereas molecules are typically treated as chemically well-defined entities with a clear identity independent of their environment. This conceptual distinction guides the choice of simulation methods, boundary conditions, and analysis tools, even though both molecules and clusters, as well as the environment in which they are embedded, are ultimately described by the same underlying physical principles.

Visualization of atomic structures is an indispensable tool for preparing, diagnosing, and interpreting simulations, yet it requires careful conceptual understanding; commonly used atomic structure visualisation software includes VMD,²⁴ OVITO,²⁵ XCRYSDEN²⁶ or VESTA.²⁷ Graphical representations often depict atoms as spheres and interactions as sticks or lines connecting pairs of atoms, but such visual elements are not themselves physical observables. In particular, the presence of a stick between two atoms in a visualisation does not guarantee that a chemical bond exists in a rigorous physical sense. Instead, these graphical connections are typically inferred from geometric criteria such as interatomic distances or coordination numbers chosen by the visualisation software. The physical meaning of an atomic bond depends on the theoretical framework used to describe the system. In classical force fields, bonds may be explicitly encoded as fixed or flexible interactions with prescribed equilibrium lengths and force constants, reflecting an underlying chemical intuition rather than a direct observable. In quantum mechanical descriptions, chemical bonding emerges from the electronic structure, where the distribution of electron density and the resulting energy lowering upon atom association provide the basis for defining bonded interactions. As a result, bonds are best understood as model-dependent constructs that summarize recurring patterns in electronic or energetic behaviour rather than as immutable geometric objects. This distinction has important consequences for the interpretation of simulation results. Changes in interatomic distances or coordination observed during a simulation do not automatically imply the formation or breaking of chemical bonds unless such events are supported by the underlying interaction model. Therefore, a critical assessment of bonding requires a thorough analysis of appropriate physical quantities, such as energy components, electron density distributions, or force constants, depending on the level of theory employed. Maintaining a clear distinction between visualisation conventions and physical definitions is paramount to develop a more robust understanding of atomic-scale phenomena. A possible way to quantify the strength of the bond between two atoms is the use of the harmonic approximation of the potential energy and the evaluation of the

corresponding force constants, which will be discussed later on in [section 8.6](#).

1.6 Crystal structure

The starting point to build a description of a crystal structure is the concept of *direct* lattice. The direct lattice is defined as a set of lattice points in the three-dimensional real space connected by the translation vector

$$\mathbf{T} = u_a \mathbf{a} + u_b \mathbf{b} + u_c \mathbf{c} \quad (1.1)$$

where u_a , u_b and u_c are arbitrary integers and $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are the *lattice vectors*, a set of non-coplanar vectors which define the *periodicity* of the lattice. The vectors form the adjacent edges of a parallelepiped *unit cell*, with volume

$$V_{\text{cell}} = |\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}| \quad (1.2)$$

where the angles between them are denoted $\angle(\mathbf{a}, \mathbf{b}) = \gamma$, $\angle(\mathbf{b}, \mathbf{c}) = \alpha$ and $\angle(\mathbf{c}, \mathbf{a}) = \beta$ respectively. By definition of the translation vector ([Equation 1.1](#)), different integer combinations in the set $\{u_a, u_b, u_c\}$ correspond to *replicas* of the unit cell at different positions in space. If the unit cell contains just one lattice point then it is called the *primitive* cell; there is no smaller volume that can reproduce the entire crystal structure through translation (infinite repetition) to fill all space. To a lattice point we assign one or more atoms, called the *basis*, with the position of an atom i in the basis given by

$$\mathbf{r}_i = v_a \mathbf{a} + v_b \mathbf{b} + v_c \mathbf{c} \quad (1.3)$$

where v_a , v_b and v_c are fractional coordinates of the atom position along each lattice vector, and vary in the range $[0, 1]$. By application of the translation vector to a lattice point, the basis is assigned identically to every other point on the lattice. Further, if the unit cell is primitive then the basis is also a *primitive basis*.

The primitive basis and the volume of the primitive cell are uniquely determined by the crystal structure, but there is a degree of freedom allowed in the choice of primitive cell since different combinations of lattice vectors can return the same parallelepiped volume V_{cell} . The degree of freedom is limited to a number of choices for arranging the lattice vectors such that infinite repetition of the unit cell will fill all space. The allowed arrangements of lattice vectors defines 7 fundamental crystal systems, where each lattice within a given system must adhere to specific restrictions on the lengths and angles between their lattice vectors. While these constraints define the shape of the unit cell, it is the symmetry operations allowed by these constraints that dictate the arrangement of lattice points in space. Imagine applying a translation vector to its corresponding unit cell: the unit cell is identical with respect to atomic arrangement and geometry regardless of its position in space, reflecting the periodicity of the lattice. This property is known as the *translational symmetry* of a crystal, which defines the arrangement of lattice points in space. Further, the basis itself may possess its own symmetries such as rotations, reflections and inversions. The set of operations which leave the basis invariant defines the *point group* of the crystal. Often, it is easier to relate the symmetry of a point group to the crystal structure using non-primitive, *conventional* unit cells composed of multiple lattice points. In order to fill all space, a conventional cell must still obey the same restrictions as primitive cells on the lengths and angles between their vectors. As a result we find in total 14 unique lattice types (belonging to 7 crystal systems), known as the *Bravais lattices*, which may be described using primitive or conventional cells. Each Bravais lattice is compatible with

different combinations of translation and point group symmetries which together form distinct *space groups*. A space group thus contains the symmetries resulting from the arrangement of the lattice and its associated basis; it provides a complete description of the atom geometry in the basis and its periodic repetition forming the crystal.

Now, how do we find the space group of a crystal in practice? While the lattice is mathematically abstract, its periodicity is observable indirectly through a Fourier transform of experimental diffraction data in *reciprocal space*, which is essentially a vectorial space whose vectors are the conjugated vectors of the atomic position vectors defined in the real space. A Fourier-based analysis of the diffraction data allows to obtain the position of the atom basis in the space, its translational and point group symmetries, hence the space group of the crystal. Note that, the utility of the Fourier transform from reciprocal to real space extends far beyond simply recovering the real-space lattice; it provides a useful wave-based description of the crystal in reciprocal space, as we discuss in [section 1.8](#).

1.7 Supercells to model defect concentration

A crystalline solid is fundamentally defined by a primitive or conventional unit cell, which represents the smallest repeating building block capable of generating the entire infinite lattice through translational symmetry. In practical atomistic simulations, however, it is often necessary to work with a larger periodic representation of the crystal, known as a *supercell*. A supercell is constructed by replicating the reference unit cell an integer number of times along the lattice vectors, thereby forming a larger simulation cell that still preserves periodic boundary conditions.

The use of a supercell is essential when modelling systems that deviate from perfect crystallinity, such as materials containing dopants, point defects, vacancies, or substitutional impurities. In these cases, the target quantity is not the absolute number of defects, but rather their concentration relative to the host lattice. Introducing a single dopant or defect into a sufficiently large supercell allows one to model a desired concentration, while maintaining periodicity. Once the defect is introduced, the resulting supercell effectively becomes the new primitive cell of the system, as it is the smallest periodically repeated representation that captures the modified chemistry and symmetry.

During the construction of defect or doped supercells, particular care must be taken to ensure a physically meaningful spatial distribution of the dopants. Whenever possible, a homogeneous distribution of dopants is preferred, as clustering or artificial ordering may introduce spurious interactions between periodic images. Such artificial correlations are rarely controllable in experimental samples, where atomic-scale disorder and thermal fluctuations lead to statistically distributed defect positions. A homogeneous arrangement within the supercell therefore provides a more representative average description of experimentally realisable materials.

Beyond defect modelling, supercells are also widely used to represent pristine crystal structures. In this context, the motivation is not chemical complexity, but rather the need to capture long-range interactions and collective phenomena that cannot be described within a minimal unit cell. For example, molecular dynamics simulations require sufficiently large supercells to accommodate long-wavelength vibrational modes, realistic phonon scattering, and spatial correlations in atomic motion. Similarly, the calculation of phonon-derived properties, such as vibrational densities of states, thermal conductivity, or free energies, often relies on supercells to represent interatomic force constants beyond nearest neighbors.

In all these applications, the choice of supercell size reflects a balance between physical accuracy and computational cost. The supercell must be large enough to suppress interac-

tions between periodic images and to capture the relevant length scales of the phenomenon under study, while remaining tractable for first-principles or classical simulations. As such, the construction and validation of an appropriate supercell is a critical step in ensuring the reliability of atomistic modelling results.

1.8 Reciprocal space

The arrangement of a crystal is invariant to translation, meaning that the positions of atoms repeat periodically throughout space. Consequently, any local physical property that depends on the atomic configuration will also exhibit the same periodic behaviour. For example, the electron density $\rho(\mathbf{r})$ is a function of the positions of the electrons $\mathbf{r}$, and inherits the periodicity of the crystal according to the translation vector $\mathbf{T}$:

$$\rho(\mathbf{r} + \mathbf{T}) = \rho(\mathbf{r}). \quad (1.4)$$

This situation is ideal for the application of Fourier analysis, which allows for the expansion of a periodic function as a sum of plane waves of the form $\exp(i\mathbf{k} \cdot \mathbf{r})$, where $\mathbf{k}$ is a three dimensional wave vector which varies in Fourier or *reciprocal* space. The reciprocal basis vectors $\{\mathbf{a}^*, \mathbf{b}^*, \mathbf{c}^*\}$ are defined as

$$\mathbf{a}^* = 2\pi \frac{\mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} \quad \mathbf{b}^* = 2\pi \frac{\mathbf{c} \times \mathbf{a}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} \quad \mathbf{c}^* = 2\pi \frac{\mathbf{a} \times \mathbf{b}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}. \quad (1.5)$$

Each reciprocal basis vector (e.g. $\mathbf{a}^*$) has a magnitude that depends on the spatial periodicity of its corresponding real lattice vector (e.g. $\mathbf{a}$), and is orthogonal to a plane spanned by the pair of non-corresponding real lattice vectors (e.g. $\mathbf{b} \times \mathbf{c}$). The orthogonality relation may be explicitly written $\mathbf{x}_i \cdot \mathbf{y}_j = 2\pi\delta_{ij}$ where $\mathbf{x}_i = \mathbf{a}^*, \mathbf{b}^*, \mathbf{c}^*$ and $\mathbf{y}_j = \mathbf{a}, \mathbf{b}, \mathbf{c}$, and δ_{ij} is the Kronecker delta. Thus, the periodicity of the real-space lattice is encoded along each reciprocal basis vector, which forms a consistent framework for describing wave vectors in periodic systems. With Equation 1.5, it is then possible to construct the *reciprocal lattice* as a set of lattice points connected by the *reciprocal lattice vectors*

$$\mathbf{G} = m_a \mathbf{a}^* + m_b \mathbf{b}^* + m_c \mathbf{c}^* \quad (1.6)$$

for a set of arbitrary integers m_a , m_b , and m_c . As a result, the set $\{\mathbf{G}\}$ for all integers m_a , m_b , and m_c precisely defines the wave vectors $\mathbf{k}$ of each plane wave in the Fourier series expansion of Equation 1.4, such that

$$\rho(\mathbf{r}) = \sum_{\mathbf{G}} \rho_{\mathbf{G}} \exp(i\mathbf{G} \cdot \mathbf{r}) \quad (1.7)$$

where $\rho_{\mathbf{G}}$ are the Fourier coefficients obtained by inversion of the Fourier series.

The utility of this formalism is that the reciprocal lattice vectors can be determined from X-Ray diffraction experiments. For an incident wave with wavelength λ matching the spacing between crystal planes d , the wave is scattered in repeating patterns of constructive and destructive interference. According to Bragg's Law, diffraction peaks occur at distinct angles of incidence θ with corresponding diffraction order n , such that

$$n\lambda = 2d \sin \theta, \quad (1.8)$$

which can be rearranged to calculate d ; the specific value depends on the choice of crystal plane. Then, the reciprocal lattice vector $\mathbf{G}$ associated with a particular d is given by

$$|\mathbf{G}| = \frac{2\pi}{d}. \quad (1.9)$$

We thus recover a set of reciprocal lattice vectors $\{\mathbf{G}\}$, which we use to construct the reciprocal lattice, and hence transform to recover the real-space lattice vectors and symmetries. If the unit cell in the real space is primitive, so is the unit cell in the reciprocal space. A primitive cell in the reciprocal space is called the *Brillouin Zone* (BZ). Due to the periodicity of the reciprocal lattice, encoded by Equation 1.6, the BZ contains all unique wave vectors $\mathbf{k}$ as all vectors $\mathbf{k} + \mathbf{G}$ are mathematically equivalent. Thus, by subtraction of the appropriate reciprocal lattice vector all wave vectors may be mapped back into the BZ.

The Brillouin zone is the Wigner–Seitz cell of the reciprocal lattice and, as such, represents the fundamental domain for wave vectors $\mathbf{k}$ in a periodic crystal. In atomistic and electronic-structure simulations, physical quantities such as the total energy, electronic density, forces, and vibrational properties are commonly expressed as integrals over the Brillouin zone or, in practice, as discrete sums over a finite set of $\mathbf{k}$ -points. The *irreducible Brillouin zone* (IBZ) is the smallest subset of the Brillouin zone that contains all symmetry-inequivalent $\mathbf{k}$ -points under the space-group symmetry operations of the crystal. Formally, if $\mathcal{G}$ denotes the symmetry group of the crystal (including point-group operations combined with reciprocal lattice translations), two wave vectors $\mathbf{k}$ and $\mathbf{k}'$ are symmetry equivalent if there exists a symmetry operation $g \in \mathcal{G}$ such that

$$\mathbf{k}' = g \mathbf{k}. \quad (1.10)$$

The irreducible Brillouin zone is defined as the set of $\mathbf{k}$ -points for which no two points are related by such symmetry operations. Any integral over the full Brillouin zone can therefore be rewritten as a weighted sum over the IBZ:

$$\int_{\text{BZ}} f(\mathbf{k}) d\mathbf{k} = \sum_{\mathbf{k} \in \text{IBZ}} w_{\mathbf{k}} f(\mathbf{k}), \quad (1.11)$$

where $f(\mathbf{k})$ is a symmetry-invariant quantity (such as the band energy or the charge density), and $w_{\mathbf{k}}$ is the weight accounting for the number of symmetry-equivalent points in the full Brillouin zone.

In atomistic simulations based on periodic boundary conditions the use of the IBZ provides a substantial computational advantage. Because many physical observables are invariant under the symmetry operations of the crystal, it is unnecessary to evaluate them at every $\mathbf{k}$ -point in the full Brillouin zone. Instead, calculations are performed only for the symmetry-inequivalent points in the IBZ, and the results are reconstructed through the appropriate weights. This reduction can be significant: for highly symmetric crystals, the IBZ may represent only a small fraction of the full Brillouin zone, leading to substantial savings in computational cost without loss of accuracy. Standard $\mathbf{k}$ -point generation schemes, such as Monkhorst–Pack grids, are therefore typically reduced automatically to the IBZ using the crystal symmetries.

Using the full Brillouin zone corresponds to sampling all $\mathbf{k}$ -points explicitly, without exploiting symmetry. While conceptually straightforward, this approach is computationally inefficient because symmetry-equivalent contributions are evaluated multiple times. It is mainly required for systems with little or no symmetry, such as disordered structures, large supercells, amorphous materials, or systems in which symmetry is explicitly broken by defects, surfaces, or external fields. By contrast, the irreducible Brillouin zone exploits the symmetry of the underlying lattice to eliminate redundant calculations. Its applicability relies on the preservation of these symmetries in the simulated system. When supercells are introduced to model defects, dopants, or thermal disorder, the crystal symmetry is generally reduced, causing the IBZ to expand and, in the limiting case, to coincide with the full Brillouin zone. In practice, the choice between sampling the full Brillouin zone or

the IBZ reflects a balance between generality and computational efficiency. For pristine, highly symmetric crystals, the IBZ is the natural and preferred choice, enabling efficient and accurate Brillouin-zone integration. For low-symmetry or large-scale atomistic models, the advantages of symmetry reduction diminish, and calculations may effectively require sampling the entire Brillouin zone. The concept of the irreducible Brillouin zone therefore plays a central role in atomistic simulations by directly linking crystallographic symmetry to computational efficiency and numerical accuracy.

Chapter 2

Geometric Optimisation

Geometric *optimisation* or *relaxation* is a fundamental computational procedure aimed at identifying atomic configurations that minimise the potential energy of a system; such configurations are said to be *optimised* or, equivalently, *relaxed*. In a relaxed geometry, the total force acting on each atom is null, that is

$$\mathbf{F}_i = \mathbf{0}, \quad \forall i \quad \Rightarrow \quad \nabla V(\mathbf{r}_1, \dots, \mathbf{r}_N) = 0. \quad (2.1)$$

In contrast to molecular dynamics, geometric optimisation is not concerned with the time evolution of the system and does not involve kinetic energy or dynamical trajectories. Instead, the problem is formulated purely as a minimisation task in the high-dimensional configurational space defined by the atomic positions. Within this framework, the system is described solely by the positions of its particles, and the relevant quantity is the potential energy surface (PES), which assigns an energy value to every possible configuration. The potential energy is determined by the chosen force field or Hamiltonian, which provides a parametric prescription for the interactions between atoms or particles.

Even when the functional form of the potential energy is analytical, the resulting forces may be highly non-linear and coupled across many degrees of freedom. As a consequence, solving the stationarity condition

$$\nabla V(\mathbf{r}_1, \dots, \mathbf{r}_N) = 0 \quad (2.2)$$

for all particles does not generally admit an analytical solution. Geometric optimisation therefore relies on numerical algorithms that iteratively update the atomic positions in order to reduce the potential energy. The simplest class of optimisation methods follows the local gradient of the potential energy surface. For example, *steepest-descent* or *tangent-based* methods update the atomic positions along the direction of the force, which locally corresponds to the direction of maximum energy decrease. Such methods are robust and easy to implement, but they typically converge slowly near a minimum and are sensitive to the initial configuration. More advanced optimisation schemes, such as *conjugate-gradient* or *quasi-Newton* methods, improve convergence by incorporating information from previous iterations to approximate the local curvature of the potential energy surface. Regardless of the specific algorithm, all local optimisation methods share the same fundamental limitation: they converge toward the nearest *local minimum* accessible from the starting configuration.

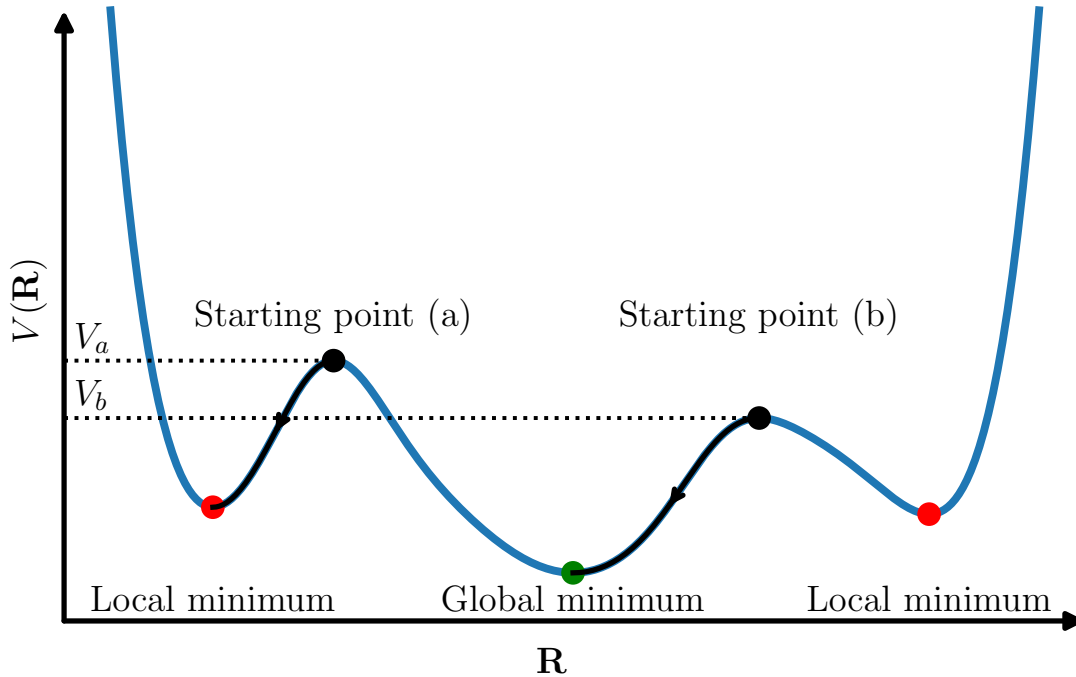


Figure 2.1: Schematic of a potential energy surface (PES) showing multiple local minima (red points) and a global minimum (green point) separated by energy barriers of varying height. The potential V is plotted as a function of the $3N$ -dimensional position vector $\mathbf{R} \equiv (\mathbf{r}_1, \dots, \mathbf{r}_N)$, where motion along the PES represents collective atomic displacements corresponding to distinct configurations. Starting from two distinct initial configurations, labelled (a) and (b) with energies V_a and V_b (and marked with black points on the plot), geometry optimisation follows the local negative gradient of the PES. The local topology of the PES therefore determines the optimisation pathway, such that different initial configurations may converge to distinct minima, and convergence to the global minimum is not guaranteed.

In complex systems, the potential energy surface is characterised by a large number of local minima separated by energy barriers of varying height, as shown in [Figure 2.1](#). Different geometric configurations may therefore correspond to distinct local minima, each associated with different structural, mechanical or electronic properties. The minimum reached by a given optimisation procedure depends critically on the initial guess for the atomic positions. Instead, the concept of *global minimum* refers to the configuration with the lowest possible potential energy over the entire configurational space. In principle, identifying the global minimum would require exploring an infinite number of configurations, which is computationally infeasible for all but the simplest systems. In practice, geometric optimisation is performed within constrained regions of phase space, defined by symmetry considerations, fixed cell parameters, chemical intuition, or prior knowledge of plausible structures. The so-called “global” minimum is therefore understood as the lowest-energy configuration found within these imposed constraints.

It is important to emphasize that a lower potential energy does not automatically imply a more stable structure in a thermodynamic sense. Geometric optimisation algorithms *try* to identify minima of the potential energy at zero temperature and neglect entropic

contributions altogether. Thermodynamic stability, by contrast, depends on free energies, which include both energetic and entropic terms and depend explicitly on external conditions such as temperature and pressure. As a result, a structure that corresponds to a shallow or even metastable minimum on the potential energy surface may be thermodynamically favoured at finite temperature due to entropic effects. Geometric optimisation should therefore be viewed as a tool for identifying possible mechanically stable or metastable configurations, which can then serve as reference structures for further analysis. Here, we say “possible” because condition 2.2 used to find a local minimum does not guarantee that the geometric configuration realising that condition is really a local minimum: in essence, the fact that the derivative of a function is zero in a specific point does not imply that point realises a minimum of the function, as it may correspond to a maximum or a saddle point. To confirm that the geometry $\{\mathbf{r}_i\}$ satisfying 2.2 is relaxed, the potential energy must be a convex function at the point $\{\mathbf{r}_i\}$; this can be confirmed by the absence of unstable vibrational modes in the *phonon spectrum* calculated by using that geometry as a reference (see chapter 8). Thermodynamic stability at a specific temperature and pressure can be confirmed only if these two quantities are included in the calculation of the phonon spectrum, as it is the case of methods like temperature dependent effective potential (TDEP)²⁸ or stochastic self-consistent harmonic approximation (SSCHA),²⁹ and others implemented in the HIPHIVE software package.³⁰ Alternatively, it is possible to monitor the evolution of the interatomic distances in molecular dynamics simulations at finite temperature and pressure, in order to capture substantial variations of the atomic positions: for example, in typical covalent solids at 300 K, variation of bond lengths not larger than ~ 0.3 Å usually indicates that the phase is stable.

In practical workflows, optimised geometries are often used as starting points for molecular dynamics simulations, vibrational analysis, or electronic structure calculations. Understanding the limitations and assumptions underlying geometric optimisation is essential for correctly interpreting its results and for placing them within a broader thermodynamic and statistical-mechanical context.

2.1 Relaxed geometry as a model benchmark

A natural question arises when introducing geometric optimisation in the context of molecular dynamics: why should we minimise a geometry if atoms are expected to move continuously in time during a simulation? The answer lies in the role of the relaxed geometry as a reference configuration that allows us to validate the physical consistency of the model before introducing dynamical effects.

In molecular dynamics, the time evolution of the system is obtained by integrating Newton’s equations of motion, which require the evaluation of interatomic forces at every time step. These forces are computed from the chosen force field or Hamiltonian and directly determine how particles accelerate and move. If the force field is poorly parameterised or inconsistent with the intended physical system, the resulting forces may drive the atoms toward unphysical configurations, leading to numerical instabilities or unrealistic trajectories. For this reason, it is common practice to perform an initial geometric optimisation before starting a molecular dynamics simulation. This procedure acts as a benchmark for the chosen interaction model by testing whether the force field admits a mechanically stable configuration under the imposed boundary conditions and constraints. In practical terms, the optimisation checks whether the forces can be reduced to zero and whether the system converges smoothly toward a minimum of the potential energy surface.

A force field can be considered appropriate, in a minimal sense, if two key conditions are satisfied. First, the optimisation must converge to a stable minimum, meaning that all residual forces fall below a predefined tolerance and no unphysical distortions of the structure occur. Failure to converge, or convergence to highly strained or fragmented geometries, often indicates deficiencies in the interaction parameters or an inconsistency between the force field and the system composition. Second, whenever experimental structural information is available, the optimised geometry should be compared against experimental observations. Such comparisons provide an essential validation step, as they assess whether the theoretical model reproduces real structural features within acceptable accuracy. One common approach is to compute structural descriptors from the relaxed configuration, such as bond lengths, bond angles, or more global quantities like the radial distribution function.

The radial distribution function, commonly denoted as $g(r)$, is a central structural descriptor in atomistic simulations and provides a statistical measure of how particle density varies as a function of distance from a reference particle. It quantifies the probability of finding a particle at a distance r from another particle relative to the probability expected for an ideal, non-interacting system at the same average density. Consider a homogeneous and isotropic system of N particles in a volume V , characterised by an average number density $\rho = N/V$. For such a system, the radial distribution function is defined so that $\rho g(r) dV$ gives the average number of particles found in a spherical shell of volume $dV = 4\pi r^2 dr$ at distance r from a tagged particle. Formally, the radial distribution function can be written as

$$g(r) = \frac{1}{\rho N} \left\langle \sum_{i=1}^N \sum_{j \neq i}^N \delta(r - |\mathbf{r}_i - \mathbf{r}_j|) \right\rangle, \quad (2.3)$$

where $\mathbf{r}_i$ and $\mathbf{r}_j$ are the positions of particles i and j , respectively, δ is the Dirac delta function, and the angular brackets denote an ensemble average or, under the ergodic hypothesis, a time average along a molecular dynamics trajectory. In practice, $g(r)$ is evaluated numerically by accumulating a histogram of interparticle distances sampled along the trajectory. Distances are binned into intervals of width $\Delta r = |\mathbf{r}_i - \mathbf{r}_j|$, and the number of particle pairs whose separation falls within each bin is counted and normalized appropriately. The normalization ensures that $g(r) \rightarrow 1$ as $r \rightarrow \infty$, corresponding to the absence of correlations at large separations.

For an ideal gas, where particles do not interact and are spatially uncorrelated, the radial distribution function is identically equal to unity for all r , i.e. $g(r) = 1$. Deviations from unity therefore directly reflect the presence of interparticle interactions and spatial correlations induced by the potential. In condensed phases, such as liquids and solids, $g(r)$ exhibits characteristic oscillatory behaviour. At short distances, $g(r)$ typically vanishes due to strong repulsive interactions that prevent particles from overlapping. The first pronounced peak corresponds to the most probable nearest-neighbor distance and provides direct information about local coordination and bonding environments. Subsequent peaks in $g(r)$ correspond to second and higher coordination shells, revealing medium-range structural order. In crystalline solids, these peaks are sharp and persist over long distances, reflecting long-range periodic order. In liquids, the peaks are broader and decay with distance, indicating short-range order but no long-range translational symmetry.

The radial distribution function also provides a direct link between microscopic structure and thermodynamic properties. For example, the *coordination number*, defined as the

average number of neighbours within a distance r_c , can be obtained by integrating $g(r)$:

$$N_c(r_c) = 4\pi\rho \int_0^{r_c} r^2 g(r) dr. \quad (2.4)$$

Choosing r_c at the first minimum of $g(r)$ yields the average number of nearest neighbours. Moreover, structural information encoded in $g(r)$ can be related to experimental observables. In particular, X-ray and neutron diffraction experiments measure structure factors, which are related to the Fourier transform of the radial distribution function. This connection allows direct comparison between simulation results and experimental data, making $g(r)$ a key quantity for validating force fields and interaction models. When computed from a relaxed, minimum-energy geometry, the radial distribution function reflects the intrinsic structural arrangement dictated by the potential energy surface at zero temperature. When averaged over a finite-temperature molecular dynamics trajectory, it incorporates thermal fluctuations and provides a more direct representation of experimentally observed structures. As such, the radial distribution function plays a crucial role in linking atomistic simulations, statistical mechanics, and experimental structural characterization

The radial distribution function can be specified for an atomic pair; in this case, it is denoted as *pair distribution function*. The pair distribution function for the atomic pair $\{A, B\}$, often denoted as $g(\mathbf{r}_1, \mathbf{r}_2)$, provides a probabilistic description of how two specific atomic types are spatially correlated; it is essentially the joint probability of finding two particles simultaneously at specified positions. To define it, we begin by considering the microscopic number density operator,

$$\hat{\rho}(\mathbf{r}) = \sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i), \quad (2.5)$$

where $\mathbf{r}_i$ denotes the position of particle i and δ is the Dirac delta distribution. The ensemble-averaged one-particle density is then given by

$$\rho^{(1)}(\mathbf{r}) = \langle \hat{\rho}(\mathbf{r}) \rangle, \quad (2.6)$$

which reduces to a constant $\rho = N/V$ for a homogeneous system. The two-particle density, or *pair density*, is defined as

$$\rho^{(2)}(\mathbf{r}_1, \mathbf{r}_2) = \left\langle \sum_{i \neq j} \delta(\mathbf{r}_1 - \mathbf{r}_i) \delta(\mathbf{r}_2 - \mathbf{r}_j) \right\rangle, \quad (2.7)$$

and represents the probability density of finding one particle at $\mathbf{r}_1$ and another at $\mathbf{r}_2$ simultaneously. The pair distribution function is then introduced by normalizing this quantity with respect to the product of single-particle densities,

$$g(\mathbf{r}_1, \mathbf{r}_2) = \frac{\rho^{(2)}(\mathbf{r}_1, \mathbf{r}_2)}{\rho^{(1)}(\mathbf{r}_1)\rho^{(1)}(\mathbf{r}_2)}. \quad (2.8)$$

For a homogeneous and isotropic system, translational and rotational invariance imply that $g(\mathbf{r}_1, \mathbf{r}_2)$ depends only on the scalar distance $r = |\mathbf{r}_1 - \mathbf{r}_2|$. In this case, the pair distribution function reduces to the radial distribution function $g(r)$. The pair distribution function is also directly related to experimentally measurable quantities. In particular, the static structure factor³¹ $S(\mathbf{k})$, obtained from X-ray or neutron scattering experiments, is given by the Fourier transform of the correlation function $g(r) - 1$,

$$S(\mathbf{k}) = 1 + \rho \int [g(r) - 1] e^{i\mathbf{k} \cdot \mathbf{r}} d\mathbf{r}, \quad (2.9)$$

which provides a direct bridge between atomistic simulations and diffraction experiments, enabling quantitative validation of interaction models. In fact, the radial distribution function can be directly related to experimental data obtained from X-ray or neutron diffraction measurements. Starting from an experimentally informed initial configuration, for example derived from X-ray diffraction data, one can perform a geometric optimisation and then compute the corresponding radial distribution function. Agreement between the calculated and experimental distributions indicates that the force field captures the essential features of the interatomic interactions. If, after minimisation, the relaxed geometry remains consistent with the expected experimental structure, the configuration can be interpreted as a reasonable zero-temperature reference state for the system. This reference geometry then serves as a starting point for finite-temperature molecular dynamics simulations, where atomic motion and thermal fluctuations are explicitly included.

It is important to note that the relaxed geometry does not represent the exact experimental structure at finite temperature, but rather the minimum-energy configuration underlying the thermally averaged structure. Nevertheless, establishing this correspondence through geometric optimisation provides confidence that subsequent dynamical simulations will explore physically meaningful regions of configuration space under the specified thermodynamic conditions.

Chapter 3

Phase transformations and reaction paths

Phase transformations and chemical reactions are central phenomena in materials science, chemistry, and condensed-matter physics, as they govern how systems evolve from one stable or metastable state to another. At the atomistic level, a phase transformation or a reaction can be understood as a collective rearrangement of atomic positions that connects two distinct configurations on the potential energy surface. These configurations may correspond to different crystal structures, different chemical compositions, or different bonding topologies, each associated with its own thermodynamic and kinetic properties.

Within atomistic simulations, phase transformations are described by identifying initial and final states and by characterizing the pathway that connects them. The initial state typically represents a reference phase or reactant configuration, while the final state corresponds to a product phase or a transformed structure. The evolution between these states is governed by the underlying potential energy surface and may involve the crossing of energy barriers, intermediate metastable states, and collective atomic motions. Understanding these pathways is essential for predicting reaction rates, transformation mechanisms, and the stability of materials under given thermodynamic conditions.

A fundamental quantity used to assess the thermodynamic driving force of a transformation or reaction is the *formation energy*. For a generic process that transforms an initial configuration into a final one, the formation energy ΔE is defined as

$$\Delta E = E_F - E_I, \tag{3.1}$$

where E_I is the total energy of the initial state and E_F is the total energy of the final state. Both energies are typically obtained from classical or quantum mechanical atomistic calculations, and correspond to relaxed configurations at zero temperature unless stated otherwise. In [Equation 3.1](#), if $\Delta E < 0$, the final state has a lower energy than the initial state, indicating that the transformation is energetically favourable and may occur spontaneously under appropriate conditions. Conversely, if $\Delta E > 0$, the transformation is energetically unfavourable and requires an external driving force, such as thermal activation, pressure, or chemical potential differences. When $\Delta E = 0$, the initial and final states

are energetically degenerate, and the transformation is governed entirely by entropy and kinetic effects.

In the context of compound formation, the formation energy is often defined relative to reference states of the constituent elements. For a compound composed of elements $\{A_i\}$ with stoichiometric coefficients $\{n_i\}$, the formation energy can be written as

$$\Delta E = E_{\text{compound}} - \sum_i n_i \mu_i, \quad (3.2)$$

where E_{compound} is the total energy of the compound and μ_i is the chemical potential (or reference energy) of element A_i in its chosen reference phase. In standard quantum mechanical calculations, μ_i is taken to be the energy of the single atom calculated with the same computational setup (e.g. unit cell, basis set used to expand the wavefunction) used to compute E_{compound} . This definition provides a quantitative measure of the thermodynamic stability of the compound with respect to phase separation into its constituent elements: the lower ΔE , the more energy is required to split the compound into single atoms. It is here important to stress that Equation 3.2 does not provide any information on the dynamical stability of the structure, which instead requires a specific study — see chapter 2 and chapter 8. Moreover, while the formation energy determines the thermodynamic feasibility of a phase transformation or reaction, it does not provide information about how the transformation occurs. The actual transformation proceeds along a *reaction path*, which describes how the atomic configuration evolves between the initial and final states. The characterization of reaction paths, energy barriers, and transition states forms the core of the kinetic description of phase transformations and will be the focus of the following sections. Together, the concepts of formation energy and reaction pathways provide a comprehensive framework for understanding both the thermodynamics and kinetics of phase transformations in atomistic simulations.

When a phase transformation does not involve a change in chemical composition, but only a rearrangement of the same atoms into a different structural or configurational state, the formation energy can be defined in an entirely analogous manner. In this case, no reference to elemental chemical potentials is required, since the number and identity of atoms are identical in the initial and final phases. The energetic cost or gain associated with the transformation is therefore obtained directly from the difference between the total energies of the two phases as in Equation 3.1. There, E_I denotes the total energy of the system in the reference phase or structure, while E_F is the total energy of the transformed phase, both evaluated for the same set of atoms and typically at the same volume and boundary conditions. This definition is particularly relevant for polymorphic phase transitions, such as solid–solid structural transformations, order–disorder transitions, or changes in crystal symmetry, where the chemical composition remains unchanged. In such cases, the formation energy directly reflects the relative stability of competing phases and provides a first criterion for identifying the thermodynamically preferred structure. As before, while the energy difference determines the thermodynamic driving force, the actual occurrence of the transformation depends on the existence of a viable reaction path and on the height of the associated energy barriers, which control the kinetics of the process.

3.1 Transformations and reaction paths

A phase transformation or chemical reaction is not fully characterised by the energies of the initial and final states alone, but also by the sequence of intermediate configurations that connects them in configuration space. This sequence is referred to as the *transformation*

path or *reaction path*, and it provides a microscopic description of how the system evolves from the initial state to the final state.

In atomistic terms, a transformation path is defined as a continuous trajectory in the $3N$ -dimensional configuration space of atomic positions, along which the atomic coordinates are gradually rearranged. Each point along the path corresponds to a specific atomic configuration with an associated potential energy. The collection of these energies defines an *energy profile* along the path, which typically exhibits one or more maxima separating local minima. Chemical reactions can be viewed as a particular class of phase transformations in which atoms occupy different positions in space and form or break chemical bonds. From this perspective, the distinction between a structural phase transition in a solid and a chemical reaction is largely semantic, as both processes involve the reorganization of the same atomic degrees of freedom under the influence of the underlying potential energy surface. In both cases, the initial and final states correspond to local minima of the potential energy, representing metastable or stable phases or molecular configurations.

Along a reaction or transformation path, the system must pass through regions of higher potential energy, corresponding to *transition states*. The highest-energy point along the minimum-energy path defines the *activation energy barrier*, which governs the kinetics of the transformation (see Figure 3.1). Even when the final state is energetically more favourable than the initial one, the presence of a large barrier can strongly inhibit the transformation on accessible time scales. This is the case, for example, of the spontaneous transformation of diamond to graphite: at standard temperature and pressure conditions, it would take billion of years, which can be reduced to minutes under suitable laboratory conditions.³²

In atomistic simulations, reaction paths are often analyzed by introducing a reaction coordinate, which parametrizes progress along the transformation. This coordinate may represent a collective variable, such as an interatomic distance, a bond angle, or a more complex function of the atomic positions like, for instance, linear combination of phonon eigenvectors.³³ By projecting the multidimensional energy landscape onto this reduced coordinate, one obtains an effective one-dimensional energy profile that facilitates the identification of metastable states, transition states, and activation energies. The explicit determination of transformation paths is central to understanding phase transitions, diffusion processes, defect migration, and chemical reactions. It provides a direct link between thermodynamic driving forces, encoded in energy differences between states, and kinetic behaviour, controlled by the structure of the energy barriers along the path.

3.2 Computational methods for identifying and characterizing reaction and transformation paths

The identification of reaction and transformation paths in atomistic simulations is fundamentally concerned with exploring the *potential energy surface* (PES) of a system and determining how distinct local minima are connected through saddle points. The PES is a high-dimensional hypersurface defined over the $3N$ atomic degrees of freedom, where each point corresponds to a specific atomic configuration with an associated potential energy. Reaction paths correspond to special trajectories on this surface that connect an initial minimum to a final minimum while passing through one or more transition states.

One of the simplest conceptual approaches to define a transformation path is the linear interpolation of atomic coordinates between initial and final states. In this approach, the

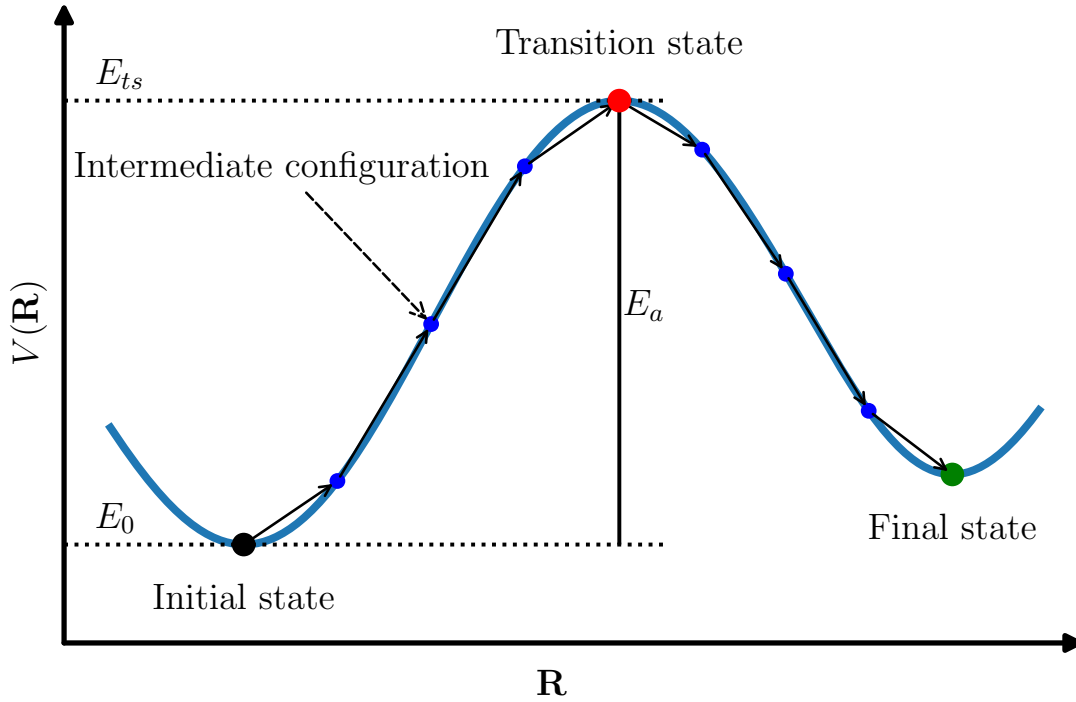


Figure 3.1: Schematic of a reaction path, illustrating how a system traverses from one potential energy minima to another through a region of higher energy. The potential V is plotted as a function of the $3N$ -dimensional position vector $\mathbf{R} \equiv (\mathbf{r}_1, \dots, \mathbf{r}_N)$, where motion along the PES represents collective atomic displacements corresponding to distinct configurations. Starting from the initial state (black point), the system is driven along the reaction path, ascending the PES through a series of intermediate configurations (blue points). The red point corresponds to the transition state, and the energy difference between this point and the initial state defines the activation energy E_a . Beyond this point, the system proceeds “downhill” towards the final state (green point).

atomic positions are interpolated as

$$\mathbf{r}_i(\lambda) = (1 - \lambda)\mathbf{r}_i^I + \lambda\mathbf{r}_i^F, \quad (3.3)$$

where $\mathbf{r}_i^I$ and $\mathbf{r}_i^F$ denote the positions of atom i in the initial and final configurations, respectively, and $\lambda \in [0, 1]$ is a reaction coordinate. Although linear interpolation provides a simple and intuitive path, it generally does not correspond to a physically meaningful reaction mechanism, as it may introduce unrealistic atomic overlaps and significantly overestimate energy barriers. Still, such linear interpolation is usually employed as initial guess for the methods briefly described in what follows.

To overcome these limitations, energy-minimization-based methods have been developed to identify the *minimum energy path* (MEP) between two states. The MEP is defined as the path that minimizes the energy at every point subject to the constraint of connecting the given endpoints. Along this path, the force perpendicular to the trajectory vanishes, and the highest-energy point corresponds to a first-order saddle point on the PES.

One of the most widely used techniques to compute MEPs is the *Nudged Elastic Band* (NEB) method. In NEB, the reaction path is discretized into a finite number of intermediate images, which are replicas of the system connecting the initial and final states. These

images are connected by fictitious elastic springs that maintain an approximately uniform spacing along the path, while the true interatomic forces derived from the potential act perpendicular to the path. By iteratively relaxing the images under these projected forces, the band converges toward the MEP. An important refinement of this approach is the climbing-image NEB method, in which the highest-energy image is driven uphill along the path while relaxing perpendicular degrees of freedom, ensuring accurate convergence to the transition state.³⁴ NEB methods are routinely applied to a wide range of atomistic problems, including diffusion of point defects in solids, migration of adatoms on surfaces, dislocation motion, and solid–solid phase transitions. For example, the migration barrier of a vacancy in a crystal lattice can be obtained by computing the MEP between two equivalent lattice sites, providing direct access to diffusion kinetics.

An alternative class of methods focuses on directly locating saddle points on the PES without requiring a predefined final state. Among these, the *dimer method* is particularly notable. The dimer method evolves a pair of nearby configurations, or “dimer”, to follow the direction of lowest curvature on the PES, thereby converging toward a nearby first-order saddle point.³⁵ Once a saddle point is identified, steepest-descent minimization on either side yields the connected initial and final minima. This approach is especially useful for exploring unknown reaction mechanisms and complex energy landscapes.

For transformations involving large configurational changes or collective atomic rearrangements, enhanced sampling techniques are often required. Methods such as *metadynamics* introduce a history-dependent bias potential acting on a set of carefully chosen collective variables, encouraging the system to escape free-energy minima and explore new regions of configuration space.³⁶ From the accumulated bias, one can reconstruct the free-energy surface as a function of the collective variables and identify reaction pathways and metastable states.

Transition path sampling (TPS) provides a fundamentally different framework by focusing on ensembles of dynamical trajectories rather than static paths on the PES. In TPS, reactive trajectories connecting initial and final states are generated using Monte Carlo sampling in trajectory space, allowing the characterization of reaction mechanisms in systems where rare events dominate the dynamics.³⁷ This approach is particularly valuable for studying activated processes in complex systems such as glasses, biomolecules, or nucleation phenomena.

In many practical applications, especially at finite temperature, reaction paths are more appropriately described in terms of free energy rather than potential energy. Free-energy methods, such as umbrella sampling, thermodynamic integration, and free-energy perturbation, allow the computation of free-energy profiles along a chosen reaction coordinate. These profiles incorporate both energetic and entropic contributions, providing a more complete thermodynamic description of phase transformations and chemical reactions.

The choice of method for identifying and characterizing reaction paths depends critically on the nature of the system, the available prior knowledge of initial and final states, and the relevant time and length scales. While zero-temperature MEP methods such as NEB are well suited for crystalline solids and diffusion processes, free-energy and trajectory-based approaches are often necessary for complex, finite-temperature transformations. Together, these computational techniques form a versatile toolkit for linking atomistic mechanisms to thermodynamic driving forces and kinetic behaviour in phase transformations and chemical reactions.

Chapter 4

Molecular Dynamics simulations

Molecular Dynamics (MD) is a computational simulation method that models the time evolution of a system of interacting particles, such as atoms or molecules, by explicitly integrating their equations of motion in time. The fundamental assumption underlying classical MD is that the nuclei can be treated as classical particles whose dynamics are governed by Newton’s laws, while the electronic degrees of freedom are incorporated implicitly through an effective interaction potential. This approximation is justified when nuclear quantum effects are negligible and forms the basis of many successful applications in physics, chemistry, materials science, and biology.

In an MD simulation, the state of the system at any given time is fully specified by the set of particle positions and velocities. The temporal evolution of these quantities is obtained by solving Newton’s equations of motion for all particles simultaneously. For a system of N particles, the equation of motion for particle i with mass m_i and position vector $\mathbf{r}_i(t)$ reads:

$$m_i \ddot{\mathbf{r}}_i(t) = \mathbf{F}_i(t), \quad (4.1)$$

where t is the simulation time, $\ddot{\mathbf{r}}_i(t)$ denotes the second time derivative of the position and $\mathbf{F}_i(t)$ is the total force acting on particle i . The total force on a given particle arises from its interactions with all other particles in the system. The forces are obtained from a potential energy function $V(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N)$, which encodes the interactions among the particles as the negative gradient of the potential with respect to the coordinates of particle i :

$$\mathbf{F}_i = -\nabla_{\mathbf{r}_i} V(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) \quad (4.2)$$

where the potential energy function V is commonly referred to as the *force field*. A force field is an analytical model designed to reproduce, at an effective level, the interactions between particles by combining physically motivated functional forms with parameters fitted to experimental data or higher-level electronic structure calculations. The choice of force field determines the accuracy, transferability and computational cost of a simulation and must be consistent with the type of system and the properties of interest.

As a simple and widely used example of a pairwise force field, one may consider the Lennard–Jones potential, which captures short-range repulsion and long-range attraction

between neutral, non-bonded particles. The Lennard–Jones potential between particles i and j separated by a distance $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ is defined as:

$$V_{ij}^{\text{LJ}}(r_{ij}) = 4\epsilon \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^6 \right], \quad (4.3)$$

where ϵ sets the depth of the potential well and thus the strength of the attractive interaction, while σ defines the characteristic length scale at which the potential approaches zero. The steep r^{-12} term represents Pauli repulsion arising from the overlap of electron clouds, whereas the r^{-6} term models dispersion interactions originating from correlated charge fluctuations.^{38,39} More complex force fields extend this framework by including bonded interactions, electrostatic terms, angular and torsional potentials and, in some cases, explicit many-body contributions. Regardless of their complexity, all classical MD force fields ultimately serve the same purpose: to provide a computable approximation to the forces required to propagate Newton’s equations of motion and thus generate trajectories from which physical observables can be extracted.

4.1 Integrating Newton’s equations of motion

Given a system of N particles interacting through a specified force field, the equations of motion constitute a set of N coupled second-order ordinary differential equations, corresponding to $3N$ independent degrees of freedom associated with the particle positions in three-dimensional space. The coupling arises because the force acting on any given particle depends, in general, on the instantaneous positions of all other particles in the system. As a consequence, the motion of each particle cannot be determined independently, but must be solved simultaneously with that of all others. The microscopic state of the system at a given time t is fully specified by the complete set of particle positions $\{\mathbf{r}_1(t), \mathbf{r}_2(t), \dots, \mathbf{r}_N(t)\}$ and velocities $\{\mathbf{v}_1(t), \mathbf{v}_2(t), \dots, \mathbf{v}_N(t)\}$. Together, these variables define a single point in the $6N$ -dimensional phase space of the system. Given the particle positions, the inter-particle forces can be evaluated from the force field through the gradient of the potential energy as defined in [Equation 4.2](#), thereby closing the system of equations of motion. The time evolution of the system is obtained by integrating the equations of motion forward in time for each particle. For classical mechanics with well-behaved interaction potentials, the equations of motion satisfy the conditions required by the existence and uniqueness theorems for ordinary differential equations. This implies that, once the initial conditions are fixed by specifying all particle positions and velocities at time $t = 0$, the subsequent evolution of the system is uniquely determined and fully deterministic.

Despite the fact that the force field may be given in terms of analytical expressions, it is generally not possible to obtain analytical solutions to the equations of motion for interacting many-particle systems. Closed-form solutions exist only for a very limited set of idealized problems, such as non-interacting particles or systems with highly symmetric and integrable interactions. In realistic atomistic systems, the nonlinear and strongly coupled nature of the forces precludes analytical integration, necessitating the use of numerical methods to approximate the time evolution of the system. Numerical integration schemes discretize time into finite intervals of length Δt , known as the *time step*, and approximate the continuous trajectories by a sequence of discrete updates. Among the most widely used algorithms in molecular dynamics are the Verlet family of integrators, which are designed to provide good numerical stability and approximate time reversibility while conserving energy over long simulation times.

The Verlet integration scheme can be derived directly from a Taylor expansion of the particle positions around a given time t , making explicit the assumptions underlying its accuracy and stability. For a particle with position $\mathbf{r}(t_0)$ at a specific time t_0 , the Taylor expansion forward and backward in time gives

$$\mathbf{r}(t_0 + \Delta t) = \mathbf{r}(t_0) + \underbrace{\frac{d\mathbf{r}(t)}{dt}}_{=\mathbf{v}_0} \bigg|_{t_0} \Delta t + \frac{1}{2} \underbrace{\frac{d^2\mathbf{r}(t)}{dt^2}}_{=\mathbf{a}_0=\mathbf{F}_0/m} \bigg|_{t_0} \Delta t^2 + O(\Delta t^3) + \dots \quad (4.4)$$

$$\mathbf{r}(t_0 - \Delta t) = \mathbf{r}(t_0) - \underbrace{\frac{d\mathbf{r}(t)}{dt}}_{=\mathbf{v}_0} \bigg|_{t_0} \Delta t + \frac{1}{2} \underbrace{\frac{d^2\mathbf{r}(t)}{dt^2}}_{=\mathbf{a}_0=\mathbf{F}_0/m} \bigg|_{t_0} \Delta t^2 - O(\Delta t^3) + \dots \quad (4.5)$$

where $\mathbf{v}_0$, $\mathbf{a}_0$ and $\mathbf{F}_0$ are the velocity, acceleration and force vectors calculated at time t_0 , respectively. By adding these two expressions, the velocity term is eliminated, resulting in

$$\mathbf{r}(t + \Delta t) = 2\mathbf{r}(t) - \mathbf{r}(t - \Delta t) + \mathbf{a}(t)\Delta t^2 + \mathcal{O}(\Delta t^4). \quad (4.6)$$

Neglecting higher-order terms leads to the standard Verlet update formula, in which the new position is computed solely from the current and previous positions and from the acceleration evaluated at time t . This formulation is time-reversible and symplectic, properties that contribute to its good long-term energy conservation in molecular dynamics simulations. A limitation of the standard Verlet algorithm is that particle velocities do not appear explicitly at integer time steps, which complicates the calculation of kinetic energies and the application of thermostats and barostats (see [section 4.2](#)). To overcome this drawback, the *velocity Verlet* algorithm reformulates the integration by retaining both positions and velocities as explicit dynamical variables. Starting from the Taylor expansion for the position at a generic time t , one first updates the position as

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \mathbf{v}(t)\Delta t + \frac{1}{2}\mathbf{a}(t)\Delta t^2; \quad (4.7)$$

the forces, and hence the accelerations, are then recalculated from the updated positions at time $t + \Delta t$. The velocity is updated using the average of the accelerations at the beginning and end of the time step:

$$\mathbf{v}(t + \Delta t) = \mathbf{v}(t) + \frac{1}{2} [\mathbf{a}(t) + \mathbf{a}(t + \Delta t)] \Delta t. \quad (4.8)$$

This staggered update ensures that positions and velocities are synchronized at the same time points while preserving the second-order accuracy of the integration scheme. As a result, the velocity Verlet algorithm combines the favourable numerical properties of the standard Verlet method with practical advantages for computing dynamical and thermodynamic observables in molecular dynamics simulations.

The choice of the time step Δt is a critical aspect of any molecular dynamics simulation because it directly controls the accuracy of the numerical integration of the equations of motion. As discussed in the derivation of the Verlet algorithm, the integration scheme is obtained by truncating the Taylor expansion of the particle positions at finite order, thereby neglecting higher-order terms of order $\mathcal{O}(\Delta t^3)$ and beyond in the position update. The magnitude of these neglected terms determines the local truncation error of the integration and thus sets an upper bound on the time step that can be used without significantly distorting the true dynamics. To ensure numerical stability and faithful reproduction of the underlying continuous-time trajectories, the time step must be small enough to resolve the

fastest motions present in the system. In atomistic simulations, these are typically high-frequency bond stretching or angle bending vibrations, which impose stringent constraints on Δt . If the time step approaches or exceeds the characteristic period of such motions, the truncated Taylor expansion no longer provides an adequate approximation, leading to growing integration errors. An excessively large time step manifests itself in practice as numerical instability, artificial heating, or systematic drift in conserved quantities such as the total energy in microcanonical simulations. These effects arise because the neglected higher-order terms in the Taylor expansion accumulate coherently over many integration steps, eventually overwhelming the nominal second-order accuracy of the Verlet scheme. Such artifacts are purely numerical in origin and do not reflect any physical property of the modeled system. Conversely, choosing a time step that is much smaller than required by the fastest relevant dynamics reduces the truncation error but increases the total number of integration steps needed to reach a given physical time. This leads to higher computational cost without a commensurate gain in physical accuracy, as other sources of error, such as force field approximations or statistical uncertainty, may already dominate. The optimal choice of Δt therefore represents a compromise between numerical accuracy and computational efficiency, guided by an understanding of the truncation errors inherent in the integration scheme and the characteristic time scales of the system under study.

It is important to emphasize that the Verlet algorithm represents only one possible choice among a broader class of numerical schemes available for integrating Newton's equations of motion. In addition to the standard and velocity Verlet formulations discussed above, widely used alternatives include the *leapfrog* integrator, *predictor-corrector* schemes, and *higher-order symplectic* integrators. The leapfrog method, for example, updates velocities at half-integer time steps and positions at integer time steps, resulting in good energy conservation properties while maintaining computational simplicity. Predictor-corrector methods attempt to improve accuracy by iteratively refining estimates of positions and velocities, although they are less commonly used in modern molecular dynamics due to weaker long-term energy conservation. Higher-order symplectic integrators extend the basic Verlet philosophy by incorporating additional force evaluations to achieve improved accuracy while preserving the geometric structure of Hamiltonian dynamics.

Regardless of the specific integration scheme employed, certain fundamental physical and numerical conditions must be satisfied to ensure that the simulated trajectories faithfully represent the underlying classical dynamics. First, the integration algorithm should preserve time reversibility, meaning that if all particle velocities are reversed at a given time and the system is integrated backward using the same time step, the trajectory should retrace itself and return to the initial configuration within numerical tolerance. This property reflects the time-reversal symmetry of Newton's equations of motion and is essential for avoiding systematic biases in phase-space sampling. Second, the integration scheme should conserve total energy in the absence of external perturbations, such as thermostats or barostats (see [section 4.2](#)). In microcanonical simulations, also known as *NVE simulations*, exact energy conservation is not achievable due to finite time-step errors, but a suitable integrator should ensure that the total energy fluctuates around a constant mean value without exhibiting systematic drift over long simulation times. Third, fundamental conservation laws associated with translational and rotational invariance must be respected. In particular, total linear momentum should be conserved in systems without external forces, and total angular momentum should remain constant when no external torques are present. Violation of these conservation laws often signals deficiencies in the integration scheme, force evaluation, or boundary condition implementation. In practice, these requirements are assessed through systematic benchmarks with respect to

the chosen time step Δt . Energy conservation is checked by monitoring the total energy as a function of time in an NVE simulation and verifying the absence of long-term drift as Δt is reduced. Momentum and angular momentum conservation can be examined by tracking the corresponding global quantities along the trajectory and ensuring that they remain constant within numerical precision.

Time reversibility is typically tested by a dedicated numerical experiment. One integrates the system forward in time for a given number of steps, then reverses all particle velocities and integrates backward for the same number of steps using the identical integration scheme and time step. If the algorithm is time reversible, the final positions and velocities should coincide with the initial ones up to small numerical deviations that decrease systematically as Δt is reduced. Deviations that grow rapidly or do not improve upon decreasing the time step indicate a breakdown of time-reversal symmetry in the integration procedure. Performing such Δt benchmarks provides a practical and quantitative way to validate both the integration scheme and the chosen time step for a given system. Only after verifying time reversibility, acceptable energy conservation, and preservation of the relevant conservation laws can one be confident that the numerical integration accurately represents the intended physical dynamics over the simulated time scales.

4.2 Thermostats and barostats

In the derivation of the equations of motions of a system, a possible choice is to keep constant the number of particles N , the volume V of the system and its total energy E ; in this case, the corresponding points in the phase space (i.e. the states of the system) constitute an *NVE* ensemble. In practical MD simulations, it is often desirable to control macroscopic variables such as temperature or pressure; this is achieved through the use of thermostats and barostats which modify the equations of motion in a controlled way to impose a fixed temperature and/or pressure value. In these latter cases, the corresponding points in the phase space constitute an *NVT* (i.e. *isothermal*) or *NPT* (i.e. *isothermal-isobaric*) ensemble, respectively. The NVE ensemble then represents a closed (constant N) isolated (constant E) system, while the NVT and NPT ensembles represent closed systems which can exchange heat with the environment. In the NVE or NVT ensembles, if the probability P_i to realise a specific value E_i of the Helmotz free energy at a temperature T is proportional to the Boltzmann factor $e^{-E_i/k_B T}$ (where k_B is the Boltzmann constant) then the ensemble is said to be *canonical*.

In the canonical (NVT) ensemble, the system of interest is in contact with an external thermal reservoir that fixes the temperature while allowing energy exchange. In this ensemble, the number of particles N and the volume V are held constant, while the total energy of the system is not conserved. Instead, a suitably defined extended energy, which includes both the physical system and the thermostat degrees of freedom, is conserved. Several thermostatting techniques exist to sample the canonical ensemble, including *Langevin dynamics*,⁴⁰ the *Andersen thermostat*,⁴¹ *velocity rescaling methods*,⁴² and the *Nosé–Hoover thermostat*.^{43–45} Among these, the Nosé–Hoover thermostat is particularly widely used because it generates continuous and deterministic equations of motion that, under appropriate conditions, rigorously sample the canonical distribution.

In the Nosé–Hoover approach, the system is coupled to an additional dynamical variable that represents an effective thermal reservoir; such variable is effectively the thermostat degree of freedom. This coupling is achieved by introducing a time-scaling factor s that rescales the particle velocities, such that the physical velocities are defined as

$$\mathbf{v}_i = s \dot{\mathbf{r}}_i, \quad (4.9)$$

where $\dot{\mathbf{r}}_i$ denotes the velocity in the extended phase space. Unlike simple velocity rescaling schemes, which periodically and explicitly modify particle velocities to enforce a target temperature, the Nosé–Hoover formulation incorporates the thermostat directly into the equations of motion. This is accomplished by redefining the system dynamics through an extended Lagrangian formalism, ensuring that temperature control emerges naturally from the time evolution rather than from ad hoc adjustments.

To construct the Nosé–Hoover equations of motion, we begin by writing the Lagrangian of the combined system consisting of the physical particles and the thermostat degree of freedom. The Lagrangian $\mathcal{L}$ includes a fictitious kinetic energy associated with the thermostat variable and a logarithmic potential term that enforces the canonical distribution:

$$\mathcal{L} = \sum_i \frac{m_i}{2} s^2 |\dot{\mathbf{r}}_i|^2 + \frac{Q}{2} |\dot{s}|^2 - \Phi(\mathbf{R}) - (f + 1)k_B T_{\text{eq}} \ln s. \quad (4.10)$$

Here, $\Phi(\mathbf{R})$ is the potential energy of the system as a function of the $3N$ particle coordinates collected in the $3N$ -dimensional vector $\mathbf{R}$. The parameter f denotes the number of degrees of freedom of the physical system, typically $f = 3N$ minus any constraints, while the quantity T_{eq} is the target equilibrium temperature imposed by the thermostat; this is the temperature specified by the user as part of the simulation input. The parameter Q plays the role of a fictitious mass associated with the thermostat variable s . Its value controls the strength and time scale of the coupling between the system and the thermal reservoir. A small value of Q leads to strong coupling and rapid temperature fluctuations, which may result in nonphysical dynamics. Conversely, a very large value of Q weakens the coupling, causing slow relaxation of the system temperature toward the target value. Choosing Q appropriately is therefore essential to achieve stable and efficient temperature control without distorting the underlying dynamics.

The equations of motion are obtained by applying the Euler–Lagrange equations to the extended Lagrangian. For the particle coordinates, this yields

$$m_i s^2 \ddot{\mathbf{r}}_i = -\frac{\partial \Phi}{\partial \mathbf{r}_i} - \frac{2\dot{s}}{s} \dot{\mathbf{r}}_i, \quad (4.11)$$

which shows that the thermostat introduces an additional friction-like term proportional to $\dot{s}/s$. This term can either remove or inject kinetic energy into the system, depending on whether the instantaneous temperature is higher or lower than the target temperature. The equation of motion for the thermostat variable is

$$Q\ddot{s} = \sum_i m_i s |\dot{\mathbf{r}}_i|^2 - \frac{(f + 1)k_B T_{\text{eq}}}{s}. \quad (4.12)$$

This equation couples the evolution of s to the instantaneous kinetic energy of the system. When the kinetic energy exceeds the value corresponding to the target temperature, the thermostat responds by increasing $\dot{s}$, which effectively damps the particle velocities; when the kinetic energy is too low, the opposite effect occurs.

In total, the Nosé–Hoover formulation introduces $3N$ equations of motion for the particles and one additional equation for the thermostat variable. These equations are integrated simultaneously at each time step using suitable numerical schemes adapted to extended systems. Although the physical energy of the system alone is not conserved, the dynamics preserve a generalized Hamiltonian defined for the extended system:

$$H = \sum_i \frac{m_i}{2} s^2 |\dot{\mathbf{r}}_i|^2 + \frac{Q}{2} \dot{s}^2 + \Phi(\mathbf{R}) + (f + 1)k_B T_{\text{eq}} \ln s \quad (4.13)$$

which remains constant along the trajectory, thus providing a useful diagnostic for the correctness of the numerical integration. The Nosé–Hoover thermostat is particularly well suited for simulations of systems at equilibrium where preservation of realistic dynamical correlations is important. Typical use cases of the NVT ensemble include simulations of bulk liquids and solids at fixed density, studies of phase stability at controlled temperature, and equilibration runs prior to production simulations in other ensembles. It is also commonly employed when computing temperature-dependent structural or thermodynamic properties, where strict control of the average temperature is required.

In addition to the canonical ensemble, another statistical ensemble of central importance in molecular dynamics simulations is the isothermal–isobaric (NPT) ensemble. In the NPT ensemble, the number of particles N is fixed, while both the temperature T and the external pressure P_{ext} are imposed through coupling to a thermostat and a barostat, respectively. This ensemble is particularly relevant for modelling systems under experimental conditions where pressure, rather than volume, is controlled. In contrast to the NVT ensemble, where the simulation box volume is fixed, the NPT ensemble allows the simulation cell to change its volume dynamically. This feature is essential for studying phenomena such as thermal expansion, compressibility, pressure-induced phase transitions and realistic liquid densities.

To briefly illustrate the derivation of the equations of motion, for simplicity, we consider a cubic simulation box with volume V and side length $L = V^{1/3}$. Particle positions are conveniently expressed in scaled, or reduced, coordinates $\mathbf{x}_i$, which are related to the physical Cartesian coordinates $\mathbf{r}_i$ through

$$\mathbf{r}_i = V^{1/3} \mathbf{x}_i. \quad (4.14)$$

This representation separates changes in particle positions due to internal motion from those arising purely from changes in the simulation box volume. To generate dynamics consistent with the NPT ensemble, the system is coupled simultaneously to a thermostat variable s and a barostat variable associated with the volume V . The dynamics of the combined system are described by an extended Lagrangian that includes the physical particles, the thermostat, and the barostat degrees of freedom. The Lagrangian can be written as

$$\mathcal{L} = \sum_i \frac{m_i}{2} s^2 V^{2/3} \dot{\mathbf{x}}_i^2 - \Phi(V^{1/3} \mathbf{x}) + \frac{Q}{2} \dot{s}^2 - (f+1) k_B T_{\text{eq}} \ln s + \frac{W}{2} \dot{V}^2 - P_{\text{ext}} V \quad (4.15)$$

where the barostat degree of freedom enters through the variable W . Applying the Euler–Lagrange equations to this extended Lagrangian yields the equations of motion for the particles, thermostat, and barostat. The equations governing the particle dynamics take the form

$$\ddot{\mathbf{x}}_i = -\frac{1}{m_i s^2 V^{1/3}} \frac{\partial \Phi}{\partial \mathbf{r}_i} - \left(\frac{2\dot{s}}{s} + \frac{2}{3} \frac{\dot{V}}{V} \right) \dot{\mathbf{x}}_i \quad (4.16)$$

where, in addition to the interatomic force term, the particle motion includes damping contributions arising from both the thermostat and the changing volume. The equation of motion for the thermostat variable remains analogous to that of the NVT ensemble:

$$Q\ddot{s} = \sum_i m_i s V^{2/3} |\dot{\mathbf{x}}_i|^2 - \frac{f+1}{s} k_B T_{\text{eq}}. \quad (4.17)$$

This equation regulates the kinetic energy of the system so that the average temperature fluctuates around the target value T_{eq} . The evolution of the simulation box volume is

governed by the barostat equation of motion:

$$W\ddot{V} = \frac{1}{3V} \left(\sum_i m_i s^2 V^{2/3} |\dot{\mathbf{x}}_i|^2 - \sum_i \mathbf{r}_i \cdot \frac{\partial \Phi}{\partial \mathbf{r}_i} \right) - P_{\text{ext}}. \quad (4.18)$$

where the term involving the dot product $\mathbf{r}_i \cdot \partial \Phi / \partial \mathbf{r}_i$ corresponds to the virial contribution, which encodes the internal pressure generated by interatomic interactions. The difference between this internal pressure and the imposed external pressure P_{ext} drives changes in the system volume.

As in the NVT case, the physical energy of the system alone is not conserved; instead, the conserved quantity is the Hamiltonian of the extended system analogous to Equation 4.13, which includes the kinetic and potential energies of the particles as well as the contributions from the thermostat and barostat. Monitoring this extended Hamiltonian provides a valuable check on the numerical stability and correctness of the integration. The parameters Q and W control the strength and time scale of the coupling between the system and the thermostat and barostat, respectively. In practice, these parameters are not chosen directly but are instead specified through characteristic coupling times, which determine how rapidly the temperature and pressure relax toward their target values. If the coupling is too strong, the system may exhibit large oscillations in temperature or volume, leading to nonphysical dynamics; instead, if the coupling is too weak, equilibration becomes inefficient and prohibitively slow. An appropriate choice of coupling times is therefore essential for stable and realistic NPT simulations.

To further improve numerical stability and sampling efficiency, it is common practice to employ chains of thermostats and barostats. In such schemes, the thermostat or barostat variable is itself coupled to additional auxiliary variables. This redistribution of energy and work along the chain reduces resonances and suppresses large fluctuations, resulting in smoother and more controlled variations of temperature and pressure. Such extended-chain methods are now standard in modern molecular dynamics simulations of the NVT and NPT ensembles; early works in this directions are, for example, those of G. J. Martyna and coworkers.^{46,47}

4.3 Sampling the phase space

The output of a molecular dynamics simulation is a *dynamical trajectory*, that is, a time-ordered sequence of configurations or *snapshots* of the system geometry and associated velocities. Each snapshot corresponds to a single point along the trajectory in phase space, and the full trajectory represents a particular realization of the system's deterministic evolution under the chosen initial conditions and force field. From a statistical mechanics perspective, the phase space of the system contains an infinite number of points corresponding to all possible combinations of positions and momenta compatible with the interaction potential and imposed constraints. A single MD trajectory samples only a subset of this phase space, determined by the initial conditions and conserved quantities such as total energy or momentum. A system is said to be *ergodic* if, over sufficiently long times, its dynamical trajectory visits all microstates compatible with the conserved quantities and their statistical weights. Under this assumption, time averages of observables computed along a molecular dynamics trajectory converge to the corresponding ensemble averages defined in statistical mechanics, thereby providing a rigorous link between microscopic dynamics and macroscopic thermodynamic properties. In practice, strict ergodicity is rarely proven for realistic many-body systems, but it is often assumed to hold approximately

for sufficiently large and complex systems over accessible simulation times. When this assumption is valid, quantities such as temperature, pressure, radial distribution functions, and transport coefficients can be estimated as time averages along a single long trajectory. The reliability of these estimates depends on adequate sampling of phase space which, in turn, requires simulations that are long compared to the relevant correlation times of the system.

An alternative approach to sampling phase space and extracting physical properties is provided by *Monte Carlo* (MC) simulation methods. In contrast to molecular dynamics, Monte Carlo simulations do not generate trajectories that correspond to a physical time evolution. Instead, they construct a stochastic sequence of configurations by proposing random changes to the system and accepting or rejecting them according to a prescribed probability distribution designed to sample a target statistical ensemble. The most widely used Monte Carlo algorithm is the *Metropolis* scheme, in which trial moves are accepted with a probability that depends on the associated change in potential energy and the desired thermodynamic ensemble. This procedure ensures that, in the long run, configurations are generated with the correct equilibrium probability, allowing ensemble averages to be computed directly from configurational sampling. Because Monte Carlo methods bypass the integration of equations of motion, they are often more efficient than molecular dynamics for sampling equilibrium properties, particularly in systems where dynamical constraints slow down phase space exploration. However, the absence of an explicit time variable means that Monte Carlo simulations cannot provide dynamical information such as time correlation functions or transport coefficients. Molecular dynamics and Monte Carlo methods are therefore complementary techniques, each with distinct strengths and limitations. While molecular dynamics offers direct access to both equilibrium and dynamical properties through time-resolved trajectories, Monte Carlo simulations provide a powerful alternative for efficient equilibrium sampling and for validating thermodynamic predictions under controlled ensemble conditions.

Having established how a molecular dynamics simulation is defined and how a dynamical trajectory is generated by numerical integration of the equations of motion, it is natural to ask why MD simulations are performed and what kind of physical information can be extracted from them. The primary purpose of MD is to connect a microscopic description of matter, expressed in terms of atomic positions and velocities, to macroscopic physical properties that are accessible experimentally. This connection is achieved through the framework of statistical mechanics, which provides the formal link between microscopic states and thermodynamic observables. A central question is how physical properties are measured, both at equilibrium and out of equilibrium. At equilibrium, one is typically interested in static thermodynamic quantities such as temperature, pressure, energy, and structural correlations. At out of equilibrium, MD can be used to study time-dependent phenomena such as relaxation after a perturbation, heat or mass transport, shear flow, or response to external fields, where the explicit time evolution generated by the simulation is essential.

At any given time t along a trajectory, the simulation provides the instantaneous positions and velocities of all particles in the system. These instantaneous quantities fluctuate strongly in time and depend sensitively on the microscopic state of the system at that moment. By integrating the equations of motion, one obtains their variation over time, producing a trajectory that samples different regions of phase space consistent with the imposed constraints. Statistical mechanics provides the theoretical framework for extracting physically meaningful quantities from this microscopic information. In equilibrium, macroscopic observables are defined as ensemble averages, that is, averages

taken over all microstates compatible with a given set of macroscopic constraints. Under the assumption of ergodicity, these ensemble averages can be replaced by time averages computed along a sufficiently long MD trajectory. This equivalence justifies the calculation of thermodynamic properties as averages over time rather than as instantaneous values. A fundamental example is the definition of temperature. In classical statistical mechanics, the temperature of a system is related to the average kinetic energy of its particles through the equipartition theorem. The instantaneous kinetic energy fluctuates in time, but its time average is directly proportional to the thermodynamic temperature. Similarly, the total energy of the system is defined as the sum of kinetic and potential energy contributions, each of which exhibits temporal fluctuations but converges to a well-defined average at equilibrium. Other important thermodynamic quantities can be defined in an analogous statistical manner. For instance, the pressure of a system can be expressed in terms of the average momentum transfer to the boundaries or, equivalently, through the virial theorem, which relates pressure to both kinetic energy and inter-particle forces. Structural properties, such as radial distribution functions, are also obtained as statistical averages that characterize the probability of finding particles at given separations.

Before meaningful averages can be computed, the system must be allowed to reach equilibrium. The equilibration phase corresponds to the initial portion of a simulation during which the system relaxes from its starting configuration toward a steady state compatible with the chosen ensemble. This process is often monitored by tracking time-dependent quantities such as temperature, pressure, or energy. When these quantities fluctuate around stable mean values within acceptable tolerances, the system is considered to be equilibrated. Different initial conditions, such as distinct initial particle arrangements or velocity distributions, generally correspond to different initial values of temperature, pressure, or energy. However, when the system is evolved under the same thermodynamic constraints, statistical mechanics predicts that equilibrium ensemble averages of macroscopic observables should be independent of the initial conditions. This property reflects the loss of memory of the initial state as the system explores the phase space. The extent to which initial conditions influence the system over time can be quantified through correlation functions, which measure how strongly microscopic variables at a later time remain correlated with their initial values. For a dynamical variable $A(t)$, the time autocorrelation function is commonly defined as

$$C_A(t) = \langle A(0)A(t) \rangle, \quad (4.19)$$

where the angular brackets denote an ensemble average or, under the assumption of ergodicity, a time average taken along a molecular dynamics trajectory. The autocorrelation function can be written more explicitly by expressing the ensemble or time average in integral form over the phase space of the system. For an equilibrium system, the time autocorrelation function of a dynamical variable $A(t)$ is defined as

$$C_A(t) = \langle A(0)A(t) \rangle = \int \rho(\Gamma) A(\Gamma) A(\Phi^t \Gamma) d\Gamma, \quad (4.20)$$

where $\Gamma = (\mathbf{r}_1, \dots, \mathbf{r}_N, \mathbf{p}_1, \dots, \mathbf{p}_N)$ denotes a point in the $6N$ -dimensional phase space, $d\Gamma$ is the corresponding phase-space volume element, and $\rho(\Gamma)$ is the equilibrium probability density associated with the chosen statistical ensemble. The mapping Φ^t represents the time evolution operator generated by the equations of motion, which propagates the phase-space point Γ forward by a time t . Under the assumption of ergodicity, the ensemble average can equivalently be expressed as a time average along a sufficiently long trajectory of total

duration T :

$$C_A(t) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T A(t') A(t' + t) dt'. \quad (4.21)$$

This formulation makes explicit how the correlation function is computed in practice from molecular dynamics data. The normalized autocorrelation function is obtained by dividing by the value of the correlation function at zero time lag:

$$\tilde{C}_A(t) = \frac{C_A(t)}{C_A(0)} = \frac{\int \rho(\Gamma) A(\Gamma) A(\Phi^t \Gamma) d\Gamma}{\int \rho(\Gamma) A(\Gamma)^2 d\Gamma}. \quad (4.22)$$

In this expression, $C_A(0) = \langle A^2 \rangle$ represents the mean-square fluctuation of the observable at equilibrium. The normalized form highlights the decay of correlations on a relative scale and facilitates comparison between different observables or systems. In practice, the normalized autocorrelation function is often used:

$$\tilde{C}_A(t) = \frac{\langle A(0)A(t) \rangle}{\langle A(0)^2 \rangle}, \quad (4.23)$$

which satisfies $\tilde{C}_A(0) = 1$ and decays toward zero as the system progressively loses memory of its initial state. Here, $A(t)$ represents any microscopic observable of interest, such as a component of a particle velocity, the total momentum or a collective structural variable. The quantity $A(0)$ is the value of the same observable evaluated at an initial reference time. The product $A(0)A(t)$ measures the degree to which the value of the observable at time t is correlated with its initial value, while the averaging procedure ensures that this measure is statistically meaningful. The decay of the autocorrelation function with time defines a characteristic correlation time, which provides a quantitative measure of how rapidly the system forgets its initial conditions. Short correlation times indicate fast relaxation and efficient sampling of phase space, whereas long correlation times signal slow dynamics and strong temporal correlations. Correlation functions therefore play a central role in assessing equilibration, estimating statistical uncertainties, and extracting dynamical properties from molecular dynamics simulations. Correlation times play a crucial role in determining how long a simulation must be run to obtain statistically independent samples and reliable estimates of physical properties.

The key conceptual message is that instantaneous measurements obtained from an MD simulation have no direct thermodynamic meaning on their own. Physical quantities are defined as statistical averages, either over time or over an ensemble, and meaningful comparisons between simulations or with experiments must always be based on such averaged quantities rather than on single snapshots of the system.

4.4 Calculating thermodynamic state variables

As a concrete illustration of the statistical-mechanical foundations underlying molecular dynamics, we consider the example of an adiabatic box, that is, an isolated system that does not exchange particles or heat with its environment. Such a system is naturally described within the microcanonical ensemble, characterised by fixed particle number N , volume V , and total energy E , i.e. the NVE ensemble. We conceptually divide the box into two subsystems, labelled 1 and 2, separated by a wall that allows energy exchange but not particle transfer. At the initial time $t = 0$, the two subsystems are prepared with different temperatures, such that $T_1(0) > T_2(0)$. The total energy of the combined system is conserved and can be written as $E = E_1 + E_2$, where E_1 and E_2 are

the instantaneous energies of subsystems 1 and 2, respectively. Because energy exchange through the separating wall is allowed, E_1 and E_2 may vary in time while their sum remains constant.

As the system evolves, energy exchanges occur between the two subsystems through particle collisions and interactions at the interface. From a macroscopic perspective, we expect that after a sufficiently long time the system reaches equilibrium, characterised by equal temperatures in the two subsystems, $T_1(t) = T_2(t)$. This equilibration process can be understood rigorously using statistical mechanics. The key quantity in the microcanonical ensemble is $\Omega(N, V, E)$, which denotes the number of distinct microscopic configurations consistent with the macroscopic constraints N , V and E . Each of these configurations is known as a *microstate*, and all microstates compatible with the given energy are assumed to be equally probable. In the context of classical mechanics, fixing all particle positions $\mathbf{r}$ and velocities $\dot{\mathbf{r}}$ uniquely specifies a microstate, corresponding to a single point in phase space. For the composite system formed by subsystems 1 and 2, the total number of accessible microstates factorizes as

$$\Omega(N, V, E) = \Omega_1(N_1, V_1, E_1) \Omega_2(N_2, V_2, E_2), \quad (4.24)$$

where Ω_1 and Ω_2 count the microstates of subsystems 1 and 2 at energies E_1 and $E_2 = E - E_1$, respectively. Taking the natural logarithm of both sides yields

$$\ln \Omega(E) = \ln \Omega_1(E_1) + \ln \Omega_2(E - E_1), \quad (4.25)$$

which simplifies the analysis of equilibrium conditions. Equilibrium corresponds to the macroscopic state that can be realized in the largest number of microscopic ways. In statistical mechanics, this is expressed by the principle that *equilibrium maximizes the total number of accessible microstates, or equivalently, the entropy of the system*. Mathematically, this condition is obtained by maximizing $\ln \Omega(E)$ with respect to E_1 at fixed total energy E , volume, and particle number. The extremum condition is given by

$$\left. \frac{\partial \ln \Omega(E_1, E - E_1)}{\partial E_1} \right|_{\text{NVE}} = 0, \quad (4.26)$$

which leads to

$$\frac{\partial \ln \Omega_1(E_1)}{\partial E_1} = \frac{\partial \ln \Omega_2(E_2)}{\partial E_2}. \quad (4.27)$$

Introducing the quantity

$$\frac{\partial \ln \Omega(E)}{\partial E} = \beta, \quad (4.28)$$

we obtain the equilibrium condition

$$\beta_1 = \beta_2, \quad (4.29)$$

which expresses the equality of a fundamental thermodynamic parameter between the two subsystems.

The statistical definition of entropy provides the physical interpretation of these results. Entropy S is defined as

$$S(N, V, E) = k_B \ln \Omega(N, V, E), \quad (4.30)$$

where k_B is the Boltzmann constant. From this definition, temperature T is introduced as a derived quantity through

$$\frac{1}{T} = \left. \frac{\partial S}{\partial E} \right|_{\text{NV}}, \quad (4.31)$$

which immediately implies

$$\beta = \frac{1}{k_B T}. \quad (4.32)$$

The parameter β therefore has units of inverse energy and provides a direct link between microscopic statistics and macroscopic temperature.

Entropy plays a central role in determining the equilibrium of a system because it quantifies the degree of disorder, or *multiplicity*, of microscopic realizations of atom positions and velocities compatible with a macroscopic state. Systems evolve spontaneously toward states of higher entropy because these states correspond to overwhelmingly larger regions of phase space. In the present example, equalization of temperatures corresponds to the most probable distribution of energy between the two subsystems.

Entropy also enters directly into thermodynamic potentials that govern equilibrium under different constraints. For example, the Gibbs free energy is defined as

$$G = U + PV - TS, \quad (4.33)$$

where U is the internal energy, P the pressure, V the volume, and T the temperature. Variation of this quantity may be used to determine, for example, if a phase transformation is spontaneous at constant temperature and pressure.

In molecular dynamics simulations, the connection between microscopic trajectories and thermodynamic temperature follows directly from classical statistical mechanics and the equipartition theorem. For a system of N classical particles with masses m_i and velocities $\mathbf{v}_i(t)$, the instantaneous kinetic energy along the MD trajectory is defined as

$$K(t) = \sum_{i=1}^N \frac{1}{2} m_i \mathbf{v}_i^2(t). \quad (4.34)$$

At equilibrium, statistical mechanics predicts that each quadratic degree of freedom contributes an average energy of $\frac{1}{2} k_B T$ to the total kinetic energy. For a three-dimensional system with $3N$ translational degrees of freedom, this leads to

$$\langle K \rangle = \frac{3N}{2} k_B T, \quad (4.35)$$

where $\langle \cdot \rangle$ denotes an ensemble average or, under the ergodic hypothesis, a time average along the MD trajectory. In practice, the temperature is therefore obtained directly from the trajectory as

$$T = \frac{2}{3Nk_B} \left\langle \sum_{i=1}^N \frac{1}{2} m_i \mathbf{v}_i^2(t) \right\rangle = \frac{1}{3Nk_B} \left\langle \sum_{i=1}^N m_i \mathbf{v}_i^2(t) \right\rangle. \quad (4.36)$$

This expression shows explicitly that temperature is not an instantaneous property but a statistical quantity defined through an average over the molecular dynamics trajectory. Instantaneous fluctuations in kinetic energy give rise to fluctuations in the instantaneous

temperature, which vanish only in the thermodynamic limit or after sufficient time averaging.

The relation between entropy and the molecular dynamics trajectory is more indirect. From statistical mechanics, entropy is defined in the microcanonical ensemble as

$$S(E, N, V) = k_B \ln \Omega(E, N, V), \quad (4.37)$$

where $\Omega(E, N, V)$ is the number of accessible microstates at energy E . Using the thermodynamic definition of temperature,

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E} \right)_{N,V}, \quad (4.38)$$

one sees that entropy is fundamentally related to how the number of accessible microstates changes with energy. In a molecular dynamics simulation, the trajectory provides direct access to the system energy and to equilibrium averages, but it does not yield $\Omega(E, N, V)$ explicitly. As a consequence, entropy cannot usually be computed as a simple time average of an instantaneous microscopic quantity. Instead, entropy differences are inferred indirectly through thermodynamic integration, free-energy calculations, or integration of measurable derivatives such as $\partial S / \partial E = 1/T$. For example, along a reversible path parametrised by a control variable λ , one may compute the entropy difference between two states by integrating

$$\Delta S = \int \frac{1}{T(\lambda)} \frac{\partial E(\lambda)}{\partial \lambda} d\lambda, \quad (4.39)$$

where $T(\lambda)$ and $E(\lambda)$ are obtained from equilibrium averages along MD trajectories at fixed λ . Concrete examples of the control parameter λ depend on the physical process or transformation of interest. One common choice is a coupling parameter that interpolates between two Hamiltonians, such that $H(\lambda) = (1 - \lambda)H_A + \lambda H_B$, where λ varies continuously from 0 to 1 to transform the system from state A to state B . In this case, λ is used extensively in free-energy and entropy calculations for alchemical transformations, such as changing the chemical identity of a particle or turning interactions on and off. Another frequent example is λ representing the volume of the simulation box, $\lambda \equiv V$, which allows entropy differences to be obtained by integrating along an isothermal expansion or compression path. In this situation, the relevant derivative involves the pressure, and the entropy change is related to the work performed during volume changes. The control parameter λ may also correspond to an external field strength, such as an electric or magnetic field applied to the system. By gradually varying the field amplitude from zero to a finite value, one can compute entropy changes associated with polarization or alignment phenomena. Finally, λ can represent a geometrical or structural parameter, such as a reaction coordinate, bond length, or angle, enabling the calculation of entropy changes along a chemical reaction path or during conformational transitions. In all cases, λ defines a reversible path in parameter space along which equilibrium molecular dynamics simulations are performed to connect microscopic trajectory data with macroscopic thermodynamic quantities.

In this way, the microscopic information contained in molecular dynamics trajectories is connected to macroscopic thermodynamic quantities, with temperature emerging directly from kinetic energy fluctuations and entropy obtained through statistically averaged thermodynamic relations rather than instantaneous measurements. This shows that the combination of microscopic dynamics with statistical-mechanical principles allows MD simulations to reproduce macroscopic thermodynamic behaviour and to provide a detailed, atomistic interpretation of equilibrium and equilibration processes.

4.4.1 Calculation of pressure in molecular dynamics simulations

The pressure is a macroscopic observable that can be obtained from the microscopic positions, velocities, and forces of the particles through statistical mechanics; its calculation is based on the *virial theorem*.⁴⁸ Essentially, for a system of N particles of masses m_i contained in a simulation box of volume V , interacting through a potential energy function $U(\{\mathbf{r}_i\})$, the instantaneous pressure $P(t)$ can be expressed as the sum of a kinetic contribution and a configurational (virial) contribution.

The starting point is the microscopic definition of pressure as the momentum flux across a surface, which, for a system with periodic boundary conditions, leads to the virial expression

$$PV = \frac{2}{3}K + \frac{1}{3} \sum_{i=1}^N \mathbf{r}_i \cdot \mathbf{F}_i, \quad (4.40)$$

where K is the total kinetic energy of the system,

$$K = \sum_{i=1}^N \frac{1}{2} m_i v_i^2, \quad (4.41)$$

and $\mathbf{F}_i$ is the total force acting on particle i . The first term in Equation 4.40 represents the ideal-gas-like kinetic contribution to the pressure. Using the definition of temperature from the equipartition theorem,

$$\langle K \rangle = \frac{3}{2} N k_B T, \quad (4.42)$$

the kinetic part of the pressure can be written as

$$P_{\text{kin}} = \frac{N k_B T}{V}. \quad (4.43)$$

The second term in Equation 4.40 accounts for interatomic interactions and is known as the *virial term*. Substituting $\mathbf{F}_i = \sum_{j \neq i} \mathbf{F}_{ij}$ and symmetrizing over particle pairs gives

$$\sum_{i=1}^N \mathbf{r}_i \cdot \mathbf{F}_i = \frac{1}{2} \sum_{i=1}^N \sum_{j \neq i} \mathbf{r}_{ij} \cdot \mathbf{F}_{ij}, \quad (4.44)$$

where $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$. This form shows that the configurational contribution depends on the relative positions and pairwise forces between particles. Combining both contributions, the instantaneous pressure becomes

$$P(t) = \frac{N k_B T(t)}{V} + \frac{1}{3V} \frac{1}{2} \sum_{i=1}^N \sum_{j \neq i} \mathbf{r}_{ij}(t) \cdot \mathbf{F}_{ij}(t). \quad (4.45)$$

In practice, the calculation proceeds step by step during a molecular dynamics simulation. At each time step, particle velocities are known, allowing the instantaneous kinetic energy and temperature to be computed. Simultaneously, forces are already evaluated to integrate the equations of motion, so the virial term can be obtained with negligible additional cost. Using these quantities, Equation 4.45 yields the instantaneous pressure.

However, pressure is a thermodynamic quantity defined as an ensemble average. Therefore, the physically meaningful pressure is obtained by averaging the instantaneous pressure over a sufficiently long, equilibrated portion of the trajectory within a time window τ :

$$\langle P \rangle = \frac{1}{\tau} \int_{t_0}^{t_0+\tau} P(t) dt, \quad (4.46)$$

or, in discrete form,

$$\langle P \rangle = \frac{1}{M} \sum_{k=1}^M P(t_k). \quad (4.47)$$

with M the number of time steps. A more complete definition of pressure should also include the stress tensor and, if the system is a crystal, the use of periodic boundary conditions. We suggest the interested reader the articles cited in references 49 and 50 as a possible starting point for a more detailed study on this matter.

4.5 Measure of average quantities

We now address the question of how likely it is to observe a given energy E_i of subsystem 2 when the total energy of the combined system is fixed. This situation naturally arises when subsystem 2 is weakly coupled to a much larger subsystem 1, which acts as an effective heat bath. In this setting, the probability of realizing a specific energy level E_i in subsystem 2 is proportional to the number of microstates available to the rest of the system at the complementary energy $E - E_i$. The probability can therefore be written as

$$P_{E_i} = \frac{\Omega_2(E - E_i)}{\sum_j \Omega_2(E - E_j)}, \quad (4.48)$$

where the denominator ensures proper normalization by summing over all accessible energy levels E_j of subsystem 2. This expression reflects the fundamental assumption of equal a priori probabilities for microstates of the isolated total system. To make further progress, we evaluate $\Omega_2(E - E_i)$ by performing a Taylor expansion of its logarithm around the total energy E , assuming that $E_i \ll E$ because subsystem 2 is small compared to the heat bath. We write

$$\ln \Omega_2(E - E_i) = \ln \Omega_2(E) - E_i \frac{\partial \ln \Omega_2(E)}{\partial E} + \mathcal{O}(E_i^2). \quad (4.49)$$

Neglecting higher-order terms and using the thermodynamic definition of temperature,

$$\frac{\partial \ln \Omega(E)}{\partial E} = \frac{1}{k_B T}, \quad (4.50)$$

we obtain

$$\ln \Omega_2(E - E_i) = \ln \Omega_2(E) - \frac{E_i}{k_B T}. \quad (4.51)$$

Exponentiating both sides yields

$$\Omega_2(E - E_i) = \Omega_2(E) \exp\left(-\frac{E_i}{k_B T}\right). \quad (4.52)$$

Substituting this result into the expression for P_{E_i} , the prefactor $\Omega_2(E)$ cancels between numerator and denominator, leading to

$$P_{E_i} = \frac{\exp(-E_i/k_B T)}{\sum_j \exp(-E_j/k_B T)}. \quad (4.53)$$

This is the Boltzmann distribution, which defines the canonical ensemble and provides the probability of observing a microstate with energy E_i at temperature T .

Once the probability distribution is known, thermodynamic averages of physical quantities can be computed. For example, the average energy of subsystem 2 is given by

$$\langle E \rangle = \sum_i E_i P_{E_i}. \quad (4.54)$$

More generally, for any observable A that takes the value A_i in microstate i , the ensemble average is defined as

$$\langle A \rangle = \sum_i A_i P_{E_i}. \quad (4.55)$$

In molecular dynamics simulations, however, we do not explicitly enumerate all microstates or their probabilities. Instead, we generate a time-ordered sequence of configurations along a single trajectory by integrating the equations of motion. The practical question then becomes how to relate averages over time to the ensemble averages defined above. This connection is provided by the ergodic theorem, which states that, for an ergodic system, time averages along a sufficiently long trajectory are equal to ensemble averages over phase space. Formally, for an observable A , the time average is defined as

$$\bar{A} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T A(t) dt, \quad (4.56)$$

and ergodicity implies

$$\bar{A} = \langle A \rangle. \quad (4.57)$$

In this sense, averaging a quantity over configurations sampled along an equilibrium MD trajectory is equivalent to averaging over the statistical ensemble defined by the corresponding thermodynamic constraints. This equivalence justifies the use of molecular dynamics simulations to compute equilibrium thermodynamic properties, provided that the trajectory is sufficiently long to explore the relevant regions of phase space and that correlations between successive configurations are properly accounted for.

An example application of [Equation 4.56](#) is the calculation of the diffusion coefficient. The diffusion coefficient is a fundamental transport property that quantifies the rate at which particles spread in space due to random thermal motion. This formalism applies broadly to atomic and molecular diffusion in gases and liquids, self-diffusion in solids at elevated temperature, ionic diffusion in electrolytes, tracer diffusion in complex fluids, and the motion of colloidal particles or macromolecules in solution. We begin from Fick's first law of diffusion, which relates the particle flux $\mathbf{J}$ to spatial gradients in the particle concentration $C(\mathbf{r}, t)$:

$$\mathbf{J} = -D \nabla C. \quad (4.58)$$

This phenomenological relation states that particles flow from regions of high concentration to regions of low concentration, with the proportionality constant D defining the diffusion coefficient. To connect this macroscopic description to microscopic particle motion, we combine Fick's first law with the continuity equation expressing particle conservation:

$$\frac{\partial C(\mathbf{r}, t)}{\partial t} + \nabla \cdot \mathbf{J}(\mathbf{r}, t) = 0; \quad (4.59)$$

substituting Fick's law into the continuity equation yields Fick's second law:

$$\frac{\partial C(\mathbf{r}, t)}{\partial t} = D \nabla^2 C(\mathbf{r}, t), \quad (4.60)$$

which governs the time evolution of the concentration field under purely diffusive transport. For an initially localized distribution of particles in an infinite, homogeneous medium, the

solution of this diffusion equation is a Gaussian whose width increases with time. From this solution, one can derive that the mean square displacement of the particles grows linearly with time. This result provides the microscopic definition of diffusion embodied in the Einstein relation.

Let $\mathbf{r}_i(t)$ denote the position of particle i at time t , and define the displacement relative to a reference time t_0 as $\Delta\mathbf{r}_i(t) = \mathbf{r}_i(t) - \mathbf{r}_i(t_0)$. The mean square displacement is then defined as

$$\langle |\mathbf{r}|^2 \rangle = \frac{1}{N} \sum_{i=1}^N \langle |\Delta\mathbf{r}_i(t)|^2 \rangle, \quad (4.61)$$

where the angular brackets denote an ensemble average or, under the ergodic assumption, a time average over the molecular dynamics trajectory. For diffusive motion in d spatial dimensions, statistical mechanics predicts that at sufficiently long times the mean square displacement satisfies

$$\frac{\partial}{\partial t} \langle |\mathbf{r}|^2 \rangle = 2dD. \quad (4.62)$$

This expression is the Einstein relation and establishes a direct connection between the macroscopic diffusion coefficient and the microscopic dynamics of particles. In the language of finite time differences, this relation can be written as

$$\frac{\Delta \langle |\mathbf{r}|^2 \rangle}{\Delta t} = 2dD, \quad (4.63)$$

$$\Delta \langle |\mathbf{r}|^2 \rangle = 2dD\Delta t, \quad (4.64)$$

which highlights the linear growth of the mean square displacement with time in the diffusive regime. In molecular dynamics simulations, the diffusion coefficient is therefore obtained by computing the mean square displacement as a function of time and identifying the time window where the behaviour is linear. A linear fit to $\langle |\mathbf{r}|^2 \rangle$ versus t yields a slope equal to $2dD$, from which the diffusion coefficient can be extracted. Care must be taken to exclude the short-time ballistic regime, where motion is dominated by inertia, and to ensure that the system has reached equilibrium so that the long-time diffusive limit is properly sampled.

4.6 Using MD trajectories to study specific properties in practice

A molecular dynamics study begins with the preparation of a representative sample, namely a microscopic model system composed of N interacting particles whose positions, velocities, interaction potential, boundary conditions and thermodynamic constraints are chosen to reflect the physical system of interest. This preparation stage includes defining the simulation box, assigning initial atomic coordinates, selecting an appropriate force field or Hamiltonian, and initializing particle velocities consistently with a target temperature or total energy. Once the system is defined, Newton's equations of motion are solved numerically and integrated in time using a suitable integration scheme and time step.

If properties at the equilibrium are to be sampled, then the first phase of the study requires that the system evolves from the imposed initial conditions toward equilibrium, a process known as *equilibration*. Equilibration is assessed by monitoring macroscopic observables such as temperature, total energy, pressure, or density and verifying that their statistical averages become stationary in time within acceptable fluctuations. After equilibration has been achieved, the simulation enters the production phase, during which

microscopic configurations are sampled along the trajectory and used to compute physical properties. Measurements are performed by evaluating appropriate microscopic observables at each time step and averaging them over time, relying on the ergodic hypothesis to relate time averages to ensemble averages. Typical equilibrium properties obtained in this way include temperature from the average kinetic energy, pressure from the virial expression, radial distribution functions characterizing structural order, and thermodynamic quantities such as internal energy or heat capacity.

Out-of-equilibrium investigations follow a similar workflow but involve applying external perturbations, such as temperature gradients, shear deformations, or external fields, and measuring the system's dynamical response in terms of transport coefficients, relaxation times, or correlation functions.

In both equilibrium and out-of-equilibrium studies, it is essential to distinguish between instantaneous quantities and statistically meaningful averages, to account for correlations between successive configurations and to estimate uncertainties associated with finite sampling. Only after adequate sampling and proper statistical analysis, can molecular dynamics simulations provide reliable and physically interpretable insight into the equilibrium and non-equilibrium behaviour of atomistic systems.

Chapter 5

Introduction to quantum mechanics

Classical molecular dynamics, based on Newtonian mechanics, provides a powerful and widely used framework for simulating the dynamical behaviour of atomistic systems. It enables the study of systems containing hundreds to millions of atoms over long time scales, allowing the extraction of statistically averaged structural, thermodynamic, and dynamical properties at relatively low computational cost. However, within this formalism atoms are treated as classical point particles interacting through parameterised force fields, whose functional form and parameters are typically fitted to experimental data or higher-level calculations. This classical description becomes increasingly inadequate when the number of atoms is small or when the physical phenomena of interest are inherently quantum mechanical. In such regimes, quantum effects such as energy quantisation, zero-point motion, tunnelling and electronic rearrangements play a significant role and cannot be faithfully captured by classical force fields. As a consequence, the predictive power of classical molecular dynamics is fundamentally limited for problems involving bond formation and breaking, charge transfer, excited states or systems where electronic structure determines the observed behaviour. Moreover, classical molecular dynamics does not treat electronic degrees of freedom explicitly. The electrons are effectively integrated out and replaced by an effective interaction between nuclei. This prevents direct access to electronic properties such as band structure, density of states, charge density or optical response, which are essential for understanding the electronic, optical, and chemical behaviour of materials.

In situations where these limitations become critical, the modelling of matter must be approached from *first principles*. In this context, first principles means that the properties of the system are derived directly from fundamental physical laws, without relying on empirical parameters fitted to a specific class of materials. The central equation governing this description is the Schrödinger equation, which provides a complete quantum mechanical description of the system. The Schrödinger equation explicitly accounts for the electronic degrees of freedom and naturally incorporates key quantum concepts such as the wave nature of matter, energy quantisation, the probabilistic interpretation of observables, spin and the superposition and interference of quantum states. Within this framework, all measurable physical quantities are obtained as expectation values of appropriate operators

acting on the quantum mechanical wavefunction.

How the Schrödinger equation can be solved in practice, and what type of physical information can be extracted from its solutions, form the central focus of the quantum mechanical part of this course. In many introductory treatments of quantum mechanics, analytical solutions are limited to highly idealised problems, most notably the hydrogen atom. In that case, the exact solution is obtained under the assumptions of a single electron moving in the Coulomb potential of a fixed point nucleus, neglecting relativistic effects, spin–orbit coupling and electron–electron interactions. For multi-electron atoms and molecules, further approximations are typically introduced, leading to hydrogen-like atomic orbitals and effective one-electron pictures. While these models provide valuable conceptual insight, they often obscure the fact that quantum mechanics is today a practical and predictive tool for determining atomic-scale interactions in real materials.

Modern computational methods based on quantum mechanics allow the calculation of structural, energetic, electronic and dynamical properties of complex systems with remarkable accuracy. These approaches are now routinely applied to problems in condensed matter physics, chemistry, materials science and nanotechnology. One of the aims of the following chapters is therefore to bridge the gap between the abstract formalism of quantum mechanics and its concrete application to realistic atomistic simulations. Within the scope of this course, we will restrict ourselves to the solution of the *time-independent* Schrödinger equation. This choice reflects the fact that many ground-state and equilibrium properties of materials can be obtained from stationary solutions. The explicit time-dependent treatment of quantum systems, while extremely important, lies beyond the scope of the present lectures and is instead suggested as a suitable topic for further study, for example within a thesis project.

5.1 Key concepts

Quantum mechanics is founded on a small set of fundamental postulates that define how physical systems are described and how measurable quantities are obtained. The complete state of a quantum system is represented by a *state function*, or *wavefunction*, which contains all accessible information about the system and belongs to a complex vector space known as a *Hilbert space*. Physical *observables*, such as position, momentum and energy, are represented by linear *operators* acting on these state functions. The possible outcomes of a measurement are given by the eigenvalues of the corresponding operator, while the probabilistic nature of quantum mechanics is reflected in the fact that only *expectation values* of observables can be predicted. Expectation values are obtained as weighted averages over the wavefunction and represent the measurable mean value of a physical quantity for a system prepared in a superposition of quantum states. For an extended discussion on these concepts, we suggest the reader to refer to the book “Quantum Mechanics, Vol. 1” by Claude Cohen-Tannoudji, Bernard Diu and Frank Laloe.³

5.2 The time independent Schrödinger equation

Let us consider a system of N particles with position vectors collected into a unique vector $\mathbf{R}$ defined as

$$\mathbf{R} \equiv (\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) \in \mathbb{R}^{3N}, \quad (5.1)$$

which is then a $3N$ -dimensional vector defined in the real space (see [section 1.5](#)). In quantum mechanics, the state of the system is no longer given by the precise positions and

momenta of the particles but by the *wavefunction* $\Psi(\mathbf{R}, t)$, which defines a probability amplitude varying in space and time. The time evolution of $\Psi(\mathbf{R}, t)$ is governed by the *time-dependent* Schrödinger equation (TDSE):

$$i\hbar \frac{\partial \Psi(\mathbf{R}, t)}{\partial t} = \hat{H}(\mathbf{R}, t) \Psi(\mathbf{R}, t), \quad (5.2)$$

where i is the imaginary unit and $\hat{H}(\mathbf{R}, t)$ is the Hamiltonian operator representing the total energy of the system

$$\begin{aligned} \hat{H}(\mathbf{R}, t) &= \hat{T}(\mathbf{R}) + \hat{V}(\mathbf{R}, t) \\ &= \sum_i^N -\frac{\hbar^2}{2m} \nabla_i^2 + V(\mathbf{R}, t), \end{aligned} \quad (5.3)$$

defined as the sum of kinetic ($\hat{T}$) and potential energy ($\hat{V}$) operators.

Instead, if the Hamiltonian is independent of time, then the total energy of the system is constant in time and the wavefunction may be written as

$$\Psi(\mathbf{R}, t) = \psi(\mathbf{R}) \chi(t), \quad (5.4)$$

that is a product of a function ψ , which depends only on the atomic positions, with a function χ , which depends only on the time. Substituting this form into Equation 5.2 yields two equations: one of these is an ordinary differential equation for the time-dependent factor,

$$i\hbar \frac{d\chi(t)}{dt} = E \chi(t), \quad (5.5)$$

whose solution is the phase factor

$$\chi(t) = \exp(-iEt/\hbar), \quad (5.6)$$

while the other is the eigenvalue-eigenvector equation

$$\hat{H}(\mathbf{R}) \psi(\mathbf{R}) = E \psi(\mathbf{R}), \quad (5.7)$$

for the Hamiltonian operator. Equation 5.7 is known as the *time-independent* Schrödinger equation (TISE). Its solution $\psi(\mathbf{R})$ is called a *stationary state* and the corresponding eigenvalue E represents the associated conserved energy.

5.3 General properties of the time-independent Schrödinger equation

For any quantum system described by a time-independent Hamiltonian, the Hamiltonian operator $\hat{H}$ admits a complete set of stationary states $\{\phi_n\}$, defined as the solutions of the time-independent Schrödinger equation,

$$\hat{H}(\mathbf{R}) \phi_n(\mathbf{R}) = E_n \phi_n(\mathbf{R}), \quad (5.8)$$

where E_n is the energy eigenvalue associated with the stationary state labeled as n . It is important to emphasise that each eigenvalue E_n represents the total energy in the corresponding stationary state ϕ_n of the *entire system*. Further, n can be a discrete or a continuous label depending on the discrete or continuous nature of the energy spectrum

$\{E_n\}$: if the Hamiltonian contains bounding interactions (i.e. non-constant potentials) and/or boundary conditions are imposed, then the spectrum is discrete and n is usually an integer number, otherwise it can be a real number or any other label indicating a generic element of an uncountable set (see [section 5.4](#) for more detail).

The set of stationary states $\{\phi_n\}$ forms a basis for the Hilbert space associated with the system. As a consequence, any admissible wavefunction $\psi(\mathbf{R})$ may be expressed as a linear combination of energy eigenstates

$$\psi(\mathbf{R}) = \sum_n c_n \phi_n(\mathbf{R}), \quad (5.9)$$

where the expansion coefficients c_n are complex numbers; if the stationary states are chosen to be orthonormal and the wavefunction is normalised, the coefficients satisfy $\sum_n |c_n|^2 = 1$. This expansion is purely a representation of the quantum state in a convenient basis. While each stationary state ϕ_n individually satisfies the time-independent Schrödinger equation [5.8](#) and corresponds to a definite energy E_n , a general linear combination of such states does not represent an energy eigenstate; instead, it represents a superposition of quantum states for which the energy is not definite.

In the quantum description, physical observables are represented by operators acting on the wavefunction. If the system is prepared in the state as defined in [Equation 5.9](#), then a measurement yields the eigenvalue E_n with probability $|c_n|^2$, after which the system is projected onto the corresponding energy eigenstate ψ_n . This motivates the introduction of the *expectation value*, which defines the average value of an observable over repeated measurements of a system prepared in a general quantum state. For an operator $\hat{A}$ associated with an observable A , the expectation value of $\hat{A}$ for a system in the state ψ is given by

$$\langle \hat{A} \rangle_\psi = \frac{\int \psi^*(\mathbf{R}) \hat{A} \psi(\mathbf{R}) d\mathbf{R}}{\int |\psi(\mathbf{R})|^2 d\mathbf{R}} \quad (5.10)$$

where $d\mathbf{R} \equiv d\mathbf{r}_1 d\mathbf{r}_2 \cdots d\mathbf{r}_N$ denotes integration over the full $3N$ -dimensional configuration space. For a stationary state, the full time-dependent wavefunction differs from $\psi(\mathbf{R})$ only by a global phase factor $\exp(-iEt/\hbar)$, which cancels in the evaluation of expectation values. Further, if the state is normalised, the denominator in [5.10](#) is equal to unity and is often omitted. In the case $\hat{A} = \hat{H}$, this expression defines the expectation value of the total energy,

$$\langle \hat{H} \rangle_\psi = \frac{\int \psi^*(\mathbf{R}) \hat{H} \psi(\mathbf{R}) d\mathbf{R}}{\int |\psi(\mathbf{R})|^2 d\mathbf{R}} \equiv E[\psi], \quad (5.11)$$

where $E[\psi]$ emphasises that the expectation value of the Hamiltonian, like any expectation value, may be regarded as a functional of the wavefunction. If ψ coincides with an exact stationary state ϕ_n , the functional $E[\psi]$ yields the corresponding energy eigenvalue E_n ; otherwise, it represents the average energy obtained from repeated measurements on identically prepared systems.

5.4 Exact solutions to the TISE

Exact analytical solutions of [Equation 5.7](#) exist only for a small number of highly idealised quantum-mechanical models, typically characterised by strong symmetry and effective separability of the Hamiltonian. In the subsections that follow, we illustrate this point by presenting several typical examples for which the time-independent Schrödinger equation can be solved exactly, namely the free particle, the particle in a box, and the hydrogen atom.

Although these examples are not exhaustive, they are representative of the limited class of systems for which closed-form solutions can be obtained and serve as important reference cases in quantum mechanics. Beyond their role as exactly solvable models, these examples also serve to demonstrate how the specific form of the Hamiltonian determines both the structure of the energy spectrum and the features of the corresponding eigenfunctions. Later, in [section 5.5](#), we will see how one proceeds when no exact analytical solution exists, and thus why approximate and numerical methods are essential for the treatment of realistic quantum systems.

5.4.1 Example 1 - Single particle in free space

Let us consider a particle of mass m moving freely in a constant potential; in this condition, the particle moves classically with constant velocity along a constant direction which we can describe with the x Cartesian axis. The Hamiltonian then reduces to the monodimensional kinetic energy operator

$$\hat{H}(x) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \quad (5.12)$$

and the TISE reads as

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x) = E\psi(x). \quad (5.13)$$

A plane wave of the form $\psi(x) = A \exp(ik_x x)$ satisfies this equation, and by substitution into [Equation 5.13](#) we obtain the result

$$E = \frac{\hbar^2 k_x^2}{2m}, \quad (5.14)$$

which can be represented in a *dispersion plot* as in [Figure 5.1](#). This solution illustrates three key features of free particle states:

- the energy is quadratic in the modulus of the wave vector k_x ,
- the state is delocalised, with the probability density $|\psi(x)|^2 = A$ constant throughout space, and
- there exists a *continuum* of possible energy eigenvalues, corresponding to the continuous range of allowed k_x values.

Further, upon measurement of momentum, the particle collapses to a definite wave vector k'_x , corresponding to the state $\psi(x) = A \exp(ik'_x x)$.

5.4.2 Example 2 - Particle in a box

Let us consider a particle of mass m confined in a one-dimensional box of length L with infinite potential walls at $x = 0$ and $x = L$ (see [Figure 5.2a](#)). The Hamiltonian inside the box reduces to the kinetic energy operator:

$$\hat{H}(x) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x), \quad V(x) = \begin{cases} \text{constant}, & 0 < x < L, \\ \infty, & x \leq 0 \text{ or } x \geq L. \end{cases} \quad (5.15)$$

subject to the boundary conditions $\psi(0) = \psi(L) = 0$. By setting $V(x) = 0$ in $0 < x < L$, the TISE is then written as

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi_n(x)}{\partial x^2} = E_n \psi_n(x), \quad (5.16)$$

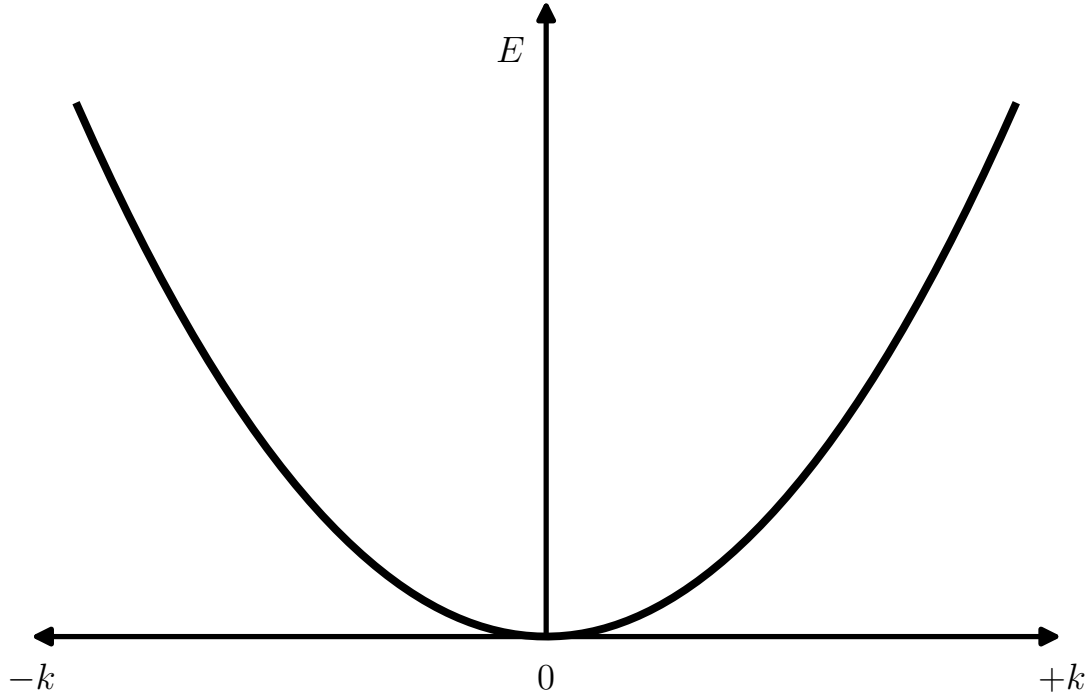


Figure 5.1: Schematic illustration of the quadratic energy-momentum dispersion relation for a free particle in space.

with solutions

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi}{L}x\right), \quad n = 1, 2, 3, \dots \quad (5.17)$$

yielding discrete energy eigenvalues

$$E_n = \frac{\hbar^2 n^2 \pi^2}{2mL^2}. \quad (5.18)$$

Key features of this solution are:

- the energy levels are discrete, i.e. *quantised*, because the particle is confined,
- the eigenfunctions are spatially localised standing waves within the box, and
- only specific wave vectors $k_n = n\pi/L$ are allowed.

This example contrasts with the free particle, illustrating how the presence of a non-constant potential transforms the energy spectrum from a continuum to a discrete set of eigenvalues (visualised in [Figure 5.2b](#)).

5.4.3 Example 3 - The hydrogen atom

Let us consider a single electron of mass m_e and charge e moving in the Coulomb potential of a proton of mass m_p with Hamiltonian

$$\hat{H}(r) = -\frac{\hbar^2}{2\mu} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r} \quad (5.19)$$

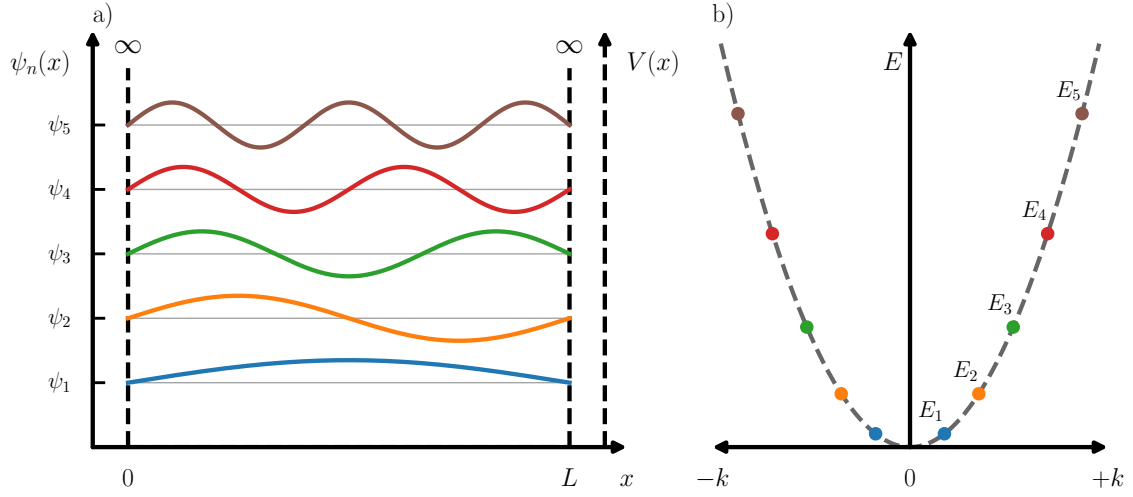


Figure 5.2: a) Schematic illustration of the first 5 eigenfunctions ψ_n ($n = 1, 2, 3, 4, 5$, solid coloured lines) of the one-dimensional infinite square well problem (“particle in a box”), for a box of length L with infinite potential walls at $x = 0$ and $x = L$ (dashed lines). b) In the presence of a confining potential, the allowed wavevectors become discretised, leading to quantised eigenvalues (coloured points) corresponding to the eigenfunctions shown in a). The dashed curve indicates the continuous quadratic free particle dispersion relation, $E \propto k^2$.

where ϵ_0 is the dielectric permittivity of the free space, $\mu = (m_p m_e)/(m_p + m_e) \approx m_e$ is the reduced mass of the two-particle system and $r = |\mathbf{r}|$ is the electron-proton separation. The spherical symmetry of the Coulomb potential suggests using spherical coordinates $\mathbf{r} = (r, \theta, \phi)$. In these coordinates, the Laplacian ∇^2 is expressed as

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) - \frac{\hat{L}^2}{\hbar^2 r^2}, \quad (5.20)$$

where

$$\hat{L}^2(\theta, \phi) = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right], \quad (5.21)$$

is the squared angular momentum operator. Using this, the Hamiltonian becomes

$$\hat{H}(\mathbf{r}) = -\frac{\hbar^2}{2\mu} \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) - \frac{\hat{L}^2}{\hbar^2 r^2} \right] - \frac{e^2}{4\pi\epsilon_0 r}. \quad (5.22)$$

The Hamiltonian commutes with $\hat{L}^2$ and $\hat{L}_z$,

$$[\hat{H}, \hat{L}^2] = 0, \quad [\hat{H}, \hat{L}_z] = 0, \quad [\hat{L}^2, \hat{L}_z] = 0, \quad (5.23)$$

where

$$\hat{L}_z(\phi) = -i\hbar \frac{\partial}{\partial \phi}. \quad (5.24)$$

The eigenfunctions φ can then be chosen to be simultaneous eigenfunctions of the energy and angular momentum:

$$\hat{H}\varphi(\mathbf{r}) = E\varphi(\mathbf{r}), \quad (5.25)$$

$$\hat{L}^2\varphi(\mathbf{r}) = \ell(\ell+1)\hbar^2\varphi(\mathbf{r}), \quad (5.26)$$

$$\hat{L}_z\varphi(\mathbf{r}) = m\hbar\varphi(\mathbf{r}), \quad (5.27)$$

with quantum numbers $n = 1, 2, 3, \dots$, $l = 0, 1, \dots, n - 1$ and $m = -l, \dots, l$.

Because the Hamiltonian separates naturally into radial and angular components, we can write the wavefunction as a product

$$\varphi(\mathbf{r}) = R_{nl}(r)Y_l^m(\theta, \phi) \quad (5.28)$$

where $R_{nl}(r)$ is the radial part and $Y_l^m(\theta, \phi)$ are the spherical harmonics. The quantum numbers (n, l, m) fully characterise each state of the hydrogen atom and specify the states (*orbitals*) in which the single electron may reside. The radial solution yields the discrete energy levels of the hydrogen atom:

$$E_n = -\frac{m_e e^4}{2(4\pi\epsilon_0)^2 \hbar^2} \frac{1}{n^2}, \quad n = 1, 2, 3, \dots \quad (5.29)$$

which depend only on the principal quantum number n , not on l nor on m . Thus, for each n , all orbitals with $l = 0, 1, \dots, n - 1$ and $m = -l, \dots, l$ are degenerate in energy.

It is also useful to examine how the wavefunctions describe the spatial distribution of the electron. The radial probability density $r^2|R_{nl}(r)|^2$ gives the likelihood of finding the electron at a distance r from the nucleus, while the angular distribution $|Y_l^m(\theta, \phi)|^2$ determines the characteristic shapes of the orbitals associated with different (l, m) .

Key features of this system are:

- hydrogen exhibits a discrete energy spectrum, unlike the free particle,
- eigenfunctions factor into radial and angular parts, reflecting the separation of the Hamiltonian into radial and angular contributions, and
- the quantum numbers (n, l, m) fully characterise each stationary state of the hydrogen atom.

The spectroscopic labels *s*, *p*, and *d* (originating historically from “sharp”, “principal”, and “diffuse” emission lines) correspond to the angular momentum quantum numbers $l = 0, 1, 2$ respectively, with higher values continuing as *f*, *g*, $\dots$. Energy degeneracy of (n, l, m) states is immediately lifted if other interactions like, for example, spin-orbit coupling, are considered. This lifting of degeneracy gives rise to the ordering of subshell energies typically used when discussing orbital filling in the periodic table.

5.5 The many particle problem

In the previous section, we presented a number of idealised systems for which exact solutions to the TISE exist. The most complicated of these was the hydrogen atom ([subsection 5.4.3](#)), containing two particles interacting via a central potential. Real materials, however, contain $N_n > 1$ nuclei and $N_e > 1$ electrons, and the corresponding Hamiltonian is written as the sum of kinetic $\hat{T}$ and potential $\hat{V}$ operators,

$$\begin{aligned} \hat{H} &= \hat{T} + \hat{V} \\ &= \hat{T}_e + \hat{T}_n + \hat{V}_{e,n} + \hat{V}_{e,e} + \hat{V}_{n,n}, \end{aligned} \quad (5.30)$$

where $\hat{T}_e$ is the electron kinetic energy operator, $\hat{T}_n$ is the nuclei kinetic energy operator, $\hat{V}_{e,n}$ is the electron-nuclei Coulombic interaction, $\hat{V}_{e,e}$ is the electron-electron Coulombic

interaction, and $\hat{V}_{n,n}$ is the nuclei-nuclei Coulombic interaction. Substituting the explicit forms of the kinetic and Coulomb interaction operators, we obtain

$$\begin{aligned} \hat{H}(\{\mathbf{r}_i\}, \{\mathbf{R}_I\}) = & -\frac{\hbar^2}{2m_e} \sum_i^{N_e} \nabla_i^2 - \frac{\hbar^2}{2} \sum_I^{N_n} \frac{1}{m_I} \nabla_I^2 + \\ & + j_0 \left[-\sum_i^{N_e} \sum_I^{N_n} \frac{Z_I}{|\mathbf{r}_i - \mathbf{R}_I|} + \frac{1}{2} \sum_{i \neq j}^{N_e} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + \frac{1}{2} \sum_{I \neq J}^{N_n} \frac{Z_I Z_J}{|\mathbf{R}_I - \mathbf{R}_J|} \right], \end{aligned} \quad (5.31)$$

where $\hbar$ is the reduced Planck constant, m_e is the electron mass, Z_I is the atomic number of nuclei I , and $j_0 = \frac{e^2}{4\pi\epsilon_0}$ with e the electric charge and ϵ_0 the permittivity of free space. The interaction terms $\hat{V}_{e,e}$ and $\hat{V}_{n,n}$ couple all degrees of freedom, such that the wavefunction $\psi(\{\mathbf{r}_i\}, \{\mathbf{R}_I\})$ cannot be factorised into independent one-particle components. As a result, the TISE

$$\hat{H}(\{\mathbf{r}_i\}, \{\mathbf{R}_I\})\psi(\{\mathbf{r}_i\}, \{\mathbf{R}_I\}) = E\psi(\{\mathbf{r}_i\}, \{\mathbf{R}_I\}) \quad (5.32)$$

becomes a high-dimensional, non-separable partial differential equation defined in a $3(N_e + N_n)$ -dimensional configuration space. This immediately precludes any closed-form analytical solution. Moreover, even if these interactions are approximated or simplified, the wavefunction would still depend on all $3(N_e + N_n)$ coordinates, and the number of degrees of freedom required to represent them grows exponentially with N_e and N_n . Thus, the fundamental obstacle to solve Equation 5.32 is not only the physics of the Coulomb coupling, but also the mathematical structure of many-body quantum mechanics itself. For realistic systems, it is infeasible to store or manipulate a wavefunction that depends on all $3(N_e + N_n)$ coordinates simultaneously. All practical electronic structure methods therefore necessarily rely on physical approximations combined with numerical techniques. The aim is not to obtain an analytical expression for ψ , but to reformulate the problem such that it can be solved numerically to controlled accuracy. Our first task will be to introduce physical approximations that decouple the particle interactions sufficiently to obtain a set of effective one-electron equations, whose solutions depend only on three spatial variables and can therefore be represented and solved numerically. Our second task will be to introduce numerical techniques for representation of the wavefunction — principally basis-set expansions and variational optimisation — that allow systematic, controllable approximation of the resulting wavefunctions. In what follows, we will restrict ourselves to search for the ground state wavefunction ψ_0 . Although excited states are essential in spectroscopy and finite-temperature physics, their treatment introduces additional conceptual and computational complexity which lies beyond the scope of this chapter. At zero temperature and in thermal equilibrium, the system is in its ground state by definition, and many equilibrium properties of materials are well described in this limit. The ground state therefore provides a natural and widely applicable starting point to obtain a reliable quantum mechanical description of the system.

5.5.1 The Born-Oppenheimer approximation

As a first step toward making the many-particle problem more tractable, we introduce the *Born-Oppenheimer* (BO) approximation, which exploits the fact that most nuclei are several thousand times more massive than electrons. This mass disparity implies that nuclei move much more slowly compared to the electrons. Consequently, the electronic subsystem is able to adapt *adiabatically* (i.e. instantaneously) to any change in nuclear geometry. Because of this intrinsic separation of timescales, the electronic and nuclear

motions may be decoupled: the nuclei evolve slowly, while the electrons remain in the electronic ground state corresponding to the *instantaneous* nuclear configuration. Thus, the nuclear coordinates may be treated as parameters rather than dynamical variables. This reduces the full quantum problem to a purely electronic one and removes the nuclear kinetic energy from the Hamiltonian. As a result, we may define a reduced electronic Hamiltonian

$$\hat{H}_e(\{\mathbf{r}_i\}) = \hat{T}_e + \hat{V}_{e,n} + \hat{V}_{e,e}, \quad (5.33)$$

where the kinetic energy operator of the nuclei $\hat{T}_n$ is omitted by construction, and the nucleus-nucleus repulsion ($\hat{V}_{n,n}$) becomes a constant for fixed nuclear positions and may therefore be excluded. The operator $\hat{H}_e$ now acts only on functions of the electronic coordinates, so the stationary electronic states are defined by the reduced eigenvalue equation known as the *electronic* TISE,

$$\hat{H}_e(\{\mathbf{r}_i\})\psi_e(\{\mathbf{r}_i\}) = E_e\psi_e(\{\mathbf{r}_i\}), \quad (5.34)$$

where $\psi_e(\{\mathbf{r}_i\})$ is the ground state *electronic* wavefunction existing now only in a $3N_e$ -dimensional configuration space, and E_e is the ground state *electronic* total energy. The BO approximation is extremely accurate for most molecular and solid-state systems under typical conditions, particular when nuclei are much heavier than electrons and electronic states are well separated in energy. It can break down when this separation of timescales fails, for example in systems with very light nuclei (notably hydrogen) or when strong non-adiabatic effects become important, such as avoided crossings, conical intersections, or ultrafast photochemical processes. At the ground state, nuclear motion is typically slow enough that the BO approximation remains valid; however, $\hat{V}_{e,e}$ still couples all electronic coordinates, such that the electronic TISE remains a high-dimensional, non-separable equation. No closed-form analytical solution exists after the BO approximation. Thus, although the BO approximation removes the nuclear degrees of freedom from the quantum problem, it does not resolve the core challenge posed by electron-electron interactions or the need for a tractable numerical representation of the electronic wavefunction. These issues motivate the next step: introducing physical approximations that replace the fully interacting electronic problem with a set of one-electron equations that can be solved numerically.

5.5.2 The mean-field approximation

Following the Born–Oppenheimer approximation, the electronic problem is governed by a reduced electronic Hamiltonian

$$\hat{H}_e(\mathbf{r}) = \hat{T}_e + \hat{V}_{e,n} + \hat{V}_{e,e}. \quad (5.35)$$

The electron-electron interaction term $\hat{V}_{e,e}$ couples all electronic coordinates, and hence prevents any factorisation of the many-electron wavefunction into products of one-electron functions. As a result, the electronic TISE remains a high-dimensional, non-separable equation that cannot be solved analytically. A natural strategy to make progress is to approximate the effect of all other electrons on a given electron by an average or *mean field* approach. Instead of allowing each electron to interact explicitly with every other electron at every point in configuration space, we replace these pairwise interactions by an effective one-body potential. Conceptually, this reduces the fully interacting problem to a set of effective one-electron equations.

In the simplest implementation of this idea, the Coulomb repulsion felt by electron i is approximated by the classical electrostatic potential generated by the total electron density,

$$V_H(\mathbf{r}) = \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' \quad (5.36)$$

where $\mathbf{r} \in \mathbb{R}^3$ is a 3-dimensional position vector. To enforce Pauli's exclusion principle, the *exchange* contribution must also be included among the electron-electron interactions. We can then define an effective one-electron operator

$$\hat{h}_{\text{eff}} = \hat{T}_e + \hat{V}_{e,n} + \hat{V}_H + \hat{V}_X \quad (5.37)$$

where $\hat{V}_X$ represents the exact exchange interaction. The crucial point is that this allows the full interacting problem to be replaced by a set of coupled one-electron equations

$$\hat{h}_{\text{eff}}\phi_n(\mathbf{r}) = \epsilon_n\phi_n(\mathbf{r}), \quad (5.38)$$

where the single electron orbitals $\{\phi_n\}$ must produce a density consistent with the mean field that generates $\hat{h}_{\text{eff}}$. Here n is an integer index which labels the infinite possible energy levels ϵ_n and the corresponding electronic states ϕ_n ; a level n is said to be *occupied* if an electron is in the corresponding state ϕ_n , otherwise the level is said to be *unoccupied* and the corresponding eigenfunction ϕ_n is a possible state accessible to the electron.

The mean-field formulation greatly simplifies the structure of the problem. Each $\phi_n(\mathbf{r})$ is now a function of only three spatial variables, not $3N_e$. However, the resulting equations still possess no closed-form analytical solution: the effective potential depends on the density, and the density depends on the orbitals. In practice, the mean-field equations are solved numerically. To do so, we must choose a finite representation for the one-electron orbitals ϕ_n and solve the equations iteratively. This motivates the next step: the introduction of finite basis sets and the variational approximation, which together provide a controlled numerical framework for representing and improving approximate solutions. Finally, we note that a mean-field description, while conceptually powerful, has intrinsic limitations. It neglects dynamical electron correlation beyond the average field and cannot reproduce all features of real many-body systems. These limitations motivate the development of alternative formulations of the electronic structure problem, in particular density functional theory (discussed later in [section 6.1](#)), which reformulates the fundamental variable from the many-electron wavefunction to the three-dimensional electron density.

5.5.3 Numerical representation of the wavefunction

Until now, we have largely focused on approximations which simplify the physics. However, we must also consider the mathematical representation of the electronic wavefunction itself, which is too high-dimensional to handle directly; this motivates the desire to obtain controlled numerical approximations to the electronic wavefunction. A general and widely used approach is to express a trial wavefunction ψ'_e as a linear combination of known functions $\{\phi_n\}$ forming a *basis set*:

$$\psi'_e(\mathbf{r}) = \sum_{n=1}^{\infty} c_n \phi_n(\mathbf{r}) \quad (5.39)$$

where the complex coefficients $\{c_n\}$ parametrise the trial state. Substitution of this expansion into [Equation 5.34](#) yields the corresponding *trial* ground state electronic total

energy

$$E'_e = \frac{\sum_{mn} c_m^* c_n H_{mn}}{\sum_{mn} c_m^* c_n S_{mn}}, \quad (5.40)$$

where the Hamiltonian and overlap matrix elements are defined as

$$H_{mn} = \int \phi_m^*(\mathbf{r}) \hat{H}_e \phi_n(\mathbf{r}) d\mathbf{r}, \quad S_{nm} = \int \phi_m^*(\mathbf{r}) \phi_n(\mathbf{r}) d\mathbf{r}. \quad (5.41)$$

If the basis were complete and, in particular, composed of exact eigenfunctions of $\hat{H}_e$, the matrix representation would be diagonal and Equation 5.40 would reproduce the exact ground state. In practice, the basis is neither infinite nor composed of eigenfunctions of $\hat{H}_e$; for this reason, the series in Equation 5.39 is truncated up to a finite number M of functions in the basis. As a result, all observables computed from it inherit approximation errors determined by the choice and completeness of the basis. This motivates the introduction of the *variational theorem*, which provides a formal criterion for assessing and systematically improving the quality of approximate wavefunctions within any finite basis.

5.5.4 The variational theorem

The variational theorem provides the formal foundation for improving approximate wavefunctions and energies within any finite basis. It states that

$$E'_e[\psi'_e] \geq E_e[\psi_e] \quad (5.42)$$

for any choice of ψ'_e . The equality may only arise when $\psi'_e(\mathbf{r})$ reproduces the exact ground state wavefunction $\psi_e(\mathbf{r})$. Hence, within a given finite basis, improving the coefficients $\{c_n\}$ to minimise E'_e yields the best possible approximation to the exact ground state electronic total energy available in that basis. The basis functions $\{\phi_n\}$ may be plane waves, localised Gaussian functions, atomic orbitals, or other suitable sets. A sufficiently large basis can represent any square-integrable wavefunction to arbitrary accuracy, although the rate and feasibility of convergence depends strongly on the basis type (e.g. plane waves *vs.* Gaussian orbitals) and the numerical method employed.

Despite its generality and conceptual elegance, the variational approach does not resolve the fundamental difficulty inherent in the many-electron Schrödinger equation. Because $\hat{V}_{e,e}$ couples all electronic coordinates, the wavefunction — whether exact or approximated — remains a functional defined in the full $3N_e$ -dimensional configuration space and cannot be factorised into independent single-electron components. Any numerical representation of ψ_e (or ψ'_e) therefore requires a number of degrees of freedom that grows exponentially with the number of electrons; for instance, discretising each spatial coordinate on a grid of M points would require M^{3N_e} values. This exponential scaling makes it impossible, in practice, to store, optimise or even meaningfully sample the many-electron wavefunction for realistic systems, and it is typical of the Hartree-Fock and post-Hartree-Fock numerical methods. Such methods are not the subject of the present discussion; the reader is referred to the suggested literature for a more detailed study on this matter. The variational principle guarantees the best approximation attainable within a finite basis, but it does not change the underlying scaling barrier. This motivates the search for alternative formulations where the central variable is not the many-electron wavefunction but a lower-dimensional quantity, most notably the ground-state electron density, leading naturally to the *density functional theory* (DFT), which we discuss in the next chapter. Nevertheless, the numerical tools introduced in this section (finite basis expansions and variational optimisation) remain essential for practical electronic structure calculations. In DFT, the problem is reformulated

in terms of the density, but its solution still requires representing orbitals in a finite basis and solving a variational eigenvalue problem.

5.6 The Self-Consistent Field procedure

The self-consistent field (SCF) procedure is a general iterative strategy used to solve quantum mechanical many-body problems within effective one-particle frameworks. It is employed in a wide range of electronic structure methods, including Hartree–Fock (HF), density functional theory (DFT), and several post-Hartree–Fock approaches. The common feature of these methods is that the effective Hamiltonian depends on quantities that are themselves determined by the solution of the eigenvalue problem, thereby requiring an iterative procedure to reach consistency.

In its most general form, the electronic structure problem can be written as in [Equation 5.38](#)

$$\hat{h}_{\text{eff}}(\mathcal{Q})\phi_n(\mathbf{r}) = \epsilon_n\phi_n(\mathbf{r}), \quad (5.43)$$

where $\hat{h}_{\text{eff}}$ is an effective Hamiltonian operator which may depend on quantities $\mathcal{Q}$ like the electronic density as in DFT, the one-particle density matrix as in Hartree–Fock theory, or more general objects such as effective potentials, Fock matrices, or correlated amplitudes in more advanced methods.

The origin of the self-consistency requirement lies in the fact that these quantities are constructed from the orbitals that are obtained as solutions of the eigenvalue problem. The SCF cycle proceeds through the following steps:

1. Initial guess. An initial approximation for the relevant quantity $\mathcal{Q}$ is chosen. This may correspond to a superposition of atomic densities, a guess density matrix, previously converged orbitals or any physically reasonable starting point.

2. Construction of the effective Hamiltonian. Using $\mathcal{Q}^{(k)}$ at iteration k , the effective Hamiltonian $\hat{h}_{\text{eff}}^{(k)}$ is constructed. In practice, this step often corresponds to building a matrix representation in the chosen basis set.

3. Solution of the eigenvalue problem. The matrix eigenvalue problem

$$\mathbf{h}_{\text{eff}}^{(k)}\mathbf{C}^{(k)} = \mathbf{S}\mathbf{C}^{(k)}\boldsymbol{\varepsilon}^{(k)} \quad (5.44)$$

is solved, where $\mathbf{h}_{\text{eff}}^{(k)}$ is the matrix representation of the effective Hamiltonian, $\mathbf{S}$ is the overlap matrix of the basis functions, $\mathbf{C}^{(k)}$ contains the expansion coefficients of the orbitals, and $\boldsymbol{\varepsilon}^{(k)}$ is the diagonal matrix of orbital energies.

4. Update of the self-consistent quantities. From the newly obtained orbitals, an updated set of quantities $\mathcal{Q}^{(k+1)}$ is computed. For example, the density matrix in a basis $\{\phi_\mu\}$ is given by

$$P_{\mu\nu} = \sum_{i=1}^{N_{\text{occ}}} C_{\mu i} C_{\nu i}^*, \quad (5.45)$$

where the sum runs over occupied orbitals. From this density matrix one can compute the electronic density, Coulomb and exchange contributions, and ultimately the total energy.

5. Convergence check and mixing. The updated quantities $\mathcal{Q}^{(k+1)}$ are compared to those from the previous iteration. Convergence can be monitored through changes in the total energy,

$$|E^{(k+1)} - E^{(k)}| < \delta_E, \quad (5.46)$$

through the norm of the difference between successive density matrices,

$$\|\mathbf{P}^{(k+1)} - \mathbf{P}^{(k)}\| < \delta_P, \quad (5.47)$$

or through changes in the effective potential or Fock matrix. The convergence tolerance, e.g. δ_E or δ_P , are an input of the SCF procedure. If convergence criteria are not satisfied, a new input $\mathcal{Q}^{(k+1)}$ is generated, often using mixing schemes such as linear mixing or more advanced methods (e.g., Pulay or Broyden schemes), and the cycle is repeated. The SCF procedure terminates when all chosen convergence thresholds are met. At this point, the orbitals, effective Hamiltonian, total energy, and all derived quantities are mutually consistent. The resulting solution corresponds to a stationary point of the chosen energy functional under the constraints imposed by the method. It is important to note that the SCF solution is not guaranteed to correspond to the global minimum of the energy functional. Different initial guesses may lead to different stationary solutions, particularly in systems exhibiting near-degeneracy or strong correlation effects. Careful convergence control and physical insight are therefore essential in practical calculations. The SCF framework provides a unifying computational backbone for a broad class of quantum mechanical methods. Whether formulated in terms of densities, density matrices, effective potentials or more general operators, the central idea remains the same: iteratively solve a nonlinear eigenvalue problem until all quantities entering the Hamiltonian are consistent with the wavefunctions that it generates.

Chapter 6

Density Functional Theory

In the previous chapter, we concluded that exact solutions of the many-particle TISE (Equation 5.32) are impossible for realistic systems. The coupling introduced by the electron-electron ($\hat{V}_{e,e}$) and nucleus-nucleus ($\hat{V}_{n,n}$) interaction terms prevents the separation of variables into independent one-particle equations. To make progress, we adopted a standard sequence of simplifications. First, we introduced the Born-Oppenheimer (BO) approximation, yielding an electronic hamiltonian $\hat{H}_e$ that depends parametrically on the fixed nuclear positions (Equation 5.33). Second, we employed numerical techniques based on finite basis sets and the variational principle (Equation 5.42) to obtain approximate solutions of the electronic TISE. However, this approach still confronts severe difficulties: the wavefunction is still a $3N_e$ -dimensional object, as the exact many-electron wavefunction remains a function defined in a $3N_e$ -dimensional configuration space. If we discretise each coordinate on a grid of M points, the total number of points required is M^{3N_e} . Even for modest values (e.g. $M = 10^2$ and $N_e = 10$), M^{3N_e} is astronomically large. For realistic systems, storing or variationally optimising such a high-dimensional object is computationally intractable. On the other hand, the electron density $\rho(\mathbf{r})$ depends only on a single spatial coordinate $\mathbf{r} \in \mathbb{R}^3$ and it is defined as

$$\rho(\mathbf{r}) = \sum_{i=1}^{N_e} \int |\psi_e(\{\mathbf{r}_i\})|^2 \delta(\mathbf{r}_i - \mathbf{r}) d\mathbf{r}_1 \dots d\mathbf{r}_{N_e}. \quad (6.1)$$

Because the electrons are indistinguishable particles, each term in the sum is identical, giving the standard expression

$$\rho(\mathbf{r}) = N_e \int |\psi_e(\mathbf{r}, \{\mathbf{r}_i\}_{i=2}^{N_e})|^2 d\mathbf{r}_2 \dots d\mathbf{r}_{N_e}. \quad (6.2)$$

The key point is that whereas the wavefunction ψ_e is a function of $3N_e$ variables, the density $\rho(\mathbf{r})$ is a function of only three variables: representing $\rho(\mathbf{r})$ on a numerical grid therefore scales as M^3 rather than M^{3N_e} . This is one of the key features of *Density functional theory* (DFT), which is based on the Hohenberg and Kohn theorems. The Hohenberg-Kohn existence theorem states that the ground state electron density uniquely determines the external potential and hence all ground-state properties of the system. This reformulation reduces the numerical complexity of the ground-state problem to the representation of a

three-dimensional function, which is computationally tractable for the implementation of numerical methods. The elaboration of the DFT approach is thus the focus of this chapter: we discuss the key ideas of Kohn-Sham DFT in its (in principle) exact representation, then introduce the Hohenberg-Kohn variational theorem and the self-consistent field procedure as the means to variationally optimise the ground state density. It is here worthy to mention that self-consistent procedures are used not only within the DFT framework but also in other schemes such as Hartree-Fock, post-Hartree-Fock, GW , and for the search of numerical solutions of the Bethe-Salpeter and Casida equations.

6.1 Exact Kohn-Sham DFT

The Kohn-Sham formulation of DFT introduces a crucial conceptual simplification: the interacting many-electron system is mapped onto an auxiliary system of non-interacting electrons, whose wavefunction is written as a *Slater determinant* of one-electron orbitals $\{\phi_n\}$,

$$\psi_e(\{\mathbf{r}_i\}) = \frac{1}{\sqrt{N_e!}} \det [\phi_n(\mathbf{r}_i)]_{n,i=1}^{N_e}, \quad (6.3)$$

where the determinant is taken over the n -th orbital evaluated at the i -th electron coordinate. These one-electron orbitals are chosen such that, by construction, the resulting electron density

$$\rho_0(\mathbf{r}) = \sum_{n=1}^{N_e} |\phi_n(\mathbf{r})|^2. \quad (6.4)$$

exactly reproduces the ground state electron density of the interacting system. Each orbital satisfies a Kohn-Sham (KS) equation

$$\hat{H}_{\text{KS}}(\mathbf{r})\phi_n(\mathbf{r}) = \epsilon_n\phi_n(\mathbf{r}), \quad (6.5)$$

with eigenvalues ϵ_n , and KS Hamiltonian defined as

$$\hat{H}_{\text{KS}}(\mathbf{r}) = \hat{T}_{\text{ref}} + \hat{V}_{en} + \hat{V}_H + \hat{V}_{\text{XC}}. \quad (6.6)$$

The terms on the right hand side represent kinetic energy of the non-interacting auxiliary/reference system, electron-nuclear potential, classical electron-electron (Hartree) repulsion, and exchange-correlation potential incorporating quantum many-body effects not captured by the previous terms, respectively. With $\hat{H}_{\text{KS}}$, it may be shown that the total energy functional becomes:

$$E_e[\rho_0] = \sum_n^{N_e} \int \phi_n^*(\mathbf{r}) \hat{T}_{\text{ref}} \phi_n(\mathbf{r}) d\mathbf{r} + E_{en}[\rho_0] + E_H[\rho_0] + E_{\text{XC}}[\rho_0], \quad (6.7)$$

where all the potential energy terms have a functional dependence *only* on the ground state electron density ρ_0 . Note that the exchange-correlation potential is the only term in the KS Hamiltonian for which we do not possess an exact analytic description.

Conceptually, the KS equations are *implicit*: the orbitals determine the density, while the density determines the KS potential that defines the orbitals. This defines a *Kohn-Sham map*

$$\mathcal{F} : \rho(\mathbf{r}) \mapsto \rho'(\mathbf{r}) \quad (6.8)$$

between an arbitrary input $\rho(\mathbf{r})$ and output $\rho'(\mathbf{r})$ density obtained by solving the KS equations. The exact ground state density $\rho_0(\mathbf{r})$ is the *fixed point* of this map: when used to construct the KS Hamiltonian, it yields orbitals whose density is exactly the same.

Exchange-correlation functional. The exchange-correlation term, V_{XC} , accounts for all the quantum many-body effects that are not captured by the classical terms or the non-interacting kinetic energy. This includes exchange interactions due to the Pauli exclusion principle, and correlation effects due to the instantaneous repulsion between electrons beyond the mean field approximation. Its exact form is unknown and must be approximated. Since the analytical form is unknown, practical calculations employ functionals which can be grouped in these main categories of approximation:⁵¹

- Local Density Approximation (LDA), which assumes that XC energy at each point depends only on the local electron density,
- Generalised Gradient Approximation (GGA), including the gradient of the local density to better account for inhomogeneity, and
- Hybrid functionals, essentially a correction to GGA, involving more accurate treatment of electron-electron repulsion.

These approximations affect *all* computed observables: energies, forces, vibrational properties, band gaps etc., as such the choice of exchange correlation functional is critical to the accuracy of our results.

Chapter 7

Quantum Simulations of Materials

In practical quantum mechanical calculations, the solution of the Schrödinger equation is obtained by expressing the electronic state functions in terms of a finite set of known functions, referred to as a *basis set*. This procedure transforms the continuous differential equation into a discrete algebraic problem that can be treated numerically. The accuracy, efficiency and computational cost of an SCF calculation are therefore strongly influenced by the choice of basis set. The completeness of the basis set controls how accurately the true solution of the Schrödinger equation can be represented. In the limit of a complete basis, the exact solution is recovered, while in practice one works with systematically improvable finite bases.

Several classes of basis sets are commonly used in atomistic simulations. Localized basis functions, such as *atomic orbitals* or *Gaussian-type* functions, are centered on atomic nuclei and are particularly efficient for describing isolated molecules or finite clusters; *plane wave* basis sets, by contrast, are delocalized functions that extend over the entire simulation domain and are especially convenient for periodic systems. Other representations, such as *real-space grids* or *wavelets*, provide alternative discretizations of the wavefunctions and are often employed in large-scale or linear-scaling approaches. The choice of basis set reflects a balance between physical transparency, numerical efficiency and systematic controllability. Localized basis sets can offer intuitive chemical insight and reduced computational cost for sparse systems, but may suffer from basis set superposition errors.^{52,53} Delocalized bases, such as plane waves, are free from such artifacts and allow straightforward convergence control, at the expense of requiring periodic boundary conditions and larger basis sizes.

When the system under investigation is periodic, as in crystalline solids or extended surfaces, translational symmetry can be exploited to simplify the problem. In this case, the most natural and convenient choice of basis functions is provided by *Bloch functions*. Bloch functions explicitly incorporate the periodicity of the lattice and form the foundation for modern electronic-structure methods applied to periodic materials. In the following sections, we will build upon this concept to describe how periodicity is usually treated in quantum mechanical simulations of solids.

7.1 Brillouin zone sampling

The key simplification which enables the numerical study (e.g. via computer simulations) of crystalline materials rely on the periodicity of the lattice. Recalling [section 1.6](#), a crystal is constructed from a basic repeating volume called the unit cell, whose edges are defined by the lattice vectors. These vectors connect abstract lattice points, to each of which we attach a basis of atoms; any physical quantity that depends on the atomic arrangement, like the electronic density or the interatomic potential, is therefore periodic in space. Since the potential is periodic, thanks to the Bloch theorem the one-electron orbitals in [Equation 5.38](#) can be written as Bloch functions

$$\phi_{n,\mathbf{k}}(\mathbf{r}) = u_{n,\mathbf{k}}(\mathbf{r}) \exp(i\mathbf{k} \cdot \mathbf{r}) \quad (7.1)$$

where $\mathbf{k}$ is a vector in the reciprocal space, n is the band index and $u_{n,\mathbf{k}}(\mathbf{r})$ is periodic with the lattice, that is

$$u_{n,\mathbf{k}}(\mathbf{r}) = u_{n,\mathbf{k}}(\mathbf{r} + \mathbf{T}) \quad (7.2)$$

with $\mathbf{T}$ being the translation vector. The periodic part $u_{n,\mathbf{k}}(\mathbf{r})$ may be expanded in a Fourier series of plane waves:

$$u_{n,\mathbf{k}}(\mathbf{r}) = \sum_{\mathbf{G}} C_{n,\mathbf{k}}(\mathbf{G}) \exp(i\mathbf{G} \cdot \mathbf{r}) \quad (7.3)$$

with $\mathbf{G}$ being a reciprocal translation vector. The expansion defines a plane wave basis which is formally infinite; however, in practice we truncate the basis by setting an *energy cutoff* E_{cut} and imposing

$$\frac{\hbar^2}{2m} |\mathbf{k} + \mathbf{G}|^2 < E_{\text{cut}}, \quad (7.4)$$

thus retaining only plane waves with kinetic energy lower than E_{cut} . Increase of spatial resolution requires increasing E_{cut} , which in turn increases the computational cost. Note that, as a choice of basis, plane waves are particularly convenient for periodic systems because they form an orthonormal basis with coefficients that can be systematically improved, and their computation can be handled efficiently using Fast Fourier Transform algorithms.⁵⁴

With the Bloch functions defined, the one-electron equations become

$$\hat{h}_{\text{eff}} \phi'_{n,\mathbf{k}}(\mathbf{r}) = \epsilon_{n,\mathbf{k}} \phi'_{n,\mathbf{k}}(\mathbf{r}) \quad (7.5)$$

where we have one equation for each electronic state $|n, \mathbf{k}\rangle$. We mark each one electron wavefunction $\phi'_{n,\mathbf{k}}$ with a prime symbol because they are *approximations* to the true one-electron wavefunctions due to truncation of the plane wave basis. Each eigenvalue $\epsilon_{n,\mathbf{k}}$ is the energy associated to the wave vector $\mathbf{k}$ and labeled with the integer index n which is said to be the *band* index. The approximate expectation value of a physical quantity is then obtained by summing over all occupied bands N_{occ} and integrating in reciprocal space (see [Equation 5.10](#)):

$$\langle \hat{A} \rangle_{\phi_n} = \sum_{n=1}^{N_{\text{occ}}} \int \left(\int \phi'^*_{n,\mathbf{k}}(\mathbf{r}) \hat{A} \phi'_{n,\mathbf{k}}(\mathbf{r}) d\mathbf{r} \right) d\mathbf{k}. \quad (7.6)$$

The advantage of the periodic description is that any wave vector $\mathbf{k}^*$ outside the first BZ is equivalent to a wave vector $\mathbf{k}$ inside the first BZ by a $\mathbf{G}$ translation: $\mathbf{k}^* = \mathbf{k} + \mathbf{G}$. Therefore, sampling the first BZ is sufficient to obtain the expectation value of any observable. In

practice, the BZ is never sampled in full: integrals over $\mathbf{k}$ are replaced by discrete sums over a finite set of $\mathbf{k}$ -points; in this way, Equation 7.6 becomes

$$\langle \hat{A} \rangle_{\phi_n} = \sum_{n=1}^{N_{occ}} \sum_{\mathbf{k} \in \text{BZ}} \int_V \phi_{n,\mathbf{k}}'^*(\mathbf{r}) \hat{A} \phi_{n,\mathbf{k}}'(\mathbf{r}) d\mathbf{r} \quad (7.7)$$

where NKPT is the number of sampled $\mathbf{k}$ -points in the first BZ and V is the volume of the unit cell. In general we can include $M > N_{occ}$ bands to represent both occupied and unoccupied energy levels, e.g. to capture conduction properties. The number of required $\mathbf{k}$ -points strongly depends on the material. In insulators and semiconductors, the electron occupancy changes smoothly with $\mathbf{k}$, so relatively coarse grids converge well; on the other hand, metals require much denser sampling because the occupation close to the Fermi level varies rapidly with $\mathbf{k}$. This difference has a direct impact on computational cost, since each additional $\mathbf{k}$ -point requires solving a separate KS-equation.

The set of $\mathbf{k}$ -points to perform integrals over the BZ is usually constructed by considering a uniform sampling based on the *Monkhorst-Pack* scheme;⁵⁵ this sampling is suitable for SCF calculations and properties derived from integrals like total energy, forces and electronic density. However, it is important to stress that the choice of the $\mathbf{k}$ -points set is arbitrary and what counts is the size of the set: in general, the larger the number of points in the set, the higher the numerical precision of the calculated quantity.

7.2 Electronic band structure

With the introduction of Bloch functions in Equation 7.1, each state $|n, \mathbf{k}\rangle$ is labelled by a band index n and a wave vector $\mathbf{k}$. Solving the one-electron equation at a given $\mathbf{k}$ yields a set of eigenvalues $\{\epsilon_{n,\mathbf{k}}\}$ which describe the allowed electronic energies for that wave vector. At fixed band index n , the variation of $\epsilon_{n,\mathbf{k}}$ with respect to $\mathbf{k}$ defines an *energy band*. The diagram showing all the eigenvalues $\{\epsilon_{n,\mathbf{k}}\}$ for a specified number of bands and set of $\mathbf{k}$ -points is called *electronic band structure*, difference examples of which are shown in Figure 7.1. Groups of energy bands may be separated by forbidden energy ranges for which the Schrödinger equation does not admit solutions; each of such forbidden ranges is called *energy gap*. A group of energy bands enclosed between two energy gaps is also called a *band*; although this might generate confusion, the use of the word *band* is usually clear from the context. Since the exact electronic spectrum contains infinitely many bands, it also contains infinitely many band gaps. In practice, however, numerical calculations consider only a finite number of bands (specified as an input of the simulation), therefore the number of calculated gaps is finite.

If the spin is not taken into account, each band represents at most two electrons at any one time. The occupation is determined by the Fermi-Dirac distribution

$$f(\epsilon_{n,\mathbf{k}}) = \frac{1}{1 + \exp \left[\frac{\epsilon_{n,\mathbf{k}} - \mu}{k_B T} \right]}, \quad (7.8)$$

where T is the temperature of the system, k_B is the Boltzmann constant and μ is the chemical potential. For a system of electrons, as it is the present case, the chemical potential is called *Fermi level*, usually indicated as E_F . At $T = 0$ K, the Fermi level is equal to the *Fermi energy* and only all states with $\epsilon_{n,\mathbf{k}} < E_F$ are fully occupied. The position of E_F relative to the band edges determines the qualitative behaviour of the material. If E_F intersects a band then the system is a conductor; if E_F lies within a gap, then the system is either a semiconductor or an insulator depending on the size of the gap. If the valence

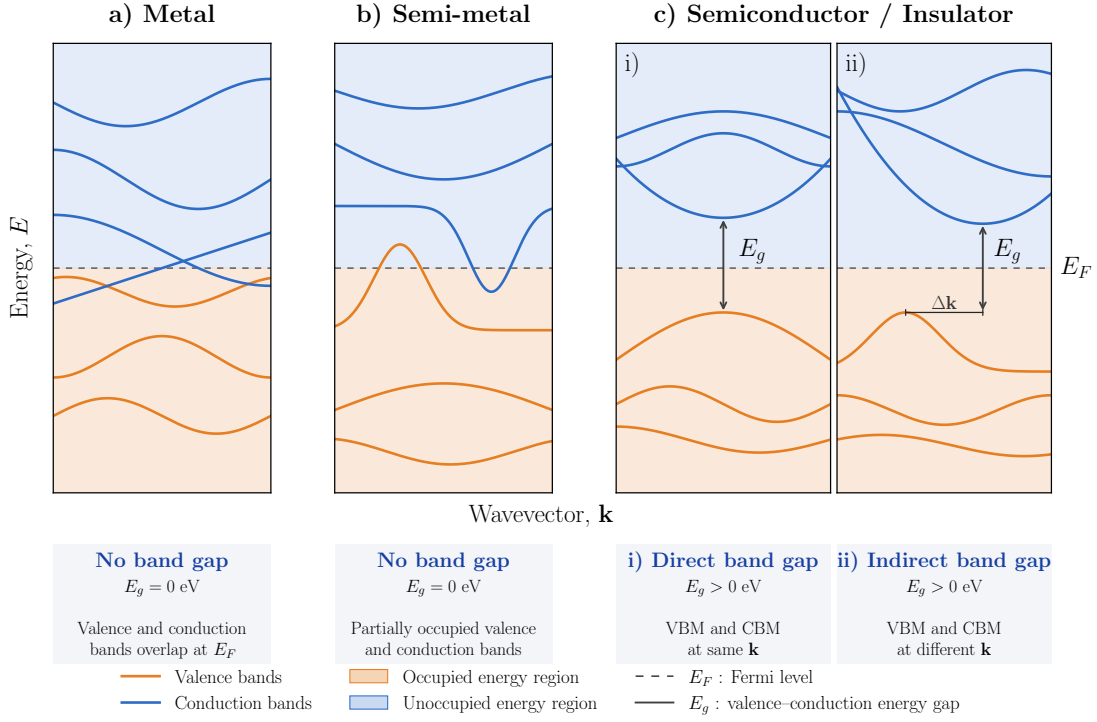


Figure 7.1: Band structure schematic, displaying typical energy-momentum dispersion relations close to the Fermi level E_F for systems with different electronic character. a) Metallic systems exhibit bands which cross the E_F , resulting in partially occupied states and no band gap. b) Semi-metals possess small occupied and unoccupied pockets close to E_F , arising from partial occupation of both valence and conduction bands. c) The valence-conduction energy gap E_g , which straddles E_F , can be either *direct* or *indirect*; the distinction between semiconductor and insulator is based on the size of the band gap.

band maximum (VBM) and the conduction band minimum (CBM) occur at the same $\mathbf{k}$ -point, the band gap is said to be *direct*, otherwise it said to be *indirect* (Figure 7.1c-d). Band structures are usually calculated on *high-symmetry paths*, that are piece-wise linear paths through the BZ joining high-symmetry points. Since these paths are a nothing else than a set of $\mathbf{k}$ -points, they can be used to perform SCF calculations or any integration in the BZ as in Equation 7.7; however, in this way, the BZ is strongly undersampled compared to the use of Monkhorst-Pack grids, resulting in a bad evaluation of quantities like forces or energies. The choice of the path for a band structure calculation is arbitrary; convention reported in Ref. 56 is usually employed.

Important electronic properties, such as VBM, CBM and hence the (valence-conduction) band gap about the Fermi level, tend to occur at high-symmetry points in the BZ; for this reason, high-symmetry paths are usually employed to determine the band gap of a material. However, for a correct determination of the band gap size, very dense sampling should be used, including both high-symmetry and unique $\mathbf{k}$ -points. If the chosen $\mathbf{k}$ -point path or grid does not include the correct extremum, the size and nature of the predicted band gap will be incorrect. Therefore, after an SCF calculation is converged, it is good practice to:

- refine the $\mathbf{k}$ -point sampling and check the stability of the VBM/CBM positions,
- compare a uniform Monkhorst-Pack sampling with a high-symmetry band-structure

path, and

- evaluate how the apparent (lack of a) band gap changes with increasing sampling density.

If the system shows a band gap, by increasing the number of $\mathbf{k}$ -points in the sampling mesh, the calculated value of the gap may remain constant or decrease but can never increase, as each $\mathbf{k}$ -point contributes with its own energy levels to the electronic band structure independently from any other $\mathbf{k}$ -point. This means that, when comparing band gap values at varying sampling mesh size or methods (e.g. Monkhorst-Pack, high-symmetry paths or else), the best estimation of the gap size is the smallest obtained value.

7.3 Reciprocal-space sampling considerations

In the following paragraphs, we analyze how reciprocal-space symmetry and real-space periodicity determine the electronic structure representation in practical calculations.

Band folding. Band folding is a direct consequence of the choice of the real-space unit cell used to describe a periodic system. When the size of the real-space cell is increased, for instance by constructing a supercell, the corresponding reciprocal lattice shrinks. As a result, electronic bands that were previously distinct in the larger Brillouin zone become “folded” into the smaller Brillouin zone associated with the larger cell.

To understand this, let us consider primitive lattice vectors $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$, with reciprocal lattice vectors $\{\mathbf{a}^*, \mathbf{b}^*, \mathbf{c}^*\}$ defined as in [Equation 1.5](#):

$$\mathbf{a}^* = 2\pi \frac{\mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} \quad \mathbf{b}^* = 2\pi \frac{\mathbf{c} \times \mathbf{a}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} \quad \mathbf{c}^* = 2\pi \frac{\mathbf{a} \times \mathbf{b}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}. \quad (7.9)$$

If we construct a supercell by scaling one lattice vector, for example

$$\mathbf{a}' = N\mathbf{a}, \quad (7.10)$$

the corresponding reciprocal vector becomes

$$\mathbf{a}'^* = \frac{1}{N}\mathbf{a}^*; \quad (7.11)$$

thus, the Brillouin zone shrinks by a factor $1/N$ along that direction. Electronic states that were located at different $\mathbf{k}$ -points in the primitive Brillouin zone may now map onto the same $\mathbf{k}$ -point in the reduced zone. This mapping is what is referred to as *band folding*.

To understand how band folding works in practice, we will refer to [Figure 7.2](#) and consider a point $\mathbf{k}_1$ with components $(k_1, 0, 0)$, where the subscript “1” indicates that it is a vector defined in the reciprocal space of the primitive unit cell. In order to understand how the k_1 coordinate is mapped due to the folding effect, let us consider, for example, a $2 \times 1 \times 1$ supercell replica of the primitive unit cell. Let us call $\mathbf{a}_1^*$ the reciprocal vector corresponding to the $\mathbf{a}_1$ direct lattice vector defining the primitive system; the reciprocal vector $\mathbf{a}_2^*$ of the $2 \times 1 \times 1$ system is then equal to $1/2\mathbf{a}_1^*$ ([Equation 7.11](#) with $N = 2$). For this reason, and thanks to the periodicity of the reciprocal lattice, the border of the Brillouin zone in the primitive system, corresponding to the point $1/2\mathbf{a}_1^*$, is mapped onto the origin of the reciprocal lattice of the doubled system, this origin being the point $\mathbf{a}_2^*$. In other words, we have the mapping $k_1 = 1/2 \rightarrow k_2 = 1$, that is, $\mathbf{k}_1 = (1/2, 0, 0)$ is mapped onto $\mathbf{k}_2 = (1, 0, 0)$, which is the Γ point of the $2 \times 1 \times 1$ system. This means that all the

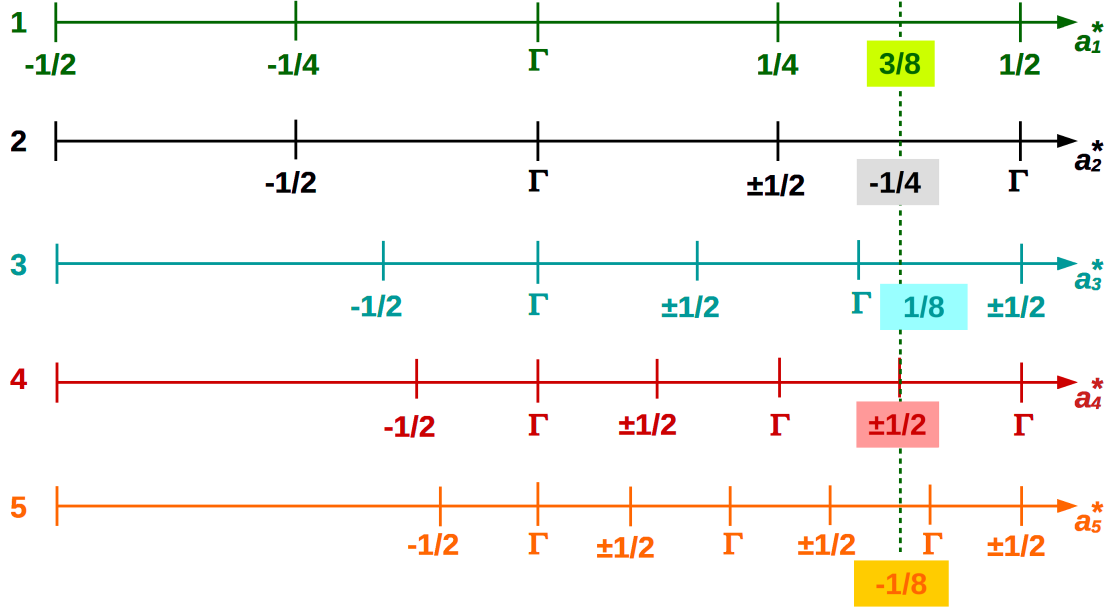


Figure 7.2: Schematic of the Brillouin zone folding. The supercell multiplicity is indicated by the number on the left side of each scheme: for example, “1” refers to the primitive unit cell while “2” refers to the $2 \times 1 \times 1$ system.

eigenvalues $\{\epsilon_{n,\mathbf{k}}\}$ of Equation 7.5 which occur at $\mathbf{k} = (1/2, 0, 0)$ in the primitive system will instead occur at $\mathbf{k} = (1, 0, 0) \equiv \Gamma$ in the $2 \times 1 \times 1$ system. Analogously, for example, the point $1/4 a_1^*$ in the primitive system corresponds to the point $\pm 1/2 a_2^*$ ($k_1 = 1/4 \rightarrow k_2 = \pm 1/2$), where the plus or minus sign applies if $1/2 a_2^*$ is approached from left or right, respectively.

As further example, we can consider a point $\mathbf{k}_1 = (3/8, 0, 0) = 3/8 a_1^*$ and calculate how it is mapped for supercells at increasing size, from example from $2 \times 1 \times 1$ up to $5 \times 1 \times 1$. We have:

$$\begin{aligned} a_2^* &= \frac{1}{2} a_1^*, & a_3^* &= \frac{1}{3} a_1^*, & a_4^* &= \frac{1}{4} a_1^*, & a_5^* &= \frac{1}{5} a_1^*; \\ a_1^* &= 2a_2^*, & a_1^* &= 3a_3^*, & a_1^* &= 4a_4^*, & a_1^* &= 5a_5^*; \end{aligned} \quad (7.12)$$

by substituting a_1^* in the definition of $\mathbf{k}_1$ we get

$$\begin{aligned} \mathbf{k}_1 &= \frac{3}{8} a_1^* \\ &= \frac{3}{8} 2a_2^* = \frac{3}{4} a_2^* = \left(\frac{1}{2} + \frac{1}{4} \right) a_2^* \equiv -\frac{1}{4} a_2^* = \mathbf{k}_2 \\ &= \frac{3}{8} 3a_3^* = \frac{9}{8} a_3^* = \left(1 + \frac{1}{8} \right) a_3^* \equiv \frac{1}{8} a_3^* = \mathbf{k}_3 \\ &= \frac{3}{8} 4a_4^* = \frac{3}{2} a_4^* = \left(1 + \frac{1}{2} \right) a_4^* \equiv \pm \frac{1}{2} a_4^* = \mathbf{k}_4 \\ &= \frac{3}{8} 5a_5^* = \frac{15}{8} a_5^* = \left(1 + \frac{1}{2} + \frac{3}{8} \right) a_5^* \equiv -\frac{1}{8} a_5^* = \mathbf{k}_5 \end{aligned} \quad (7.13)$$

This result is generalised as follows. A generic reciprocal-space vector $\mathbf{k}$ can be written in reduced coordinates as

$$\mathbf{k} = k_a \mathbf{a}^* + k_b \mathbf{b}^* + k_c \mathbf{c}^*, \quad (7.14)$$

with $k_a, k_b, k_c \in [-1/2, 1/2)$. Now consider a supercell constructed by repeating the primitive cell $m \times n \times l$ times along the $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ directions, respectively. The direct lattice vectors

of the supercell are

$$\mathbf{a}' = m\mathbf{a}, \quad \mathbf{b}' = n\mathbf{b}, \quad \mathbf{c}' = l\mathbf{c}. \quad (7.15)$$

The corresponding reciprocal lattice vectors of the supercell are therefore

$$\mathbf{a}^{*'} = \frac{1}{m}\mathbf{a}^*, \quad \mathbf{b}^{*'} = \frac{1}{n}\mathbf{b}^*, \quad \mathbf{c}^{*'} = \frac{1}{l}\mathbf{c}^*. \quad (7.16)$$

Expressing the same reciprocal-space vector $\mathbf{k}$ in terms of the supercell reciprocal basis,

$$\mathbf{k} = k'_a \mathbf{a}^{*'} + k'_b \mathbf{b}^{*'} + k'_c \mathbf{c}^{*'}, \quad (7.17)$$

and substituting the relations between primitive and supercell reciprocal vectors, one immediately obtains

$$k'_a = m k_a, \quad k'_b = n k_b, \quad k'_c = l k_c. \quad (7.18)$$

Thus, when moving from a primitive cell to an $m \times n \times l$ supercell, the reduced components of a $\mathbf{k}$ -point are scaled by the corresponding supercell replication factors. As a consequence, multiple $\mathbf{k}$ -points of the primitive Brillouin zone may map onto the same reduced $\mathbf{k}$ -point of the supercell Brillouin zone, which is the mathematical origin of band folding.

Equivalence of $+1/2$ and $-1/2$ reduced coordinates. Recalling definition 7.14 of a generic $\mathbf{k}$ vector, we can see that the reduced coordinates $+1/2$ and $-1/2$ along a given reciprocal direction are equivalent because they differ by a reciprocal lattice vector. For example,

$$\left(+\frac{1}{2}\right) \mathbf{a}^* = \left(-\frac{1}{2}\right) \mathbf{a}^* + \mathbf{a}^* \quad (7.19)$$

where, in this case, the translational vector in the reciprocal space is $\mathbf{G} = 1\mathbf{a}^*$. For this reason, Brillouin zones are conventionally defined as half-open intervals $[-1/2, 1/2)$, that is, to avoid double counting of boundary points.

Equivalence of $+\mathbf{k}$ and $-\mathbf{k}$. In systems that preserve time-reversal symmetry and in the absence of spin-orbit coupling, the Hamiltonian satisfies

$$\hat{H}(\mathbf{k}) = \hat{H}^*(-\mathbf{k}), \quad (7.20)$$

which implies that

$$E_n(\mathbf{k}) = E_n(-\mathbf{k}). \quad (7.21)$$

Therefore, electronic bands are symmetric with respect to inversion of $\mathbf{k}$. In such cases, sampling only half of the Brillouin zone is sufficient, since the other half can be generated by symmetry. However, when spin-orbit coupling is included, the simple relation between $+\mathbf{k}$ and $-\mathbf{k}$ may no longer hold in the same way, especially in systems lacking inversion symmetry, and both points must be sampled.

Relation between k -point components and real-space replicas. The components of a k -point are directly related to the periodic boundary conditions imposed on the wavefunction. A Bloch phase factor

$$e^{i\mathbf{k}\cdot\mathbf{r}} \quad (7.22)$$

implies that translating the system by a lattice vector $\mathbf{R}$ introduces a phase

$$\psi_{n\mathbf{k}}(\mathbf{r} + \mathbf{R}) = e^{i\mathbf{k}\cdot\mathbf{R}} \psi_{n\mathbf{k}}(\mathbf{r}). \quad (7.23)$$

Thus, the allowed k -points are determined by the size of the simulation cell. For a finite supercell containing m , n , l repetitions along the directions $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, the spacing between allowed $\mathbf{k}$ -points along the corresponding reciprocal vectors is proportional to $1/m$, $1/n$, $1/l$, respectively (see discussion on band folding above). Increasing the real-space cell size therefore reduces the required $\mathbf{k}$ -point sampling density in reciprocal space.

Brillouin zone sampling for finite or partially periodic systems. In practice, periodic boundary conditions are often applied even to systems that are physically finite, such as molecules, nanowires, or slabs. To avoid artificial interactions between periodic images, large vacuum regions are introduced along the non-periodic directions; in this way, the system is effectively non-periodic and any sampling which represents replicas along such directions is not necessary. Therefore, one sets the number of $\mathbf{k}$ -points to sample equal to 1 along the non-periodic direction. In practice:

- a bulk crystal requires dense $\mathbf{k}$ -point sampling in all three directions,
- a surface (periodic in two directions) requires dense sampling in-plane, but 1 $\mathbf{k}$ -point along the non-periodic direction,
- a nanowire (periodic in one direction) requires sampling only along the wire axis, and
- an isolated molecule requires a Γ -point calculation.

Using more than one $\mathbf{k}$ -point along a non-periodic direction is not wrong, as in the Bloch theorem there is no restriction about the presence of vacuum in the system; however, since replicas along non-periodic directions are not interacting, solutions of Equation 7.5 are identical at any $\mathbf{k}$ -point representing those replicas. This property can be used to check if two images are far enough to be considered non-interacting: if the band dispersion is flat along a $\mathbf{k}$ -path made by points representing replicas along a non-periodic direction, then the distance between images is large enough to prevent interactions, and that direction can be effectively considered non-periodic.

7.4 Electronic Density of States

The electronic density of states $D(E)$ (DOS) is a fundamental quantity that describes how electronic energy levels are distributed in energy. More precisely, it measures the number of available electronic states per unit energy interval at a given energy E . The DOS plays a central role in determining many physical properties of materials, including electronic transport, optical response, and thermodynamic behaviour.

A rigorous and formal definition of the density of states can be found in standard solid-state physics textbooks, such as *Introduction to Solid State Physics* by Charles Kittel.³¹ Here, we adopt an operative definition that is particularly convenient in the context of electronic-structure calculations. For a periodic system described by energy bands $\epsilon_{n,\mathbf{k}}$, where n is the band index and $\mathbf{k}$ is a wave vector in the Brillouin zone, the DOS is defined as

$$D(E) = \sum_n \int_{\text{BZ}} \delta(E - \epsilon_{n,\mathbf{k}}) d\mathbf{k}. \quad (7.24)$$

This expression counts all electronic states whose energy equals E , summing over all bands and integrating over the Brillouin zone. The Dirac delta function ensures that only states with energy $\epsilon_{n,\mathbf{k}} = E$ contribute to the DOS. In practical calculations, the integral over the Brillouin zone is replaced by a finite sum over a discrete set of $\mathbf{k}$ -points, and the delta

function is approximated by a smearing or broadening function. Numerically, the DOS is therefore obtained by collecting the eigenvalues $\epsilon_{n,\mathbf{k}}$ from a suitably dense $\mathbf{k}$ -point grid and binning them into energy intervals. The choice of smearing method and width affects the smoothness and resolution of the computed DOS and must be chosen consistently with the $\mathbf{k}$ -point sampling.

The density of states provides direct insight into key features of the electronic structure. Band edges appear as the onset of nonzero DOS and are crucial for identifying band gaps in insulators and semiconductors. Van Hove singularities manifest as peaks or divergences in the DOS and originate from critical points in the band dispersion where the gradient $\nabla_{\mathbf{k}}\epsilon_{n,\mathbf{k}}$ vanishes. The overall width and shape of the DOS reflect the dispersion and overlap of electronic bands.

Beyond the total DOS, it is often useful to decompose the electronic states into contributions associated with specific atoms or orbitals. Two commonly used quantities are:

- *atom-projected density of states* (PDOS), where the contribution of each atom or group of atoms to the DOS is evaluated, and
- *orbital-resolved density of states*, where the DOS is further decomposed into atomic-like orbitals, such as *s*, *p*, or *d* spherical harmonics.

These projections provide valuable chemical insight, allowing one to identify which atoms or orbitals dominate particular energy regions, such as valence bands, conduction bands, or localized states. It is important to note, however, that such projections are not unique and depend on the chosen projection scheme and basis. In practice, the number of functions used in the projection is finite; as a consequence, part of the information contained in the full wavefunction is discarded. In terms of the DOS, the sum of all projected DOS components does not necessarily reconstruct the total DOS exactly. This limitation must be kept in mind when interpreting projected or orbital-resolved DOS plots.

An analogous decomposition can also be performed for the band structure itself. Atom-projected and/or orbital-resolved band structures allow one to visualize how the chemical character of electronic states evolves with $\mathbf{k}$, providing a complementary perspective to the DOS. Such analyses are particularly useful for identifying bonding and antibonding states, hybridization effects, and the origin of band dispersion.

The DOS is especially powerful for analyzing the effects of defects, dopants, or vacancies. The introduction of impurities or structural defects can give rise to localized or quasi-localized states within the band gap or near band edges. These states often appear as sharp peaks or additional features in the DOS, commonly referred to as *impurity bands*. Their presence can dramatically modify the electronic properties of a material, for example by enabling carrier doping or trap-assisted conduction.

The quantitative accuracy of the computed DOS, as well as of related quantities such as band structures, charge densities, and total energies, depends primarily on two numerical parameters. The first is the plane-wave energy cutoff E_{cut} , which controls the spatial resolution of the wavefunctions and the completeness of the basis set.^a The second is the $\mathbf{k}$ -point sampling density, which determines the resolution of integrations over the Brillouin zone. Increasing either parameter improves the accuracy of the calculated DOS but also increases its computational cost. For this reason, systematic convergence testing with

^aHere, we are assuming that the solution of the TISE is expanded in the basis set of Bloch functions; however, the discussion holds for any other kind of basis set.

respect to both E_{cut} and $\mathbf{k}$ -point density is an essential part of any reliable electronic-structure calculation. Only after convergence has been established can features observed in the DOS be interpreted as physically meaningful rather than as numerical artifacts.

7.5 Geometry optimisation in ab initio simulations

As we saw in [chapter 2](#), a geometry optimisation is an iterative procedure which aims to find a geometric configuration which realises a local minimum of the potential energy. This procedure then requires the evaluation of the forces on the atoms: the iterative loop is interrupted and convergence is said to be achieved when the forces on the atoms are below a specified threshold. In ab initio simulations, the evaluation of the forces can be performed only after convergence of the SCF procedure: only after the wavefunction of the system has been obtained, atomic forces can be calculated by means of the Hellmann–Feynman theorem.⁵⁷

To understand how a geometry optimisation is performed in practice, let us start by recalling the Born–Oppenheimer approximation ([subsection 5.5.1](#)): the nuclei are treated as classical point charges and their kinetic energy is neglected during the electronic structure calculation. In this approximation, the electronic Hamiltonian depends parametrically on the nuclear coordinates $\{\mathbf{R}_I\}$ through the electron–nucleus interaction potential. As a result, the total energy of the system for a given nuclear geometry is obtained by adding the classical nucleus–nucleus Coulomb repulsion to the electronic energy E_e . The nucleus–nucleus interaction energy is given by

$$E_{nn} = \frac{1}{2} \sum_{I \neq J} \frac{Z_I Z_J}{|\mathbf{R}_I - \mathbf{R}_J|}, \quad (7.25)$$

where Z_I and Z_J are the charges of nuclei I and J , respectively. The approximate total energy E_{tot} corresponding to a given nuclear configuration is therefore

$$E_{\text{tot}}(\{\mathbf{R}_I\}) = E_e(\{\mathbf{R}_I\}) + E_{nn}(\{\mathbf{R}_I\}). \quad (7.26)$$

This expression is completely general and applies to Hartree–Fock, density-based methods, and post-Hartree–Fock approaches alike, as long as the BO approximation is adopted. A natural question then arises: is E_{tot} the best possible estimate of the ground-state total energy of the system? The answer is that this is only true for the specific nuclear geometry at which the electronic problem was solved. Any change in the nuclear positions modifies the electron–nucleus interaction, alters the electronic Hamiltonian, leads to a different converged electronic state, and therefore changes E_e . At the same time, the nucleus–nucleus repulsion energy E_{nn} is also modified. For this reason, the total energy $E_{\text{tot}}(\{\mathbf{R}_I\})$ can be viewed as defining a *potential energy surface* (PES) over the $3N_{\text{nuc}}$ -dimensional space of nuclear coordinates, where N_{nuc} is the number of nuclei. Each point on this surface corresponds to the total energy of the system for a particular arrangement of the nuclei.

In the practice, the geometry optimisation in ab initio simulations can be summarised as follows.

- 1) A trial nuclear geometry is chosen, either from experimental information, chemical intuition, or a previously optimised structure. For this geometry, the electronic structure problem is solved self-consistently to obtain the converged electronic energy E_e and the corresponding total energy E_{tot} .
- 2) The forces acting on the nuclei are computed. Within the Born–Oppenheimer approximation, the force acting on nucleus I is defined as the negative derivative of the total

energy with respect to its nuclear coordinate,

$$\mathbf{F}_I = -\frac{\partial E_{\text{tot}}}{\partial \mathbf{R}_I}. \quad (7.27)$$

Using the Hellmann–Feynman theorem, this derivative can be expressed explicitly in terms of the electronic wavefunction. If $\Psi(\mathbf{r}; \{\mathbf{R}_I\})$ denotes the many-electron wavefunction obtained from the converged SCF solution and $\hat{H}_e(\{\mathbf{R}_I\})$ is the electronic Hamiltonian, the electronic contribution to the force reads

$$\mathbf{F}_I^{(e)} = -\int \Psi^*(\mathbf{r}) \frac{\partial \hat{H}_e}{\partial \mathbf{R}_I} \Psi(\mathbf{r}) d\mathbf{r}. \quad (7.28)$$

This expression shows that the force can be evaluated directly as the expectation value of the gradient of the Hamiltonian with respect to the nuclear coordinate, without requiring explicit derivatives of the wavefunction itself. Adding the classical nucleus–nucleus contribution,

$$\mathbf{F}_I^{(nn)} = -\frac{\partial E_{nn}}{\partial \mathbf{R}_I}, \quad (7.29)$$

the total force acting on nucleus I becomes

$$\mathbf{F}_I = -\int \Psi^*(\mathbf{r}) \frac{\partial \hat{H}_e}{\partial \mathbf{R}_I} \Psi(\mathbf{r}) d\mathbf{r} - \frac{\partial E_{nn}}{\partial \mathbf{R}_I}. \quad (7.30)$$

3) Nuclear positions are updated using the computed forces. In principle, the optimal update direction is determined by the Hessian matrix of the PES, defined as the second derivative of the total energy with respect to the nuclear coordinates,

$$H_{I,J}^{\alpha\beta} = \frac{\partial^2 E_T}{\partial R_I^\alpha \partial R_J^\beta}, \quad (7.31)$$

where α and β label Cartesian components. Because explicit evaluation of the Hessian is computationally expensive, practical geometry optimisation algorithms typically rely on quasi-Newton methods, which build an approximate Hessian iteratively from information gathered during previous optimisation steps.

The procedure of solving the electronic problem, computing forces, and updating the nuclear positions is repeated until convergence is achieved. Convergence is usually defined by requiring that the magnitude of the force on each nucleus falls below a predefined tolerance and that changes in the total energy and atomic positions are sufficiently small. The final result of this iterative process is an optimised nuclear geometry corresponding to a local minimum of the potential energy surface. This geometry usually serves as a reference structure for further analysis, such as vibrational calculations, molecular dynamics simulations, or comparisons between different isomers, phases, or reaction intermediates.

Chapter 8

Phonons

Phonons are quantized vibrational motions of a system of atoms oscillating about their equilibrium position. The study of phonons is essential for understanding a broad range of physical materials properties such as internal energy, free energy, entropy, heat transport and heat capacity. In addition, phonons play a key role in electron–phonon coupling, influencing electrical resistivity, superconductivity, and carrier mobility in semiconductors and metals.

From a practical point of view, phonon calculations provide a powerful link between atomistic simulations and experiments. Vibrational spectra obtained from phonon calculations can be directly compared with infrared, Raman, and inelastic neutron or X-ray scattering measurements. The study of the phonon properties may reveal dynamical instabilities of an atomic structure, offering a rigorous criterion for assessing whether a given geometry corresponds to a mechanically stable phase. This makes phonon analysis an indispensable tool for studying phase stability and structural transformations. The effectiveness of the phonon description is also evident from a theoretical perspective: it allows to transform the coupled Newton equations of motions into a linear eigenvalue problem whose solution yields the atomic motion in terms of vibrational frequencies and displacements. In what follows, we will show the essential steps to develop the phonon theory and how to use it for the numerical calculation of physical quantities.

8.1 The harmonic approximation

The phonon theory is general and can be applied to any kind of system, regardless of whether it is periodic or no, i.e. crystal (3D), surface (2D), atomic chain (1D) or cluster/molecule (0D). We will develop the discussion by considering the most general case of 3D systems; we will see that structures with lower dimensionality can be described with the same formalism by a proper selection of vectors in the reciprocal space (see [section 7.3](#), paragraph “Brillouin zone sampling for finite or partially periodic systems.”).

For a general 3D bulk crystal, the system is defined by the periodic repetition of a unit cell. The equilibrium position vector $\mathbf{r}_{kl}^0$ of an atom k in the replica l is defined as

$$\mathbf{r}_{kl}^0 = \mathbf{r}_k^0 + \mathbf{r}_l, \quad (8.1)$$

where $\mathbf{r}_k^0$ is the position of the atom with respect to the origin of the unit cell and $\mathbf{r}_l$ is the position of the origin of the cell replica l (see [Figure 8.1](#) for reference). The full set of position vectors $\{\mathbf{r}_{kl}\}$ is therefore a $3N \times M$ -dimensional object, where N is the number of

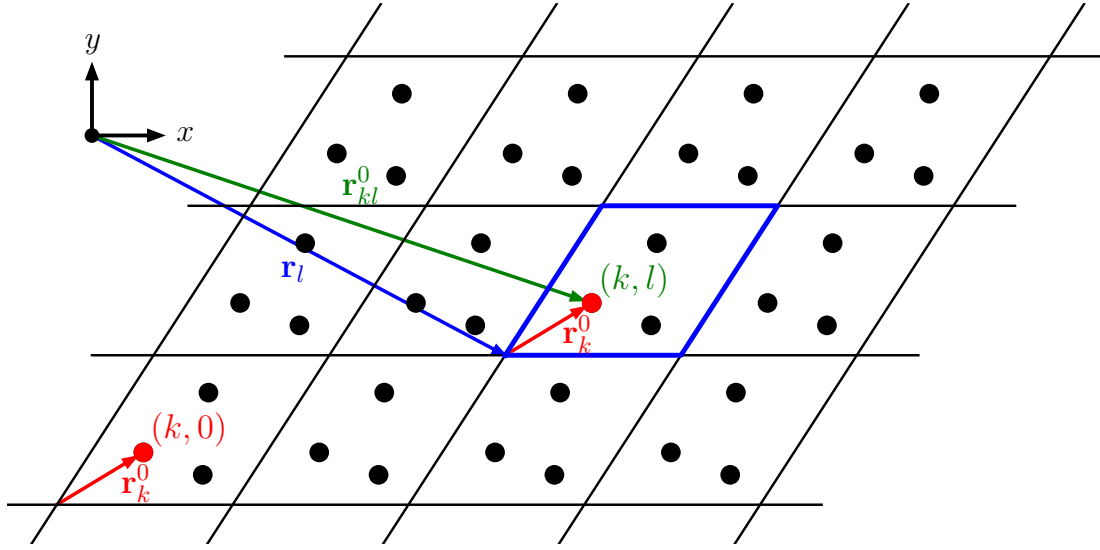


Figure 8.1: 2D schematic showing the replication of a unit cell in the (x, y) plane. The borders of each cell is drawn using black lines, and the basis (represented by black filled circles) is reproduced identically in each replica. The red and green filled circles highlight atom k in replica 0 (the reference unit cell) and replica l , labelled $(k, 0)$ and (k, l) accordingly, and the particular replica cell is highlighted in blue. $\mathbf{r}_k^0$ is the position of atom k with respect to the origin of the unit cell ($\mathbf{r}_k^0$ is independent of the specific cell replica we select), $\mathbf{r}_l$ is the position of the replica l with respect to the origin of the Cartesian reference frame, and $\mathbf{r}_{kl}^0$ is the vector position of atom k in replica l with respect to the origin of the Cartesian reference frame.

atoms per unit cell and M is the number of replicas. At time t , the instantaneous atomic position is

$$\mathbf{r}_{kl}(t) = \mathbf{r}_k(t) + \mathbf{r}_l; \quad (8.2)$$

accordingly, we can define

$$\mathbf{u}_{kl}(t) = \mathbf{r}_{kl}(t) - \mathbf{r}_{kl}^0 \quad (8.3)$$

as the displacement of the atom with respect to the equilibrium position. Let us consider the Taylor series expansion of the potential energy about the equilibrium positions:

$$V(\{\mathbf{u}_{kl}\}) = V_0 + \sum_{kl} \left. \frac{\partial V}{\partial \mathbf{u}_{kl}} \right|_0 \mathbf{u}_{kl}(t) + \frac{1}{2} \sum_{kl, k'l'} \left. \frac{\partial^2 V}{\partial \mathbf{u}_{kl} \partial \mathbf{u}_{k'l'}} \right|_0 \mathbf{u}_{kl}(t) \mathbf{u}_{k'l'}(t) + \dots, \quad (8.4)$$

where each displacement vector contains three Cartesian components. At the equilibrium, the first-order derivatives vanish; if we set the reference energy value $V_0 = 0$, the expansion truncated at the second order in matrix notation reads

$$V_2 = \frac{1}{2} \sum_{kl, k'l'} \mathbf{u}_{kl}^T(t) \mathbb{V}_{kl, k'l'} \mathbf{u}_{k'l'}(t), \quad \mathbb{V}_{kl, k'l'} = \left. \frac{\partial^2 V}{\partial \mathbf{u}_{kl} \partial \mathbf{u}_{k'l'}} \right|_0 \quad (8.5)$$

which is called the *harmonic approximation* of the potential; here, $\mathbb{V}_{kl, k'l'}$ is a 3×3 matrix containing the second order derivatives of the potential with respect to the atom displacements, which are called the *second order interatomic force constants* (IFC). Up to this point no assumption has been made about the nature of the potential; the only requirement

is that the potential energy surface is smooth near equilibrium. The framework therefore applies equally to empirical interatomic potentials, quantum mechanical descriptions or any other method providing atomic forces. Inserting the harmonic potential in Equation 8.5 into the classical equations of motion yields

$$m_{kl}\ddot{\mathbf{u}}_{kl}(t) = - \sum_{k'l'} \mathbb{V}_{kl,k'l'} \mathbf{u}_{k'l'}, \quad (8.6)$$

giving a system of $3N \times M$ coupled linear differential equations. In other words, we are making a model in which atoms interact by means of ideal harmonic springs; in fact, Equation 8.6 is the multidimensional form of Hook's law usually known simply as $F = -kx$, where k is the force constant of the spring and x its elongation or compression. In this picture, each IFC defined in Equation 8.5 is the force constant k of the corresponding ideal spring. For a periodic crystal, it is natural to seek plane-wave solutions of Equation 8.6 of the form

$$\mathbf{u}_{kl}(t) = \frac{1}{\sqrt{Mm_{kl}}} Q \mathbf{e}_k(\mathbf{q}) \exp(i\mathbf{q} \cdot \mathbf{r}_{kl}^0) \exp(-i\omega_{\mathbf{q}} t), \quad (8.7)$$

where the vector $\mathbf{q}$ lies in the first Brillouin zone, $\mathbf{e}_k(\mathbf{q})$ is a vector associated to the atom k , Q is a complex amplitude and $\omega_{\mathbf{q}}$ is an angular frequency. Substituting Equation 8.7 into Equation 8.6 gives

$$\omega^2(\mathbf{q}) \mathbf{e}_k(\mathbf{q}) = \sum_{k'} \sqrt{\frac{1}{m_k m_{k'}}} \left[\sum_{l'} \mathbb{V}_{kl,k'l'} \exp(i\mathbf{q} \cdot [\mathbf{r}_{kl}^0 - \mathbf{r}_{k'l'}^0]) \right] \mathbf{e}_{k'}(\mathbf{q}), \quad (8.8)$$

where the sum over replicas l' is the Fourier transform of the second order IFCs; for each k, k' couple of atoms, the Fourier elements can be collected into a matrix $\mathbb{F}_{kk'}$

$$\mathbb{F}_{kk'}(\mathbf{q}) = \sum_{l'} \mathbb{V}_{kl,k'l'} \exp(i\mathbf{q} \cdot [\mathbf{r}_{kl}^0 - \mathbf{r}_{k'l'}^0]). \quad (8.9)$$

With this definition, Equation 8.8 can be rewritten as

$$\omega^2(\mathbf{q}) \mathbf{e}_k(\mathbf{q}) = \sum_{k'} \sqrt{\frac{1}{m_k m_{k'}}} \mathbb{F}_{kk'}(\mathbf{q}) \mathbf{e}_{k'}(\mathbf{q}) \quad (8.10)$$

or, in compact form,

$$\omega^2(\mathbf{q}, j) \mathbf{e}(\mathbf{q}, j) = \mathbb{D}(\mathbf{q}) \mathbf{e}(\mathbf{q}, j), \quad (8.11)$$

where $\mathbb{D}(\mathbf{q}) \in \mathbb{R}^{3N \times 3N}$ is called *dynamical matrix*. Equation 8.11 is an eigenvalue-eigenvector equation for the dynamical matrix. For each wave vector $\mathbf{q}$, diagonalising $\mathbb{D}(\mathbf{q})$ yields $3N$ eigenvalues $\omega^2(\mathbf{q}, j)$ and corresponding eigenvectors $\mathbf{e}(\mathbf{q}, j)$, labeled by the wave vector $\mathbf{q}$ and the index $j = 1, \dots, 3N$. Each eigenvector $\mathbf{e}(\mathbf{q}, j)$ is a $3N \times 1$ matrix whose components specify the displacement pattern of all atoms in the unit cell, while the square root of the corresponding eigenvalue is the frequency of oscillation of the atom along that displacement pattern. Thus, every pair $(\mathbf{q}, j)$ uniquely defines a *phonon mode*, also called *normal* or *vibrational* mode: a collective harmonic motion of atoms which displace along the directions specified by $\mathbf{e}(\mathbf{q}, j)$, in a synchronous way, and at frequency $\omega(\mathbf{q}, j)$.

If a single displacement as in Equation 8.7 at wave vector $\mathbf{q}$ is a solution, then a superposition of such solutions at all $\mathbf{q}$ vectors in the first Brillouin zone,

$$\mathbf{u}_{kl}(t) = \sum_{\mathbf{q}, j} Q_{\mathbf{q}, j} \frac{\mathbf{e}_k(\mathbf{q})}{\sqrt{Mm_{kl}}} \exp(i\mathbf{q} \cdot \mathbf{R}_{kl}^0) \exp(-i\omega(\mathbf{q})t), \quad (8.12)$$

is also solution of Newton's equations in the harmonic approximation. This reflects the completeness of the phonon eigenmodes: they form a basis for the atomic displacements, in close analogy to how Bloch waves form a basis for the electronic problem (section 7.1).

8.2 Limitations of the harmonic approximation

The harmonic approximation relies on the truncation of the potential energy expansion at the second order in the atomic displacements. It is then valid only when atoms vibrate close to their equilibrium positions and the potential energy surface can be approximated as locally quadratic. Under these conditions, the dynamics reduce to a set of non-interacting normal modes with fixed frequencies. This description is not able to capture several important physical phenomena such as:

- **Anharmonicity.** In real materials, third and higher (i.e. *anharmonic*) terms of the Taylor expansion of the potential are necessary to account for phenomena like thermal resistivity and temperature-dependent frequency shifts. Anharmonic effects become increasingly important at elevated temperatures and are entirely neglected within the harmonic framework.
- **Large-amplitude motions and structural instabilities.** When atomic displacements are no longer small, for example near structural phase transitions, the harmonic approximation breaks down. In such cases the potential energy surface cannot be represented by a single quadratic minimum.
- **Soft modes.** A soft mode is a phonon whose frequency decreases towards zero as an external parameter (e.g. temperature, pressure, or composition) is varied. The appearance of a soft mode signals a dynamical instability of the crystal and often precedes a structural phase transition. Within the harmonic approximation, such instabilities manifest as imaginary phonon frequencies, indicating that the assumed equilibrium structure is not dynamically stable.
- **Thermal expansion.** In the harmonic approximation, phonon frequencies are independent of volume, implying zero thermal expansion. Real solids expand upon heating due to anharmonicity, which couples lattice vibrations to changes in volume.
- **Strong electron–phonon coupling.** In some materials, lattice vibrations significantly modify the electronic structure, and the electronic response feeds back onto the phonons. While phonons can still be defined, their frequencies and lifetimes are renormalised by the electronic degrees of freedom, an effect not captured by purely harmonic lattice dynamics.

A commonly used extension of the harmonic approximation is the *quasi-harmonic approximation* (QHA).⁵⁸ In the QHA, harmonic phonon calculations are performed at several different crystal volumes. This allows phonon frequencies to acquire an implicit volume dependence, enabling the calculation of temperature-dependent free energies, equilibrium volumes, and thermal expansion. While the QHA captures important thermodynamic effects, it still neglects explicit phonon–phonon interactions and therefore remains limited. More advanced treatments of lattice dynamics require explicit inclusion of anharmonic terms beyond second order; examples include the theory implemented in software packages such as HiPhive,³⁰ A-TDEP²⁸ and SSCHA.²⁹

8.3 Phonon dispersion and density of states

Having established the harmonic eigenvalue problem in Equation 8.11, we now examine the behaviour of the phonon frequencies across the first Brillouin zone (BZ). For each wave vector $\mathbf{q}$, diagonalisation of the dynamical matrix $\mathbb{D}(\mathbf{q})$ yields $3N$ normal-mode frequencies

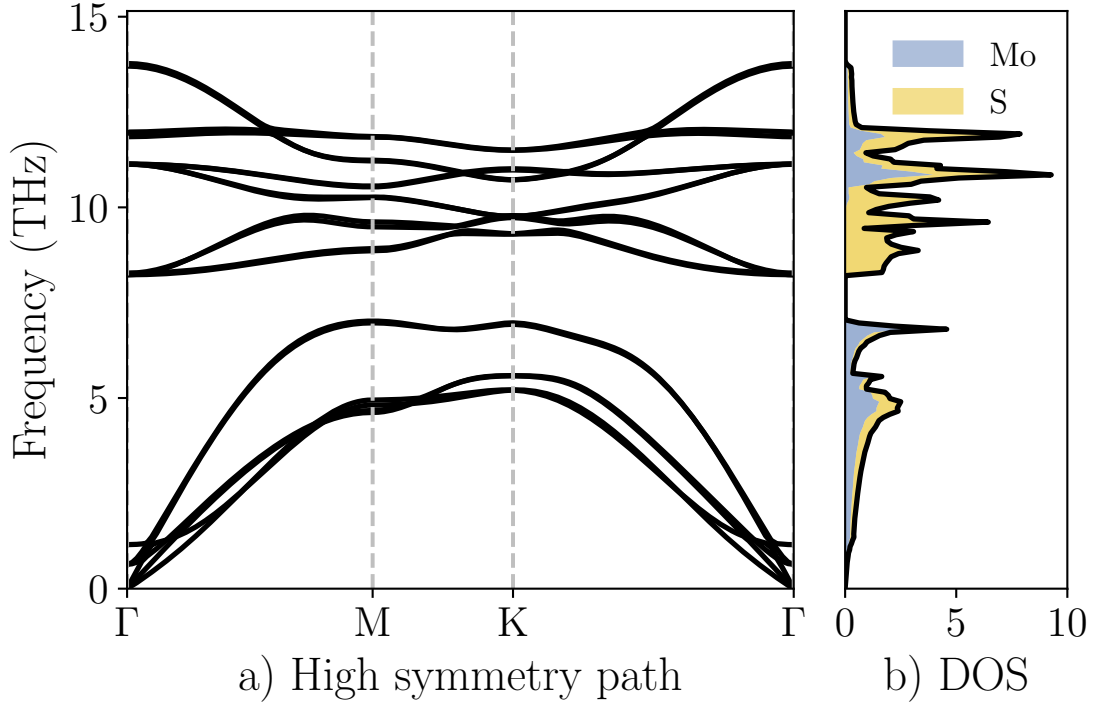


Figure 8.2: Example a) band structure and b) density of states (DOS) of bilayer MoS_2 calculated using PHONOPY.⁵⁹ a) Phonon band structure along the Γ -M-K- Γ high-symmetry path. b) Phonon density of states $g(\omega)$, representing the number of vibrational modes per unit frequency interval, obtained by counting the allowed phonon frequencies over the Brillouin zone and binning them into a frequency histogram; Mo and S contribution to the total density of states are indicated by the corresponding coloured areas.

$\omega(\mathbf{q}, j)$ and associated eigenvectors $\mathbf{e}(\mathbf{q}, j)$. As $\mathbf{q}$ varies through the first Brillouin zone, these frequencies trace out $3N$ curves $\omega(\mathbf{q}, j)$, collectively known as the *phonon dispersion* or *phonon band structure* (see Figure 8.2a). Analogously to what we discussed about the electronic states $\epsilon_{n\mathbf{k}}$ in section 7.2, at fixed index j , the variation of $\omega(\mathbf{q}, j)$ with respect to $\mathbf{q}$ defines a phonon *band* or *branch*. The slope of a phonon band defines the *phonon group velocity*,

$$\mathbf{v}_g(\mathbf{q}, j) = \nabla_{\mathbf{q}} \omega(\mathbf{q}, j), \quad (8.13)$$

which describes the speed and direction at which vibrational energy associated with a given mode propagates through the crystal. For a unit cell containing N atoms, the $3N$ phonon bands naturally separate into 3 *acoustic* modes, and $3N - 3$ *optical* modes. Near Γ , the band dispersion of the acoustic modes is approximately linear,^a and the group velocities equal to the speeds of sounds along the corresponding crystallographic directions.

In analogy to the electronic DOS (section 7.4), the phonon DOS counts the number of vibrational modes per unit frequency:

$$g(\omega) = \sum_{j=1}^{3N} \int_{\text{BZ}} \delta(\omega - \omega(\mathbf{q}, j)) d\mathbf{q}. \quad (8.14)$$

^aIn the presence of a free surface, one of the three acoustic modes has quadratic dispersion — for example, see L. Landau and E. Lifshitz, *Theory of Elasticity* (Pergamon, Oxford, 1995).

As shown in [Figure 8.2b](#), the phonon DOS reveals several important physical features like the energetic distribution of vibrational modes, or the presence of van Hove singularities arising from stationary points of $\omega(\mathbf{q}, j)$. A *projected* phonon DOS (e.g. atom-resolved) can also be computed by weighting the contribution of each mode by $|e_k(\mathbf{q}, j)|^2$. This provides insight into which atoms participate the most to form the displacement pattern of specific phonon modes or frequency ranges.

8.4 Brillouin-Zone sampling and supercell commensurability

In practical phonon calculations, lattice vibrations are represented in a finite real space supercell constructed by replicating the primitive unit cell N_i times along each lattice vector $\mathbf{a}_i$, such that the supercell lattice vectors are $\mathbf{L}_i = N_i \mathbf{a}_i$. A phonon with wave vector $\mathbf{q}$ can be represented exactly in this supercell only if its displacement pattern is periodic over the supercell, which requires

$$\exp(i\mathbf{q} \cdot \mathbf{L}_i) = 1 \quad \text{for all } i. \quad (8.15)$$

Writing $\mathbf{q}$ in the reciprocal lattice basis $\{\mathbf{b}_i\}$,

$$\mathbf{q} = \sum_i^3 \alpha_i \mathbf{b}_i, \quad (8.16)$$

and using $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}$, [Equation 8.15](#) is satisfied when

$$\mathbf{q} \cdot \mathbf{L}_i = \sum_{j=1}^3 \alpha_j N_i (\mathbf{b}_j \cdot \mathbf{a}_i) = 2\pi N_i \alpha_i = 2\pi n_i, \quad n_i \in \mathbb{Z}. \quad (8.17)$$

Therefore,

$$\alpha_i = \frac{n_i}{N_i}, \quad (8.18)$$

so that the set of exact wave vectors accessible in an $N_1 \times N_2 \times N_3$ supercell is

$$\mathbf{q} = \sum_i^3 \frac{n_i}{N_i} \mathbf{b}_i. \quad (8.19)$$

Although this condition is satisfied for any integer n_i , phonon wave vectors that differ by a reciprocal lattice vector of the primitive cell are physically equivalent — we already discussed this aspect in [section 1.8](#). Consequently, only values $n_i = 0, 1, \dots, N_i - 1$ correspond to distinct phonon wave vectors. Physically, the supercell size discretises the first Brillouin zone into a finite set of $\mathbf{q}$ -points commensurate with the supercell periodicity, which constitute the natural reciprocal space representation of the truncated real space force constant model (i.e. the potential). In other words, a primitive unit cell allows to calculate exactly all the phonons at the Γ point, while any supercell allows to calculate exactly all the phonons with $\mathbf{q}$ commensurate with the supercell; diagonalisation of the dynamical matrix at any other $\mathbf{q}$ -point corresponds to a Fourier interpolation of the real space IFCs obtained from the available supercell, and yields an approximate evaluation of the eigenvectors and eigenfrequencies at that point. This makes it possible to compute phonon frequencies on arbitrarily dense grids, or along high-symmetry paths in reciprocal space.

The physical accuracy of phonon spectra obtained from supercell-based calculations is determined by the quality and range of the real-space interatomic force constants. A finite supercell truncates these force constants beyond a characteristic distance, which limits how accurately long-wavelength behaviour and dispersive features of the phonon spectrum can be described; convergence with respect to supercell size is therefore required to ensure that the relevant force-constant interactions are adequately captured. Fourier interpolation does not increase this underlying physical accuracy; it provides a practical means of diagonalising the dynamical matrix at arbitrary wave vectors $\mathbf{q}$ using the real-space force constants obtained from the finite supercell. Within this framework, the numerical accuracy of quantities derived from the phonon spectrum further depends on the density and choice of the sampled $\mathbf{q}$ -points. Meaningful phonon densities of states or dispersions require that the chosen $\mathbf{q}$ -grid or path be sufficiently dense to resolve the relevant spectral features, with convergence assessed independently of the supercell size.

Finally, it is important to stress that all the considerations we made about the reciprocal sampling in [section 7.3](#) for the electronic case hold also for the phonon calculations, as the definition of the reciprocal space is independent on the physical property we want to calculate.

8.5 Numerical evaluation of Interatomic Force Constants

All phonon dispersions, densities of states and any other derived phonon-based quantities ultimately depend on the interatomic force constants entering the dynamical matrix. These force constants must themselves be obtained numerically from an explicit representation of the potential energy surface. Two broad approaches are commonly used:

- **Finite-displacement methods**, where small atomic displacements are applied and the derivatives of the potential are computed, either from classical force fields or from electronic structure calculations, and
- **Density functional perturbation theory (DFPT)**, a linear-response method that computes IFCs from perturbations of the wavefunction, without requiring supercells.

The two methods differ substantially in formulation, computational workflow, and practical advantages.

In the finite displacement approach, phonons are obtained by explicitly displacing atoms from their equilibrium positions and computing the resulting forces. From these forces, the second derivatives of the total energy with respect to atomic displacements are constructed numerically, yielding the force-constant matrix. This method is conceptually simple and largely independent of the underlying method to calculate the forces, whether it be based on classical force fields or electronic-structure formalism, making it applicable to classical dynamics, Hartree–Fock, density functional theory and post-Hartree–Fock methods alike. A major advantage of the finite displacement method is its generality. It can be applied to systems with reduced symmetry, large supercells, defects, surfaces, interfaces, and even non-periodic or partially periodic systems. Furthermore, it naturally lends itself to extensions beyond the harmonic approximation, such as the explicit calculation of higher-order force constants needed for anharmonic phonon properties. However, the finite displacement approach can be computationally expensive. The number of required force calculations scales with the number of atoms in the supercell, and large supercells are often necessary to capture long-range interactions and to accurately sample phonons at small wave vectors. Care must also be taken in choosing the displacement amplitude, which must

be small enough to remain in the harmonic regime while large enough to avoid numerical noise.

Density-functional perturbation theory, by contrast, is a linear-response method that computes phonons directly as the response of the electronic ground state to infinitesimal atomic displacements. Within this framework, the second derivatives of the total energy are obtained by solving a set of perturbative equations, without the need for explicit atomic displacements. As a result, phonon frequencies can be computed directly at arbitrary wave vectors in the Brillouin zone using the primitive unit cell. The principal advantage of density-functional perturbation theory is its efficiency and accuracy for crystalline systems. Because it avoids supercells and numerical differentiation, it typically requires fewer electronic-structure calculations and yields highly precise force constants. It is particularly well suited for calculating phonon dispersions, dielectric properties and electron–phonon coupling in periodic solids. The main limitation of density-functional perturbation theory lies in its restricted scope. It is inherently formulated within density functional theory, hence quantum description, and is not available for Hartree–Fock or correlated post-Hartree–Fock methods. In addition, its application to systems with large unit cells, defects, strong disorder or reduced dimensionality can be computationally demanding.

In summary, the finite displacement method offers broad applicability and conceptual simplicity at the cost of higher computational effort for large systems. Density-functional perturbation theory provides an elegant and efficient route to phonons, but is limited by its reliance on density functional theory and by practical constraints in complex or low-symmetry systems. The choice between the two methods therefore depends on the nature of the system under study, the required level of accuracy and the available computational resources. We now briefly discuss some aspects of both methods which must be carefully considered.

8.5.1 Finite displacements approach

Reliable phonon calculations based on the finite displacement approach require systematic convergence testing with respect to several numerical and methodological parameters. Because phonon frequencies are second derivatives of the total energy with respect to atomic displacements, they are particularly sensitive to small numerical inaccuracies in the underlying force calculations. Failure to properly converge these parameters can lead to spurious features in the phonon spectrum, including artificial soft modes or imaginary frequencies. In what follows, we mention few aspects which require attention to achieve proper convergence of phonon calculations based on the finite displacement method.

- **Supercell size.** The supercell must be sufficiently large to ensure that interatomic force constants decay to negligible values before interacting with their periodic images. If the supercell is too small, long-range interactions are artificially truncated, leading to incorrect force constants and distorted phonon dispersions, especially at small wave vectors. Convergence with respect to supercell size is therefore essential, particularly for polar materials or systems with long-range elastic interactions.
- **$\mathbf{k}$ -point mesh for the electronic SCF calculation.** The forces used to construct the force-constant matrix depend directly on the quality of the electronic ground state. An insufficiently dense $\mathbf{k}$ -point mesh can lead to poorly converged forces, which in turn propagate into significant errors in the phonon frequencies. This effect is especially pronounced in metals and small-gap systems, where accurate Brillouin-zone sampling is crucial.

- **Plane-wave energy cutoff.** In plane-wave-based electronic structure methods, the plane-wave cutoff controls the completeness of the basis set and the spatial resolution of the wavefunctions. Because phonon calculations rely on accurate force differences between displaced configurations, the cutoff must be high enough to ensure that forces are converged to a much tighter tolerance than typically required for total energies. Inadequate cutoffs often manifest as noisy or nonphysical phonon modes.
- **$\mathbf{q}$ -point sampling for phonon dispersions and phonon DOS.** Once the force constants are obtained, phonon dispersions and phonon densities of states are computed by sampling wave vectors $\mathbf{q}$ in the Brillouin zone. The density and uniformity of the $\mathbf{q}$ -point mesh determine the resolution, smoothness, and accuracy of the phonon band structure and phonon DOS. Insufficient $\mathbf{q}$ -point sampling can obscure fine features and lead to inaccurate thermodynamic quantities derived from the phonon spectrum.

The appearance of unstable or imaginary phonon frequencies must be interpreted with care. Such features may indicate insufficient numerical convergence with respect to one or more of the parameters listed above. They may also arise from an inadequate supercell size that fails to capture the true range of interatomic interactions. Only after thorough convergence testing can imaginary phonon modes be reliably attributed to a genuine physical instability of the structure, such as a soft mode associated with a phase transition.

8.5.2 Density Functional Perturbation Theory

Phonon calculations performed using density functional perturbation theory (DFPT) also require careful convergence testing, although the relevant parameters and sources of numerical error differ in part from those of finite-displacement approaches. Because DFPT computes second derivatives of the total energy through a linear-response formalism, the accuracy of the phonon spectrum depends critically on the quality of the underlying electronic ground state and on the resolution of reciprocal-space sampling. In the following, we highlight several key aspects that require careful attention in order to achieve properly converged phonon calculations within density functional perturbation theory.

- **$\mathbf{k}$ -point mesh for the electronic ground state.** DFPT phonons are obtained as perturbations around the converged electronic ground state. An insufficiently dense $\mathbf{k}$ -point mesh leads to inaccurate charge-density responses and perturbed wavefunctions, which directly affects the calculated dynamical matrix. This issue is particularly severe in metals and small-gap systems, where fine Brillouin-zone sampling and appropriate smearing schemes are essential for convergence.
- **Plane-wave energy cutoff.** As in finite-displacement calculations, the plane-wave cutoff controls the completeness of the basis set. In DFPT, both the ground-state wavefunctions and their linear response to atomic displacements must be accurately represented. As a consequence, phonon frequencies often converge more slowly with respect to the cutoff than total energies, and higher cutoffs are typically required to achieve reliable results.
- **$\mathbf{q}$ -point mesh for phonon properties.** DFPT allows phonons to be computed directly at arbitrary wave vectors $\mathbf{q}$ in the Brillouin zone without the use of supercells. The choice of the $\mathbf{q}$ -point mesh determines the resolution of the phonon dispersion relations and the accuracy of derived quantities such as the phonon density of states

and vibrational free energies. Dense and uniform $\mathbf{q}$ -point grids are required to obtain smooth dispersions and well-converged thermodynamic properties.

- **Treatment of long-range interactions.** In polar materials, long-range dipole–dipole interactions give rise to non-analytic contributions to the dynamical matrix near the Brillouin-zone center. Accurate DFPT calculations therefore require proper treatment of these long-range effects, typically through the inclusion of Born effective charges and dielectric constants. Neglecting or poorly converging these quantities can lead to incorrect longitudinal–transverse optical (LO–TO) splittings and spurious phonon features.

As in the finite-displacement approach, the presence of imaginary phonon frequencies in DFPT calculations must be interpreted cautiously. Such modes may arise from insufficient convergence with respect to $\mathbf{k}$ -point sampling or plane-wave cutoff. They may also reflect numerical instabilities in the solution of the linear-response equations. Only after all relevant parameters have been carefully converged can imaginary frequencies be confidently associated with genuine dynamical instabilities of the crystal structure.

Overall, DFPT offers an efficient and accurate framework for phonon calculations in periodic systems, provided that convergence with respect to electronic and reciprocal-space parameters is thoroughly established. Its strengths lie in the direct access to phonons at arbitrary wave vectors and in the avoidance of large supercells, while its reliability depends critically on rigorous convergence testing analogous in spirit, though not identical in detail, to that required for finite-displacement methods.

8.6 Evaluation of bond strength

To quantify the strength of a bond between two atoms A and B we can exploit the second-order (harmonic) interatomic force constants. Focusing on two specific atoms A and B, the interatomic force constants between them form a 3×3 matrix (see Equation 8.5),

$$\mathbb{V}_{A,B} = \begin{pmatrix} \frac{\partial^2 V}{\partial u_{x,A} \partial u_{x,B}} & \frac{\partial^2 V}{\partial u_{x,A} \partial u_{y,B}} & \frac{\partial^2 V}{\partial u_{x,A} \partial u_{z,B}} \\ \frac{\partial^2 V}{\partial u_{y,A} \partial u_{x,B}} & \frac{\partial^2 V}{\partial u_{y,A} \partial u_{y,B}} & \frac{\partial^2 V}{\partial u_{y,A} \partial u_{z,B}} \\ \frac{\partial^2 V}{\partial u_{z,A} \partial u_{x,B}} & \frac{\partial^2 V}{\partial u_{z,A} \partial u_{y,B}} & \frac{\partial^2 V}{\partial u_{z,A} \partial u_{z,B}} \end{pmatrix}, \quad (8.20)$$

which describes how a displacement of atom B along a given Cartesian direction induces a force on atom A along another Cartesian direction. This matrix contains both longitudinal and transverse components of the interaction and therefore encodes the full directional character of the bond. A possible easy descriptor for the average character of the bond is the absolute value of the average of the trace of such matrix: by taking the absolute value, we focus on the magnitude of the restoring interaction regardless of its sign, which may depend on conventions or on the relative orientation of the atoms.

We consider the trace as a relevant quantity because it remains invariant under spatial transformations of the system coordinates, unlike the full matrix, and thus contains the physically meaningful information on the overall coupling strength. Indeed, while individual matrix elements depend on the position of the atoms in the Cartesian frame, the trace is invariant under orthogonal rotations of the coordinate system and therefore provides a robust scalar descriptor of the bond. In this sense, the average of the trace can be interpreted as an orientation-independent measure of the stiffness associated with relative displacements of atoms A and B. Within the harmonic approximation, larger values of

trace average correspond to stronger restoring forces upon relative displacement of the atoms A and B, and therefore to a stronger bond; conversely, smaller values indicate weaker mechanical coupling between the two atoms. This definition allows for a consistent and quantitative comparison of bond strengths across different atomic pairs, chemical environments, or structural configurations within the same theoretical framework.

Characterisation of the mechanical strength of a bond is related to its chemical nature. In particular, one may wish to determine to what extent a bond between two atoms A and B is covalent or ionic and relate this to the bond strength. At a qualitative level, these two limiting bonding mechanisms can be distinguished in terms of the distribution of electronic charge between the atoms.

A *covalent bond* arises when two atoms share valence electrons. In this situation the electronic density is distributed between the atoms, forming a region of enhanced charge density along the line connecting the nuclei. Conversely, an *ionic bond* corresponds to a situation in which electrons are transferred from one atom to another. This produces oppositely charged ions which interact primarily through electrostatic Coulomb attraction. In real materials, however, bonds rarely fall exactly into one of these two ideal categories; quantifying the bond character therefore requires assigning portions of the electronic distribution to individual atoms. A fundamental difficulty arises because there is no unique way to partition the electronic density of a system into atomic contributions. The electronic density is a continuous quantity extending throughout space, and any decomposition into atomic fragments necessarily involves an arbitrary choice of partitioning scheme. As a consequence, any quantitative definition of bond covalency or ionicity is inherently method-dependent.

One class of approaches is based on projecting the electronic wavefunctions onto atomic-like orbitals. In this framework the electronic states are expanded in terms of basis functions centred on individual atoms, and the resulting coefficients are used to estimate atomic charges and bond populations. A widely known example is the *Mulliken population analysis*. Within Mulliken analysis, the electronic population associated with an atom or a bond is obtained by summing contributions from the expansion coefficients of the wave function in the atomic orbital basis. In particular, the so-called *bond population* provides a measure of the electron sharing between two atoms and is often interpreted as an indicator of covalent bonding. However, Mulliken and related projection-based analyses suffer from several important drawbacks. The main difficulty arises from the presence of overlap integrals between atomic basis functions. Because atomic orbitals centred on different atoms are generally not orthogonal, the overlap population must be distributed between the atoms according to a chosen convention. To see why this introduces an ambiguity, consider the expansion of a molecular or crystalline electronic state in terms of atomic orbitals,

$$\psi = \sum_i c_i \phi_i, \quad (8.21)$$

where ϕ_i are basis functions centred on atoms and c_i are the corresponding expansion coefficients. The electronic population associated with two orbitals ϕ_i and ϕ_j involves the overlap integral

$$S_{ij} = \langle \phi_i | \phi_j \rangle. \quad (8.22)$$

If the basis functions were orthonormal, one would have $S_{ij} = \delta_{ij}$ and the population associated with each orbital would be uniquely determined by the coefficients $|c_i|^2$. However, when the orbitals belong to different atoms they typically overlap in space and therefore $S_{ij} \neq 0$. In this situation, the electronic population associated with a given orbital cannot be defined independently of the others, because cross terms proportional to $c_i c_j S_{ij}$ appear

in the total population. These cross terms represent electronic density located in the spatial region where the orbitals overlap, that is, in the region between atoms. Since this overlap density cannot be uniquely attributed to either atom, a convention must be introduced to partition it. In the Mulliken scheme, this overlap contribution is divided equally between the two atoms involved. As a result, the population assigned to orbital i contains a term of the form

$$P_i = |c_i|^2 + \frac{1}{2} \sum_{j \neq i} c_i c_j S_{ij}. \quad (8.23)$$

While this prescription is simple and symmetric, it is not unique. Other partitioning choices would lead to different atomic populations and therefore different estimates of bond populations and atomic charges. Moreover, the magnitude of the overlap integrals depends strongly on the specific form and number of basis functions used in the calculation. Changing the basis set therefore changes the amount of overlap density and consequently the populations assigned to atoms and bonds. This explains why Mulliken populations can vary significantly between calculations that use different basis sets, even when the underlying electronic structure remains essentially unchanged. This lack of robustness makes it difficult to interpret the results quantitatively or to compare them across different calculations.

More reliable approaches instead rely directly on the *electronic charge density*. The charge density $\rho(\mathbf{r})$ is a physical observable; for this reason, methods based on the charge density generally provide a more physically meaningful description of bonding. Several schemes have been developed to partition the charge density into atomic regions. One widely used approach is the *Hirshfeld analysis*,⁶⁰ in which the density at each point in space is assigned to atoms according to weighting functions derived from the densities of isolated atoms. Another important method is the *Bader analysis*, also known as the *atoms-in-molecules* partitioning.^{61,62} In the Bader approach, space is divided into atomic basins bounded by surfaces across which the flux of the density gradient vanishes. Each basin contains exactly one nucleus and corresponds to a region in which gradient paths of the charge density lead toward that nucleus. Integration of the density within these basins yields well-defined atomic charges that can be used to assess the degree of charge transfer between atoms. By comparing the charge associated with each atom to its neutral reference value, one may estimate the amount of electron transfer occurring in a bond. Large charge transfer typically indicates a more ionic bond, whereas smaller charge transfer combined with significant charge accumulation in the interatomic region indicates a stronger covalent contribution. Although these methods still involve a certain level of convention in defining atomic regions, their reliance on the charge density rather than on basis-dependent wave function projections makes them significantly more robust and transferable. For this reason, charge-density-based analyses such as Hirshfeld or Bader partitioning are generally preferred when assessing the ionic or covalent character of chemical bonds in atomistic simulations.

Another simple and widely used qualitative indicator of the covalent or ionic character of a bond is based on the difference in electronegativity between the bonded atoms. Electronegativity is a measure of the tendency of an atom to attract electrons toward itself when forming a chemical bond. Atoms with higher electronegativity tend to draw electronic charge toward their nucleus more strongly than atoms with lower electronegativity. One of the most commonly used electronegativity scales is the *Pauling electronegativity* scale, introduced by Linus Pauling.⁶³ In this scale, electronegativity values are derived empirically from bond energies and provide a relative measure of the electron-attracting ability of different elements. If two bonded atoms have similar electronegativities, the

electrons are shared more equally between them, leading to a bond with predominantly covalent character. Conversely, if the electronegativity difference between the atoms is large, the electronic charge tends to be transferred toward the more electronegative atom, producing partial charges and therefore a bond with stronger ionic character. Although this approach provides only a qualitative estimate and does not explicitly account for the detailed electronic structure of the system, the electronegativity difference remains a useful guideline for anticipating trends in bond polarity and charge distribution in molecules and solids.

It is then interesting to discuss possible relations between the covalent or ionic character of a bond and its mechanical strength as quantified by the interatomic force constants. Covalent bonds often display relatively large force constants. This can be understood by recalling that covalent bonding involves the sharing of electrons in a localized region between the atoms. When the atoms are displaced from their equilibrium positions, the overlap between the atomic orbitals forming the bond changes rapidly, leading to a strong variation of the electronic energy and therefore to large restoring forces. As a consequence, many strongly covalent materials, such as network-forming solids like diamond, exhibit large interatomic force constants and high vibrational frequencies. Ionic bonds, on the other hand, are primarily governed by long-range electrostatic interactions between charged ions. Although these interactions can also produce strong binding, the restoring forces are typically more isotropic and depend less sensitively on the precise overlap of electronic orbitals. For this reason, ionic systems may exhibit somewhat smaller directional stiffness compared to strongly covalent bonds, even when the overall binding energy remains large. Finally, it is worthy to note that in compounds formed by only one atomic type, electrons are shared in equal amount along the bond, which then shows a covalent character; however, such compounds may exhibit very different bond stiffness, like for example diamond and graphite.

Therefore, it is important to emphasize that there is no universal one-to-one correspondence between bond covalency or ionicity and the magnitude of the interatomic force constants. Bond strength measured through the harmonic force constants reflects the curvature of the potential energy surface with respect to atomic displacements, whereas covalency and ionicity describe the distribution of the electronic charge. These two aspects are related but conceptually distinct. For example, some ionic crystals can exhibit very stiff bonds due to strong Coulomb interactions, while certain covalent bonds may appear softer if the bonding orbitals allow significant structural flexibility. Therefore, analysing both the interatomic force constants and the charge distribution provides complementary information. The former quantifies the mechanical stiffness of the bond, while the latter characterises its electronic nature. Together, these analyses offer a more complete description of chemical bonding in atomistic simulations.

Chapter 9

Electron-phonon coupling

Electron–phonon coupling describes the interaction between electronic degrees of freedom and lattice vibrations and constitutes a fundamental mechanism behind many key properties of solids, including electrical resistivity, carrier mobility, superconductivity, temperature-dependent band renormalization and phonon-assisted optical processes. Microscopically, this interaction originates from the dependence of the electronic Hamiltonian on the nuclear positions: any displacement of the atoms modifies the electronic potential and therefore the electronic states and energies.

Within the Born–Oppenheimer approximation, the electronic Hamiltonian $\hat{H}_e$ depends parametrically on the set of nuclear coordinates $\{\mathbf{R}_I\}$.

Expanding $\hat{H}_e$ around the equilibrium configuration $\{\mathbf{R}_I^0\}$ in terms of small atomic displacements $\mathbf{u}_I = \mathbf{R}_I - \mathbf{R}_I^0$ yields

$$\hat{H}_e(\{\mathbf{R}_I\}) = \hat{H}_e(\{\mathbf{R}_I^0\}) + \sum_I \mathbf{u}_I \cdot \frac{\partial \hat{H}_e}{\partial \mathbf{R}_I} + \mathcal{O}(\mathbf{u}^2). \quad (9.1)$$

The term linear in the displacements defines the electron–phonon interaction, which couples electronic states to lattice vibrations. In a crystalline solid, atomic displacements are conveniently expressed in terms of phonon normal modes. For an atom κ in the unit cell located at lattice vector $\mathbf{R}$, the displacement can be written in terms of the creation and annihilation operators $\hat{a}_{\mathbf{q}\nu}$ and $\hat{a}_{\mathbf{q}\nu}^\dagger$ as

$$\mathbf{u}_\kappa(\mathbf{R}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{q}\nu} \sqrt{\frac{\hbar}{2m_\kappa\omega_{\mathbf{q}\nu}}} \mathbf{e}_{\kappa\nu}(\mathbf{q}) (\hat{a}_{\mathbf{q}\nu} e^{i\mathbf{q}\cdot\mathbf{R}} + \hat{a}_{\mathbf{q}\nu}^\dagger e^{-i\mathbf{q}\cdot\mathbf{R}}), \quad (9.2)$$

where N is the number of unit cells, $\mathbf{q}$ the phonon wave vector, ν the branch index, $\omega_{\mathbf{q}\nu}$ the phonon frequency, m_κ the atomic mass and $\mathbf{e}_{\kappa\nu}(\mathbf{q})$ the phonon polarization vector. Substituting this expression into the linearized Hamiltonian leads to the electron–phonon interaction Hamiltonian

$$\hat{H}_{\text{ep}} = \sum_{\mathbf{q}\nu} \left(\hat{a}_{\mathbf{q}\nu} + \hat{a}_{-\mathbf{q}\nu}^\dagger \right) \hat{V}_{\mathbf{q}\nu}, \quad (9.3)$$

where the perturbation operator associated with a phonon mode $(\mathbf{q}, \nu)$ is given by

$$\hat{V}_{\mathbf{q}\nu} = \sum_{\kappa} \sqrt{\frac{\hbar}{2M_\kappa\omega_{\mathbf{q}\nu}}} \mathbf{e}_{\kappa\nu}(\mathbf{q}) \cdot \frac{\partial \hat{H}_e}{\partial \mathbf{R}_\kappa} e^{i\mathbf{q}\cdot\mathbf{R}}. \quad (9.4)$$

The strength of the electron–phonon interaction is quantified by matrix elements between electronic Bloch states. For electronic eigenstates $\psi_{n\mathbf{k}}$ with energies $\epsilon_{n\mathbf{k}}$, the electron–phonon matrix element reads

$$g_{mn\nu}(\mathbf{k}, \mathbf{q}) = \left\langle \psi_{m, \mathbf{k}+\mathbf{q}} \left| \hat{V}_{\mathbf{q}\nu} \right| \psi_{n, \mathbf{k}} \right\rangle, \quad (9.5)$$

and represents the probability amplitude for an electron to scatter from state $|n, \mathbf{k}\rangle$ to $|m, \mathbf{k} + \mathbf{q}\rangle$ by absorbing or emitting a phonon of wave vector $\mathbf{q}$ and branch ν . These matrix elements enter directly into scattering rates, which can be evaluated using Fermi’s golden rule. The electron linewidth (inverse lifetime) due to phonon scattering is given by

$$\Gamma_{n\mathbf{k}} = \frac{2\pi}{\hbar} \sum_{m, \mathbf{q}, \nu} |g_{mn\nu}(\mathbf{k}, \mathbf{q})|^2 [(n_{\mathbf{q}\nu} + 1 - f_{m, \mathbf{k}+\mathbf{q}}) \delta(\epsilon_{n\mathbf{k}} - \epsilon_{m, \mathbf{k}+\mathbf{q}} - \hbar\omega_{\mathbf{q}\nu}) + (n_{\mathbf{q}\nu} + f_{m, \mathbf{k}+\mathbf{q}}) \delta(\epsilon_{n\mathbf{k}} - \epsilon_{m, \mathbf{k}+\mathbf{q}} + \hbar\omega_{\mathbf{q}\nu})], \quad (9.6)$$

where $n_{\mathbf{q}\nu}$ and $f_{n\mathbf{k}}$ are the Bose–Einstein and Fermi–Dirac occupation factors, respectively. These processes govern temperature-dependent transport properties and quasiparticle lifetimes.

For metallic systems, it is often useful to introduce a dimensionless measure of the overall coupling strength, the electron–phonon coupling constant λ , defined as

$$\lambda = \sum_{\mathbf{q}\nu} \frac{2}{\hbar\omega_{\mathbf{q}\nu} N(E_F)} \sum_{n, m, \mathbf{k}} |g_{mn\nu}(\mathbf{k}, \mathbf{q})|^2 \delta(\epsilon_{n\mathbf{k}} - E_F) \delta(\epsilon_{m, \mathbf{k}+\mathbf{q}} - E_F), \quad (9.7)$$

where $N(E_F)$ is the electronic density of states at the Fermi level. This quantity plays a central role in the theory of conventional superconductivity and in describing mass enhancement and band renormalization effects.

In practical first-principles calculations, electron–phonon coupling is most efficiently evaluated within linear-response frameworks, where both phonon modes and the induced variation of the electronic potential are obtained self-consistently. Accurate results require dense sampling of electronic $\mathbf{k}$ -points and phononic $\mathbf{q}$ -points, as well as careful convergence testing and interpolation schemes. Despite these computational challenges, electron–phonon coupling calculations provide a powerful and predictive link between microscopic lattice dynamics and macroscopic electronic, optical, and transport phenomena.

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